One Identity Manager 9.3

Administration Guide for Connecting to Microsoft Entra

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One Identity Manager Administration Guide for Connecting to Microsoft Entra
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For the most recent documents and product information, see Online product documentation.

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Managing Microsoft Entra ID environments

One Identity Manager offers simplified user account administration for Microsoft Entra ID. One Identity Manager concentrates on setting up and editing user accounts and providing the required permissions. To equip users with the required permissions, One Identity Manager maps subscriptions, service plans, groups, and administration roles. This makes it possible to use Identity and Access Governance processes, including attestation, Identity Audit, user account management and system entitlements, IT Shop, or report subscriptions for Microsoft Entra ID tenants.

One Identity Manager provides company identities with the necessary user accounts. There are different ways for you to connect identities to their user accounts. You can also manage user accounts independently of identities and thus set up administrator user accounts.

Additional information about the Microsoft Entra ID core directory, such as tenants and verified domains, is loaded into the One Identity Manager database by data synchronization. There are limited options for customizing this information in One Identity Manager due to the complex dependencies and far-reaching effects of any changes.

For more information about the Microsoft Entra ID structure, see the *Microsoft Entra ID documentation* from Microsoft.

NOTE: The Microsoft Entra ID module must be installed as a prerequisite for managing One Identity Manager in Microsoft Entra ID Module For more information about installing, see the *One Identity Manager Installation Guide*.

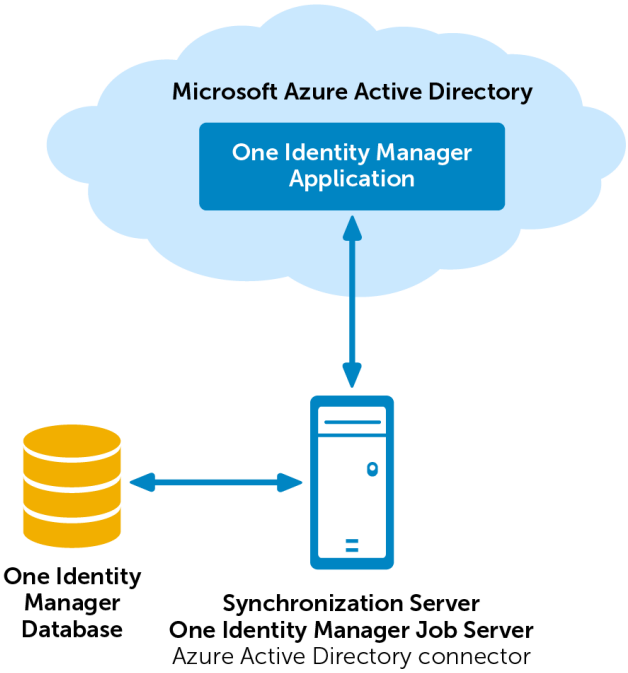
Architecture overview

To access Microsoft Entra ID tenant data, the Microsoft Entra ID connector is installed on a synchronization server. The synchronization server ensures data is compared between the One Identity Manager database and Microsoft Entra ID. The Microsoft Entra ID connector uses the Microsoft Graph API for accessing Microsoft Entra ID data.

The Microsoft Entra ID connector must authenticate itself on the Microsoft Entra ID tenant to access Microsoft Entra ID tenant data. Authentication is carried out by an application for

One Identity Manager that is integrated in the Microsoft Entra ID tenant and equipped with the respective access permissions.

Figure 1: Architecture for synchronization



One Identity Manager users for managing Microsoft Entra ID

The following users are involved in the administration of Microsoft Entra ID.

Table 1: Users

User	Tasks
Target system administrators	Target system administrators must be assigned to the **Target systems | Administrators** application role. Users with this application role: • Administer application roles for individual target system types. • Specify the target system manager. • Set up other application roles for target system managers if required.

User	Tasks
	• Specify which application roles for target system managers are mutually exclusive. • Authorize other identities to be target system administrators. • Do not assume any administrative tasks within the target system.
Target system managers	Target system managers must be assigned to the **Target systems | Microsoft Entra ID** application role or a child application role. Users with this application role: • Assume administrative tasks for the target system. • Create, change, or delete target system objects. • Edit password policies for the target system. • Prepare groups to add to the IT Shop. • Can add identities that do not have the **Primary identity** identity type. • Configure synchronization in the Synchronization Editor and define the mapping for comparing target systems and One Identity Manager. • Edit the synchronization's target system types and outstanding objects. • Authorize other identities within their area of responsibility as target system managers and create child application roles if required.
One Identity Manager administrators	One Identity Manager administrator and administrative system users Administrative system users are not added to application roles. One Identity Manager administrators: • Create customized permissions groups for application roles for role-based login to administration tools in the Designer as required. • Create system users and permissions groups for non role-based login to administration tools in the Designer as required. • Enable or disable additional configuration parameters in the Designer as required. • Create custom processes in the Designer as required.

User	Tasks
	• Create and configure schedules as required. • Create and configure password policies as required.
Administrators for the IT Shop	Administrators must be assigned to the **Request & Fulfillment | IT Shop | Administrators** application role. Users with this application role: • Assign groups to IT Shop structures.
Product owners for the IT Shop	Product owners must be assigned to the **Request & Fulfillment | IT Shop | Product owners** application role or a child application role. Users with this application role: • Approve through requests. • Edit service items and service categories under their management.
Administrators for organizations	Administrators must be assigned to the **Identity Management | Organizations | Administrators** application role. Users with this application role: • Assign groups to departments, cost centers, and locations.
Business roles administrators	Administrators must be assigned to the **Identity Management | Business roles | Administrators** application role. Users with this application role: • Assign groups to business roles.

Configuration parameters for managing Microsoft Entra ID environments

Use configuration parameters to configure the behavior of the system's basic settings. One Identity Manager provides default settings for various configuration parameters. Check the configuration parameters and modify them as necessary to suit your requirements.

Configuration parameters are defined in the One Identity Manager modules. Each One Identity Manager module can also install configuration parameters. In the Designer, you can find an overview of all configuration parameters in the **Base data > General > Configuration parameters** category.

For more information, see Configuration parameters for managing a Microsoft Entra ID environment on page 283.

2

Synchronizing a Microsoft Entra ID environment

NOTE: Synchronization of the following national cloud deployments with the Microsoft Entra ID connector is not supported.

- Microsoft Cloud for US Government (L5)
- Microsoft Cloud Germany
- Microsoft Entra ID and Microsoft 365 operated by 21Vianet in China

For more information, see https://support.oneidentity.com/KB/312379.

The One Identity Manager Service is responsible for synchronizing data between the One Identity Manager database and the Microsoft Entra ID tenant.

This sections explains how to:

- Set up synchronization to import initial data from Microsoft Entra ID tenant to the One Identity Manager database.
- Adjust a synchronization configuration to synchronize different Microsoft Entra ID tenants with the same synchronization project, for example.
- Start and deactivate the synchronization.
- Analyze synchronization results.

TIP: Before you set up synchronization with a Microsoft Entra ID tenant, familiarize yourself with the Synchronization Editor. For more information about this tool, see the *One Identity Manager Target System Synchronization Reference Guide*.

Detailed information about this topic

- Setting up initial synchronization with a Microsoft Entra ID tenant on page 17
- Adjusting the synchronization configuration for Microsoft Entra ID environments on page 36
- Running synchronization on page 51
- Tasks following synchronization on page 55

- Troubleshooting on page 59
- Editing Microsoft Entra ID system objects on page 290

Setting up initial synchronization with a Microsoft Entra ID tenant

The Synchronization Editor provides a project template that can be used to set up the synchronization of user accounts and permissions for the Microsoft Entra ID environment. You use these project templates to create synchronization projects with which you import the data from a Microsoft Entra ID tenant into your One Identity Manager database. In addition, processes are created that are required to provision changes to target system objects from the One Identity Manager database into the target system.

To load Microsoft Entra ID tenant objects into the One Identity Manager database for the first time

1. Ensure the Microsoft Entra ID tenant has a license for the **SharePoint Online** service.

   NOTE: If no such license is available, an error will occur when loading the Microsoft Entra ID user accounts. For more information, see Possible errors when synchronizing an Microsoft Entra ID tenant on page 281.

2. Register a One Identity Manager application in your Microsoft Entra ID tenant.

   Depending on how the One Identity Manager application is registered in the Microsoft Entra ID tenant, either a user account with sufficient permissions or the secret key is required.

3. The One Identity Manager components for managing Microsoft Entra ID tenants are available if the **TargetSystem | AzureAD** configuration parameter is set.
   - In the Designer, check if the configuration parameter is set. Otherwise, set the configuration parameter and compile the database.

     NOTE: If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.

   - Other configuration parameters are installed when the module is installed. Check the configuration parameters and modify them as necessary to suit your requirements.

4. Install and configure a synchronization server and declare the server as a Job server in One Identity Manager.
5. Create a synchronization project with the Synchronization Editor.

Detailed information about this topic

- Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18
- Users and permissions for synchronizing with Microsoft Entra ID on page 22
- Setting up the Microsoft Entra ID synchronization server on page 24
- Creating a synchronization project for initial synchronization of a Microsoft Entra ID tenant on page 28
- Customizing synchronization projects to invite guest users on page 38
- Configuration parameters for managing a Microsoft Entra ID environment on page 283
- Default project template for Microsoft Entra ID on page 287

Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant

To synchronize data between One Identity Manager and Microsoft Entra ID, you must register an application in the Microsoft Entra ID tenants. The Microsoft Entra ID connector uses the One Identity Manager application to authenticate itself to the Microsoft Entra ID tenant.

- Register the One Identity Manager application in the Microsoft Azure management portal (https://portal.azure.com/) or in the Microsoft Entra ID admin center (https://admin.microsoft.com/).

  > NOTE: An application ID is created when you add One Identity Manager as an application to Microsoft Entra ID. You need the application ID for setting up the synchronization project.

  For more information about registering an application, see https://learn.microsoft.com/en-us/entra/identity-platform/quickstart-register-app?tabs=certificate.

- There are two different ways to authenticate the application.
  - Authentication in the directory user context (delegated permissions)

    If you use authentication in the directory user context, you need a user account with sufficient permissions when setting up the synchronization project.
  - Authentication in the application context (application entitlements)

    If you use authentication in the context of an application, you need the value of the secret when setting up the synchronization project. The secret is generated when the One Identity Manager application is registered with the Microsoft Entra ID tenant.

> NOTE: The key is only valid for a limited period and must be renewed when it expires.

To configure authentication in the directory user context (delegated permissions)

1. In the Microsoft Azure management portal, select your application under **App registrations**.
2. Configure the following settings under **Manage > Authentication**.
   a. In the **Platform configurations** section, click **Add a platform** and, under **Configure platforms**, select the **Mobile and desktop applications** tile.
      i. Under **Custom redirect URIs**, you can specify any URI.
      ii. Click **Configure**.
   b. In the **Supported account types** section, select **Accounts in this organization directory only (single tenant)**.
   c. In the **Advanced settings** section, enable the **Allow public client flows** option.
3. Configure the permissions under **Manage > API permissions**.
   a. In the **Configured permissions** section, click **Add a permission**.
      i. Under **Request API permissions > Microsoft APIs**, select the tile **Microsoft Graph**.
      ii. Select **Delegated permissions** and select the following permissions:
         - **Directory.AccessAsUser.All** (Access directory as the signed in user)
         - **Directory.ReadWrite.All** (Read and write directory data)
         - **AuditLog.Read.All** (Read all login times)
         - **CustomSecAttributeDefinition.Read.All** (Read all custom security attribute definitions)
         - **CustomSecAttributeAssignment.ReadWrite.All** (Read and write all custom security assignments for all users and applications)
         - **UserAuthenticationMethod.ReadWrite.All** (Read and write authentication methods for all users)

           > NOTE: This permission is not available in B2C tenants.
         - **User.ReadWrite.All** (Read and write all users' full profile)
         - **Group.ReadWrite.All** (Read and write all groups)
         - **RoleManagement.ReadWrite.Directory** (Read and write all directory RBAC settings)
         - **RoleAssignmentSchedule.ReadWrite.Directory** (Read, update, and delete all policies for privileged role assignments of

your company's directory)

- **RoleEligibilitySchedule.ReadWrite.Directory** (Read, update, and delete all eligible role assignments and schedules for your company's directory)
- **Policy.Read.All** (Policy.Read.All)
- **openid** (Sign users in)
- **ChannelMember.Read.All** (Read the members of channels)

> NOTE: This is only required if you are synchronizing Microsoft Teams.

iii. Click **Add permissions**.

b. In the **Configured permissions** section, click **Grant admin consent for ...** and confirm the security prompt with **Yes**.

This enables the configured permissions.

To configure authentication in the application context (application entitlements)

1. In the Microsoft Azure management portal, under **App registrations**, select your application.
2. Configure the following settings under **Manage > Authentication**.
   a. In the **Platform configurations** section, click **Add platform**, and under **Configure platforms**, select the **Web** tile.
      i. Under **Redirect URIs**, you can specify any URI.
      ii. Click **Configure**.
   b. In the **Supported account types** section, select **Accounts in this organization directory only (single tenant)**.
   c. In the **Advanced settings** section, enable the **Allow public client flows** option.
3. Configure the permissions under **Manage > API permissions**.
   a. In the **Configured Permissions** section, click **Add a permission**.
      i. Under **Request API permissions > Microsoft APIs**, select the tile **Microsoft Graph**.
      ii. Select **Application entitlements** and select the following permissions:
         - **Application.ReadWrite.All** (Read and write all applications)
         - **CustomSecAttributeDefinition.Read.All** (Read all custom security attribute definitions)
         - **CustomSecAttributeAssignment.ReadWrite.All** (Read and write all custom security assignments for all users and applications)
         - **Directory.ReadWrite.All** (Read directory data)

- **Group.ReadWrite.All** (Read and write all groups)
- **Policy.Read.All** (Read your organization's policies)
- **RoleManagement.ReadWrite.Directory** (Read and write all directory RBAC settings)
- **RoleAssignmentSchedule.ReadWrite.Directory** (Read, update, and delete all policies for privileged role assignments of your company's directory)
- **RoleEligibilitySchedule.ReadWrite.Directory** (Read, update, and delete all eligible role assignments and schedules for your company's directory)
- **User.ReadWrite.All** (Read and write all users' full profile)
- **UserAuthenticationMethod.ReadWrite.All** (Read and write authentication methods for all users)

  NOTE: This permission is not available in B2C tenants.
- **AuditLog.Read.All** (Read all login times)
- **ChannelMember.Read.All** (Read the members of all channels)

  NOTE: This is only required if you are synchronizing Microsoft Teams.

iii. Click **Add permissions**.

b. In the **Configured permissions** section, click **Grant admin consent for ...** and confirm the security prompt with **Yes**.

This enables the configured permissions.

4. Under **Manage > Certificates & secrets**, create a secret or use a certificate.

a. Using a secret:

i. In the **Client secrets** section, click **New client secret**.

ii. Enter a description and the validity period for the secret.

iii. Click **Add**.

iv. The secret is generated and displayed in the **Client secrets** section.

b. Using a connection certificate:

i. You require an **X.509 certificate** including a private key as a `*.CER` or a `*.PFX` file.

ii. You can use a self-signed certificate. For information on how to create it, see **https://learn.microsoft.com/en-us/azure/active-directory/develop/howto-create-self-signed-certificate**.

iii. Import the certificate (`*.PFX`) into the certificate store of the Job server and the administrative workstation used to set up synchronization.

- OR -

Open the `'.CER` file and copy the "Thumbprint" value from the properties. This is required in the connection dialog-

5. In the **Microsoft Entra ID** portal, assign the user administrator role.
   a. Select **Roles and admins** and select the following roles:
      - **User Administrator**
      - **Privileged Authentication Administrator**
   b. Under **Add assignments**, select the application you want to assign.
   c. Click **Assign**.

Related topics

- Users and permissions for synchronizing with Microsoft Entra ID on page 22
- Creating a synchronization project for initial synchronization of a Microsoft Entra ID tenant on page 28

Users and permissions for synchronizing with Microsoft Entra ID

The following users play a role in synchronizing One Identity Manager with a Microsoft Entra ID tenant.

Table 2: Users for synchronization

User	Permissions
User for accessing Microsoft Entra ID or The secret's value	Depending on how the One Identity Manager application is registered in the Microsoft Entra ID tenant, either a user account with sufficient permissions or the secret is required. • If you use authentication in the context of a directory user (delegated permissions), you require a user account that is a member in the **Global administrator** Microsoft Entra ID administration role when you set up the synchronization project. Use the Microsoft Entra ID Admin Center to assign the Microsoft Entra ID administrator role to the user account. For more information on managing permissions in Microsoft Entra ID, see the *Microsoft documentation*. NOTE: The user account used to access Microsoft Entra ID must not use multifactor

User	Permissions
	authentication to allow automated logins in a user context. • If you use authentication in the context of an application (application entitlements), you need the value of the secret when you set up the synchronization project. The secret is generated when the One Identity Manager application is registered with the Microsoft Entra ID tenant. NOTE: The key is only valid for a limited period and must be renewed when it expires.
One Identity Manager Service user account	The user account for the One Identity Manager Service requires user permissions to carry out operations at file level (adding and editing directories and files). The user account must belong to the **Domain users** group. The user account must have the **Login as a service** extended user permissions. The user account requires permissions for the internal web service. NOTE: If the One Identity Manager Service runs under the network service (**NT Authority\NetworkService**), you can grant permissions for the internal web service with the following command line call: `netsh http add urlacl url=http://<IP address>:<port number>/ user="NT AUTHORITY\NETWORKSERVICE"` The user account needs full access to the One Identity Manager Service installation directory in order to automatically update One Identity Manager. In the default installation, One Identity Manager is installed under: • `%ProgramFiles(x86)%\One Identity` (on 32-bit operating systems) • `%ProgramFiles%\One Identity` (on 64-bit operating systems)
User for accessing the One Identity Manager database	The **Synchronization** default system user is provided to run synchronization using an application server.

Related topics

- Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18

Setting up the Microsoft Entra ID synchronization server

All One Identity Manager Service actions are run against the target system environment on the synchronization server. Data entries required for synchronization and administration with the One Identity Manager database are processed by the synchronization server.

The One Identity Manager Service with the Microsoft Entra ID connector must be installed on the synchronization server.

Detailed information about this topic

- System requirements for the Microsoft Entra ID synchronization server on page 24
- Installing One Identity Manager Service with a Microsoft Entra ID connector on page 25

System requirements for the Microsoft Entra ID synchronization server

To set up synchronization with a Microsoft Entra ID tenant, a server must be available with the following software installed on it:

- Windows operating system

  The following versions are supported:

  - Windows Server 2025
  - Windows Server 2022
  - Windows Server 2019
  - Windows Server 2016
  - Windows Server 2012 R2
  - Windows Server 2012
- .NET 8 Desktop Runtime

  NOTE: Take the target system manufacturer's recommendations into account.

Installing One Identity Manager Service with a Microsoft Entra ID connector

The One Identity Manager Service with the Microsoft Entra ID connector must be installed on the synchronization server. The synchronization server must be declared as a Job server in One Identity Manager.

Table 3: Properties of the Job server

Property	Value
Server function	Microsoft Entra ID connector
Machine role	Server | Job Server | Microsoft Entra ID

NOTE: If several target system environments of the same type are synchronized under the same synchronization server, it is recommended that you set up a Job server for each target system for performance reasons. This avoids unnecessary swapping of connections to target systems because a Job server only has to process tasks of the same type (re-use of existing connections).

To set up a Job server, perform the following steps.

1. Create a Job server and install and configure the One Identity Manager Service.

   Use the One Identity Manager Service to install the Server Installer. The program runs the following steps:

   - Sets up a Job server.
   - Specifies machine roles and server function for the Job server.
   - Installs One Identity Manager Service components corresponding to the machine roles.
   - Configures the One Identity Manager Service.
   - Starts the One Identity Manager Service.

   Use the Server Installer to install the One Identity Manager Service locally or remotely.

   To remotely install the One Identity Manager Service, provide an administrative workstation on which the One Identity Manager components are installed. Ensure that the One Identity Manager components are installed on the server before installing locally. For more information about installing One Identity Manager components, see the *One Identity Manager Installation Guide*.

2. If you are working with an encrypted One Identity Manager database, declare the database key in the One Identity Manager Service. For more information about working with an encrypted One Identity Manager database, see the *One Identity Manager Installation Guide*.

3. To generate processes for the Job server, you need the provider, connection parameters and the authentication data. By default, this information is determined

from the database connection data. If the Job server runs through an application server, you must configure extra connection data in the Designer. For more information about connection data, see the *One Identity Manager Configuration Guide*.

To install and configure the One Identity Manager Service on a server

1. Start the Server Installer program.

   NOTE: To install remotely, start the Server Installer program on your administrative workstation. To install locally, start the program on the server.

2. On the **Database connection** page, enter the valid connection credentials for the One Identity Manager database.

   You can connect via the application server or directly to connect to the database.

3. On the **Server properties** page, specify the server on which you want to install the One Identity Manager Service.

   a. Select a Job server from the **Server** drop-down.

      - OR -

      To create a new Job server, click **Add**.

   b. Enter the following data for the Job server.

      - **Server**: Name of the Job server.
      - **Queue**: Name of the queue to handle the process steps. Each Job server within the network must have a unique queue identifier. The process steps are requested by the Job queue using this exact queue name. The queue identifier is entered in the One Identity Manager Service configuration file.
      - **Full server name**: Full server name in accordance with DNS syntax.

        Syntax:

        ```
        <Name of servers>.<Fully qualified domain name>
        ```

   NOTE: You can use the **Extended** option to make changes to other properties for the Job server. You can also edit the properties later with the Designer.

4. On the **Machine roles** page, select **Microsoft Entra ID**.
5. On the **Server functions** page, select **Microsoft Entra ID connector (via Microsoft Graph)**.
6. On the **Service Settings** page, enter the connection data and check the One Identity Manager Service configuration.

   NOTE: The initial service configuration is predefined. If further changes need to be made to the configuration, you can do this later with the Designer. For more information about configuring the service, see the *One Identity Manager Configuration Guide*.

   For a direct connection to the database:

a. In the module list, select **Process collection > sqlprovider**.
b. Click the **Connection parameter** entry, then click the **Edit** button.
c. Enter the connection data for the One Identity Manager database.
d. Click **OK**.

For a connection to the application server:

a. In the module list, select the **Process collection** entry and click the **Insert** button.
b. Select **AppServerJobProvider** and click **OK**.
c. In the module list, select **Process collection > AppServerJobProvider**.
d. Click the **Connection parameter** entry, then click the **Edit** button.
e. Enter the address (URL) for the application server and click **OK**.
f. Click the **Authentication data** entry and click the **Edit** button.
g. In the **Authentication method** dialog, select the authentication module for logging in. Depending on the authentication module, other data may be required, such as user and password. For more information about One Identity Manager authentication modules, see the *One Identity Manager Authorization and Authentication Guide*.
h. Click **OK**.

7. To configure the installation, click **Next**.
8. Confirm the security prompt with **Yes**.
9. On the **Select installation source** page, select the directory with the install files. Change the directory if necessary.
10. On the **Service access** page, enter the service's installation data.
    - **Computer**: Select the server, on which you want to install and start the service, from the drop-down or enter the server's name or IP address.

      To run the installation locally, select `Local installation` from the drop-down.
    - **Service account**: Enter the details of the user account that the One Identity Manager Service is running under. Enter the user account, the user account's password and password confirmation.

    The service is installed using the user account with which you are logged in to the administrative workstation. If you want to use another user account for installing the service, you can enter it in the advanced options.

    You can also change the One Identity Manager Service details, such as the installation directory, name, display name, and the One Identity Manager Service description, using the advanced options.
11. Click **Next** to start installing the service.

    Installation of the service occurs automatically and may take some time.
12. Click **Finish** on the last page of the Server Installer.

NOTE: In a default installation, the service is entered in the server's service management with the name **One Identity Manager Service**.

Creating a synchronization project for initial synchronization of a Microsoft Entra ID tenant

Use the Synchronization Editor to configure synchronization between the One Identity Manager database and Microsoft Entra ID. The following describes the steps for initial configuration of a synchronization project. For more information about setting up synchronization, see the *One Identity Manager Target System Synchronization Reference Guide*.

After the initial configuration, you can customize and configure workflows within the synchronization project. Use the workflow wizard in the Synchronization Editor for this. The Synchronization Editor also provides different configuration options for a synchronization project.

Related topics

- Information required for Microsoft Entra ID synchronization projects on page 28
- Creating an initial synchronization project for a Microsoft Entra ID tenant on page 30
- Customizing synchronization projects to invite guest users on page 38

Information required for Microsoft Entra ID synchronization projects

Have the following information available for setting up a synchronization project.

Table 4: Information required to set up a synchronization project

Data	Explanation
Application ID	The application ID is generated when registering the One Identity Manager application in the Microsoft Entra ID tenant.
Login domain	Microsoft Entra ID name of the domain for logging in to Microsoft Entra ID. You can use the base domain or your Microsoft Entra ID tenant's verified domain.
User account and password for logging in	Depending on how the One Identity Manager application is registered in the Microsoft Entra ID tenant, either a user account with sufficient permissions or the secret is required.

Data	Explanation
or The secret's value	For more information, see Users and permissions for synchronizing with Microsoft Entra ID on page 22.
Synchronization server for Microsoft Entra ID	All One Identity Manager Service actions are run against the target system environment on the synchronization server. Data entries required for synchronization and administration with the One Identity Manager database are processed by the synchronization server. The One Identity Manager Service with the Microsoft Entra ID connector must be installed on the synchronization server. The synchronization server must be declared as a Job server in One Identity Manager. Use the following properties when you set up the Job server. • Server function: **`Microsoft Entra ID connector (via Microsoft Graph)`** • Machine role: **`Server | Job Server | Microsoft Entra ID`**
One Identity Manager database connection data	• Database server • Database name • SQL login and password • Specifies whether integrated Windows authentication is used Use of the integrated Windows authentication is not recommended. If you decide to use it anyway, ensure that your environment supports Windows authentication.
Remote connection server	To configure synchronization with a target system, One Identity Manager must load the data from the target system. One Identity Manager communicates directly with the target system to do this. Sometimes direct access from the workstation, on which the Synchronization Editor is installed, is not possible. For example, because of the firewall configuration or the workstation does not fulfill the necessary hardware and software requirements. If direct access is not possible from the workstation, you can set up a remote connection. Remote connection server configuration: • One Identity Manager Service is started • **RemoteConnectPlugin** is installed and an authentication method is set up • Microsoft Entra ID connector is installed

Data	Explanation
	The remote connection server must be declared as a Job server in One Identity Manager. The Job server name is required. TIP: The remote connection server requires the same configuration as the synchronization server (with regard to the installed software and entitlements). Use the synchronization server as remote connection server as well by installing the RemoteConnectPlugin. For more detailed information about setting up a remote connection, see the *One Identity Manager Target System Synchronization Reference Guide*.

Related topics

- Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18
- Setting up the Microsoft Entra ID synchronization server on page 24

Creating an initial synchronization project for a Microsoft Entra ID tenant

NOTE: The following sequence describes how to configure a synchronization project if the Synchronization Editor is both:

- Run in default mode
- Started from the Launchpad

If you run the project wizard in expert mode or directly from the Synchronization Editor, additional configuration settings can be made. Follow the project wizard instructions through these steps.

NOTE: Just one synchronization project can be created per target system and default project template used.

To set up an initial synchronization project for a Microsoft Entra ID tenant

1. Start the Launchpad and log in on the One Identity Manager database.

   NOTE: If synchronization is run by an application server, connect the database through the application server.

2. In the **Installation overview > Data synchronization** section, select the **Target system type Microsoft Entra ID** and click **Run**.

   This starts the Synchronization Editor's project wizard.

3. On the wizard's start page, click **Next**.
4. On the **System access** page, specify how One Identity Manager can access the target system.
   - If access is possible from the workstation on which you started the Synchronization Editor, do not change any settings.
   - If access is not possible from the workstation on which you started the Synchronization Editor, you can set up a remote connection.

     Select the **Connect using remote connection server** and enter the remote connection properties.

     Remote connection properties

     - **Access parameters**
       - **Server**: Full server name or IP address of the server.

         To select an existing Job server as the remote connection server, click and select the server from the drop-down. This displays all the Job servers that have the **One Identity Manager Service installed** server function selected.
       - **Port**: Port that is configured for the RemoteConnectPlugin.
     - **Authentication**

       If **SecretAuthentication** is configured for the RemoteConnectPlugin:
       - **Secret**: Secret used by the Synchronization Editor to authenticate on the RemoteConnectPlugin.

       If **ADGroupAuthentication** is configured for the RemoteConnectPlugin, no data is required.
     - **Options**
       - **Request timeout**: Maximum time allowed for a server query in seconds. If the time is exceeded, the request is canceled.
       - **Accept self-signed certificates**: Specifies whether self-signed certificates can be accepted.
5. On the **Microsoft Entra ID tenant** page, enter the following information:
   - **Deployment**: Select your cloud deployment. Select from `Microsoft Graph global service` or `Microsoft Graph for US Government (L4)`.
   - **Application ID**: Enter the application ID. The application ID was generated when registering the One Identity Manager application in the Microsoft Entra ID tenant.
   - **Login domain**: Enter the base domain or a verified domain of your Microsoft Entra ID tenant.
6. On the **Authentication** page, select the type of login and enter the required login data. The information is required depends on how the One Identity Manager application is registered with the Microsoft Entra ID tenant.

- If you have integrated the One Identity Manager as a mobile device and desktop application in your Microsoft Entra ID tenant, select **Authenticate as mobile device or desktop application** and enter the user account and password for logging in.
- If you have integrated One Identity Manager as a web application in your Microsoft Entra ID tenant, select the option **Authenticate as web application** and enter the value in the secret.

  The secret was generated when the One Identity Manager application was registered with the Microsoft Entra ID tenant.

7. On the last page of the system connection wizard, you can save the connection data.
   - Set the **Save connection locally** option to save the connection data. This can be reused when you set up other synchronization projects.
   - Click **Finish**, to end the system connection wizard and return to the project wizard.
8. On the **One Identity Manager Connection** tab, test the data for connecting to the One Identity Manager database. The data is loaded from the connected database. Reenter the password.

   NOTE:
   - If you use an unencrypted One Identity Manager database and have not yet saved any synchronization projects to the database, you need to enter all connection data again.
   - This page is not shown if a synchronization project already exists.
9. The wizard loads the target system schema. This may take a few minutes depending on the type of target system access and the size of the target system.
10. On the **Select project template** page, select the **Microsoft Entra ID Synchronization** template.
11. On the **Restrict target system access** page, specify how system access should work. You have the following options:

Table 5: Specify target system access

Option	Meaning
Read-only access to target system.	Specifies that a synchronization workflow is only to be set up for the initial loading of the target system into the One Identity Manager database. The synchronization workflow has the following characteristics: • Synchronization is in the direction of `One Identity Manager`. • Processing methods in the synchronization steps are only defined for synchronization in the direction

Option	Meaning
	of `One Identity Manager`.
Read/write access to target system. Provisioning available.	Specifies whether a provisioning workflow is set up in addition to the synchronization workflow for the initial loading of the target system. The provisioning workflow displays the following characteristics: • Synchronization is in the direction of the `Target system`. • Processing methods are only defined in the synchronization steps for synchronization in the direction of the `Target system`. • Synchronization steps are only created for such schema classes whose schema types have write access.

12. On the **Synchronization server** page, select the synchronization server to run the synchronization.

    If the synchronization server is not declared as a Job server for this target system in the One Identity Manager database yet, you can add a new Job server.

    a. Click to add a new Job server.

    b. Enter a name for the Job server and the full server name conforming to DNS syntax.

    > TIP: You can also implement an existing Job server as the synchronization server for this target system.
    >
    > • To select a Job server, click .
    >
    > This automatically assigns the server function matching this Job server.

    c. Click **OK**.

    The synchronization server is declared as Job server for the target system in the One Identity Manager database.

    d. > NOTE: After you save the synchronization project, ensure that this server is set up as a synchronization server.

13. To close the project wizard, click **Finish**.

    This creates and allocates a default schedule for regular synchronization. Enable the schedule for regular synchronization.

    This sets up, saves and immediately activates the synchronization project.

NOTE:

- If enabled, a consistency check is carried out. If errors occur, a message appears. You can decide whether the synchronization project can remain activated or not.

  Check the errors before you use the synchronization project. To do this, in the **General** view on the Synchronization Editor's start page, click **Verify project**.
- If you do not want the synchronization project to be activated immediately, disable the **Activate and save the new synchronization project automatically option**. In this case, save the synchronization project manually before closing the Synchronization Editor.
- The connection data for the target system is saved in a variable set and can be modified in the Synchronization Editor in the **Configuration > Variables** category.

Related topics

- Users and permissions for synchronizing with Microsoft Entra ID on page 22
- Information required for Microsoft Entra ID synchronization projects on page 28
- Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18
- Setting up the Microsoft Entra ID synchronization server on page 24
- Configuring the synchronization log on page 34
- Adjusting the synchronization configuration for Microsoft Entra ID environments on page 36
- Running synchronization on page 51
- Tasks following synchronization on page 55
- Possible errors when synchronizing an Microsoft Entra ID tenant on page 281
- Project template for Microsoft Entra ID tenants on page 287
- Project template for Microsoft Entra ID B2C tenants on page 288
- Microsoft Entra ID connector settings on page 292

Configuring the synchronization log

All the information, tips, warnings, and errors that occur during synchronization are recorded in the synchronization log. You can configure the type of information to record separately for each system connection and synchronization workflow.

To configure the content of the synchronization log for a system connection

1. To configure the synchronization log for target system connection, in the Synchronization Editor, select the **Configuration > Target system** category.

   - OR -

   To configure the synchronization log for the database connection, in the Synchronization Editor, select the **Configuration > One Identity Manager connection** category.
2. In the **General** section, click **Setup**.
3. In the **Synchronization log** section, set **Create synchronization log**.
4. Enable the data to be logged.

   NOTE: Some content generates a particularly large volume of log data. The synchronization log should only contain data required for error analysis and other analyzes.
5. Click **OK**.

To configure the content of the synchronization log for a synchronization workflow

1. In the Synchronization Editor, select the **Workflows** category.
2. Select a workflow in the navigation view.
3. In the **General** section, click **Edit**.
4. Select the **Synchronization log** tab.
5. Enable the data to be logged.

   NOTE: Some content generates a particularly large volume of log data. The synchronization log should only contain data required for error analysis and other analyzes.
6. Click **OK**.

Synchronization logs are stored for a fixed length of time.

To modify the retention period for synchronization logs

- In the Designer, enable the **DPR | Journal | LifeTime** configuration parameter and enter the maximum retention period.

Related topics

- Displaying synchronization results on page 54

Adjusting the synchronization configuration for Microsoft Entra ID environments

Having used the Synchronization Editor to set up a synchronization project for initial synchronization of a Microsoft Entra ID tenant, you can use the synchronization project to load Microsoft Entra ID objects into the One Identity Manager database. If you manage user accounts and their authorizations with One Identity Manager, changes are provisioned in the Microsoft Entra ID environment.

You must customize the synchronization configuration to be able to regularly compare the database with the Microsoft Entra ID environment and to synchronize changes.

- To use One Identity Manager as the primary system during synchronization, create a workflow with synchronization in the direction of the **`Target system`**.
- You can use variables to create generally applicable synchronization configurations that contain the necessary information about the synchronization objects when synchronization starts. Variables can be implemented in base objects, schema classes, or processing method, for example.
- Use variables to set up a synchronization project for synchronizing different tenants. Store a connection parameter as a variable for logging in to the tenants.
- To specify which Microsoft Entra ID objects and database objects are included in synchronization, edit the scope of the target system connection and the One Identity Manager database connection. To prevent data inconsistencies, define the same scope in both systems. If no scope is defined, all objects will be synchronized.
- Update the schema in the synchronization project if the One Identity Manager schema or target system schema has changed. Then you can add the changes to the mapping.
- To synchronize additional schema properties, update the schema in the synchronization project. Include the schema extensions in the mapping.

For more information about configuring synchronization, see the *One Identity Manager Target System Synchronization Reference Guide*.

Detailed information about this topic

- Configuring synchronization with Microsoft Entra ID tenants on page 37
- Configuring synchronization of different Microsoft Entra ID tenants on page 37
- Customizing synchronization projects to invite guest users on page 38
- Supporting custom Microsoft Entra ID extensions on page 39
- Changing system connection settings of Microsoft Entra ID tenants on page 40
- Updating schemas on page 42

- Configuring the provisioning of memberships on page 48
- Configuring single object synchronization on page 49
- Accelerating provisioning and single object synchronization on page 50

Configuring synchronization with Microsoft Entra ID tenants

The synchronization project for initial synchronization provides a workflow for initial loading of target system objects (initial synchronization) and one for provisioning object modifications from the One Identity Manager database to the target system (provisioning). To use One Identity Manager as the primary system during synchronization, you also require a workflow with synchronization in the direction of the `Target system`.

To create a synchronization configuration for synchronizing in Microsoft Entra ID tenants

1. In the Synchronization Editor, open the synchronization project.
2. Check whether the existing mappings can be used to synchronize into the target system. Create new maps if required.
3. Create a new workflow with the workflow wizard.

   This creates a workflow with `Target system` as its direction of synchronization.
4. Create a new start up configuration. Use the new workflow to do this.
5. Save the changes.
6. Run a consistency check.

Related topics

- Configuring synchronization of different Microsoft Entra ID tenants on page 37

Configuring synchronization of different Microsoft Entra ID tenants

If you want to customize a synchronization project to synchronize another Microsoft Entra ID tenant, make sure that you use the same type of authentication on the application when registering it in the Microsoft Entra ID tenant.

Depending on how the One Identity Manager application is registered in the Microsoft Entra ID tenant, either a user account with sufficient permissions or the secret key is required. For more information, see Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18.

To customize a synchronization project for synchronizing another Microsoft Entra ID tenant

1. In the Synchronization Editor, open the synchronization project.
2. Create a new base object for every other client.
   - Use the wizard to attach a base object.
   - In the wizard, select the Microsoft Entra ID connector.
   - Declare the connection parameters. The connection parameters are saved in a special variable set.

   A start up configuration is created that uses the newly created variable set.
3. Change other elements of the synchronization configuration as required.
4. Save the changes.
5. Run a consistency check.

Related topics

- Configuring synchronization with Microsoft Entra ID tenants on page 37
- Registering an enterprise application for One Identity Manager in the Microsoft Entra ID tenant on page 18

Customizing synchronization projects to invite guest users

For more information about guest users in Microsoft Entra ID, see the *Microsoft Entra ID documentation* from Microsoft.

In One Identity Manager you can set up user account with the following user types:

- **`Member`**: Normal Microsoft Entra ID user account.
- **`Guest`**: User account for guest users. The Microsoft Entra ID connector creates a user account for guest users and ensures that an invitation is sent by email to the given email address.

To send guest user invitations, you must alter the variables in the synchronization project.

Variable	Description
GuestInviteSendMail	Specifies whether the guest user invitation will be sent. Default: **`True`**
GuestInviteLanguage	Language to use for sending the guest user invitation. Default: **`en-us`**

Variable	Description
GuestInviteCustomMessage	Personal welcome greeting for the guest user.
GuestInviteRedirectUrl	URL to reroute guest users after they have accepted the invitation and registered. Default: `http://www.office.com`

To edit a variable

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > Variables** category.
3. Select the variable and edit its value.
4. Save the changes.

Related topics

- General main data of Microsoft Entra ID user accounts on page 216
- Editing connection parameters in the variable set on page 40

Supporting custom Microsoft Entra ID extensions

In Microsoft Entra ID, you can add schema extensions for Microsoft Entra ID applications that are registered in the company. Schema extensions in Microsoft Entra ID have the format `extension_<appId>_<propertyName>`. For more information about schema extensions, see the Microsoft Graph API under https://docs.microsoft.com/en-us/graph/extensibility-overview.

The Microsoft Entra ID connector can read and write Microsoft Entra ID schema extensions.

To map and synchronize Microsoft Entra ID schema extensions in One Identity Manager

1. Extend the One Identity Manager schema by the custom columns. Use the Schema Extension program to do this.

   For more information about extending the One Identity Manager schema, see the *One Identity Manager Configuration Guide*.

2. Use the Synchronization Editor to update the target system schema in your synchronization project and the One Identity Manager connection's schema.

   For more information about updating schema in the Synchronization Editor, see the *One Identity Manager Target System Synchronization Reference Guide*.

3. In the Synchronization Editor, extend the mappings in your synchronization project by the respective property mapping rules for schema extensions.

   For more information about editing property mapping rules in the Synchronization Editor, see the *One Identity Manager Target System Synchronization Reference Guide*.

Changing system connection settings of Microsoft Entra ID tenants

When you set up synchronization for the first time, the system connection properties are set to default values that you can modify. There are two ways to do this:

a. Specify a specialized variable set and change the values of the affected variables.

   The default values remain untouched in the default variable set. The variables can be reset to the default values at any time. (Recommended action).

b. Edit the target system connection with the system connection wizard and change the effected values.

   The system connection wizard supplies additional explanations of the settings. The default values can only be restored under particular conditions.

Detailed information about this topic

- Editing connection parameters in the variable set on page 40
- Editing target system connection properties on page 41
- Customizing synchronization projects to invite guest users on page 38
- Microsoft Entra ID connector settings on page 292

Editing connection parameters in the variable set

The connection parameters were saved as variables in the default variable set when synchronization was set up. You can change the values in these variables to suit you requirements and assign the variable set to a start up configuration and a base object. This means that you always have the option to use default values from the default variable set.

NOTE: To guarantee data consistency in the connected target system, ensure that the start-up configuration for synchronization and the base object for provisioning use the same variable set. This especially applies if a synchronization project is used for synchronizing different Microsoft Entra ID tenants.

To customize connection parameters in a specialized variable set

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > Target system** category.
3. Open the **Connection parameters** view.

   Some connection parameters can be converted to variables here. For other parameters, variables are already created.
4. Select a parameter and click **Convert**.
5. Select the **Configuration > Variables** category.

   All specialized variable sets are shown in the lower part of the document view.
6. Select a specialized variable set or click on in the variable set view's toolbar.
   - To rename the variable set, select the variable set and click the variable set view in the toolbar . Enter a name for the variable set.
7. Select the previously added variable and enter a new value.
8. Select the **Configuration > Start up configurations** category.
9. Select a start up configuration and click **Edit**.
10. Select the **General** tab.
11. Select the specialized variable set in the **Variable set** menu.
12. Select the **Configuration > Base objects** category.
13. Select the base object and click .

    - OR -

    To add a new base object, click .
14. Select the specialized variable set in the **Variable set** menu.
15. Save the changes.

For more information about using variables and variable sets, or restoring default values and adding base objects, see the *One Identity Manager Target System Synchronization Reference Guide*.

Related topics

- Editing target system connection properties on page 41

Editing target system connection properties

You can also use the system connection wizard to change the connection parameters. If variables are defined for the settings, the changes are transferred to the active variable set.

NOTE: In the following circumstances, the default values cannot be restored:

- The connection parameters are not defined as variables.
- The default variable set is selected as an active variable set.

In both these cases, the system connection wizard overwrites the default values. They cannot be restored at a later time.

To edit connection parameters using the system connection wizard

1. In the Synchronization Editor, open the synchronization project.
2. In the toolbar, select the active variable set to be used for the connection to the target system.

   NOTE: If the default variable set is selected, the default values are overwritten and cannot be restored at a later time.
3. Select the **Configuration > Target system** category.
4. Click **Edit connection**.

   This starts the system connection wizard.
5. Follow the system connection wizard instructions and change the relevant properties.
6. Save the changes.

Related topics

- Editing connection parameters in the variable set on page 40

Updating schemas

All the schema data (schema types and schema properties) of the target system schema and the One Identity Manager schema are available when you are editing a synchronization project. Only a part of this data is really needed for configuring synchronization. If a synchronization project is finished, the schema is compressed to remove unnecessary data from the synchronization project. This can speed up the loading of the synchronization project. Deleted schema data can be added to the synchronization configuration again at a later point.

If the target system schema or the One Identity Manager schema has changed, these changes must also be added to the synchronization configuration. Then the changes can be added to the schema property mapping.

To include schema data that have been deleted through compression and schema modifications in the synchronization project, update each schema in the synchronization project. This may be necessary if:

- A schema was changed by:
  - Changes to a target system schema
  - Customizations to the One Identity Manager schema

    - A One Identity Manager update migration
- A schema in the synchronization project was shrunk by:
    - Enabling the synchronization project
    - Saving the synchronization project for the first time
    - Compressing a schema

To update a system connection schema

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > Target system** category.

   - OR -

   Select the **Configuration > One Identity Manager connection** category.
3. Select the **General** view and click **Update schema**.
4. Confirm the security prompt with **Yes**.

   This reloads the schema data.

To edit a mapping

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Mappings** category.
3. Select a mapping in the navigation view.

   Opens the Mapping Editor. For more information about mappings, see the *One Identity Manager Target System Synchronization Reference Guide*.

NOTE: The synchronization is deactivated if the schema of an activated synchronization project is updated. Reactivate the synchronization project to synchronize.

Speeding up synchronization

The Microsoft Entra ID connector supports delta synchronization to speed up Microsoft Entra ID synchronization. The method is based on the delta query function from Microsoft Graph. It supports the schema types **User** (user account), **Group** (group), and **DirectoryRole** (administrator role). Delta synchronization is not enabled by default. It is a custom setup.

Implementing delta synchronization

1. Set up a regular Microsoft Entra ID synchronization project.
2. Run initial synchronization.
3. Modify the **TargetSystem | AzureAD | DeltaTokenDirectory** configuration parameter.

The configuration parameter contains the directory where the delta token files are stored. In the Designer, modify the value of the configuration parameter. Ensure that the One Identity Manager Service user account has write access to the directory.

4. (Optional) Modify the `AAD_Organization_DeltaSync` process.

   The process is made up of three process steps. Each process step handles on of the three supported schema types. Each process step is configured such that it synchronizes all supported delta properties of the schema types. Furthermore, each of these process step adds its own delta token file. The order of process steps is as follows:

   - Synchronize user accounts (`Synchronize User` process step)
   - Synchronize groups (`Synchronize Group` process step)
   - Synchronize administration roles (`Synchronize DirectoryRole` process step)

   If customized, ensure that the process is only generated if there is no other similar process is in the Job queue. In the same way, the process may not start during a regular synchronization.

5. (Optional) Customize processing scripts for supporting schema types.

   - Process user accounts (`AAD_ProcessDeltaQueryUser` script)
   - Process groups (`AAD_ProcessDeltaQueryGroup` script)
   - PRocess administration roles (`AAD_ProcessDeltaQueryDirectoryRole` script)

   The `AAD_ProcessDeltaQueryGroup` script has had comments added to it to simplify editing and custom development.

6. Adjust and enable the **Microsoft Entra ID delta synchronization** schedule

   The schedule ensures regular delta synchronization of the Microsoft Entra ID tenants. The schedule is run by default at **15** minute intervals. You can change this interval in the Designer if necessary. Enable the schedule.

The delta synchronization sequence

1. An initial query is run for a schema type (user account, group, administrator role). The initial query returns a complete list for the schema type, such as all user accounts, including the queried properties. This also returns a state token. This token represents the state of the data at the time of the query in Microsoft Entra ID.

   The state token and the queried properties are written to a delta token file. By default, there is no initial processing of the data.

   Delta token file storage structure

   ```
   <Directory in TargetSystem | AzureAD | DeltaTokenDirectory configuration parameter>\<UID_AADOrganization>_<SchemaTyp>Query.token
   ```

   Example:

   ```
   C:\Temp\OneIM\DeltaToken\2da43fd4-ce7b-48af-9a00-686e5e3fb8a5_UserQuery.token
   ```

2. The rest of the queries use the state token of the previous query. Apart from the new state token, they only return the objects that have changed since the last query.

- Tries to add new objects if all mandatory properties have been queried.
- Objects deleted in the target system are generally marked as **`Outstanding`**.

Objects that fail during processing are logged in the process step's messages.

The new state token is written in the delta token file.

Restrictions

With respect to the stability of repetitions, the difference query method has certain limitations. If a state token has been used once, it is generally invalid and the query cannot be run again. If an error occurs processing the return date, the respective change cannot be loaded until the next time synchronization is scheduled to run. For example, this happens to new group memberships if the member themselves has not been loaded yet.

Another disadvantage is the runtime of the initial query and initial data processing. This process is not recommended. However, because initial processing is meant to be carried out during scheduled synchronization, it is recommended to set the `DoNotProcessOffset` parameter in the process steps to **`True`** (default).

You should also take into account that not all properties can be queried using the Microsoft Graph API delta query.

If the data in the delta token file does not match the calling parameters of a query, the existing file is renamed to `<alterName>.backup` in order not to lose the state token and a new file is created. In this case, a new initial query is run. This also happens if the file does not exist or is empty.

Supported schema types

The following tables contain the supported schema types and their supported properties. As long as new objects are imported into the database, the mandatory properties in the delta synchronization must be queried.

Table 6: Supported properties for user accounts (schema type: User)

Property	Mandatory	Remark
AccountEnabled		
AgeGroup		
BusinessPhones		
City		
CompanyName		
ConsentProvidedForMinor		
Country		
Department		

Property	Mandatory	Remark
DisplayName	X	
ExternalUserState		
ExternalUserStateChangeDateTime		
GivenName		
ID	X	
JobTitle		
LastPasswordChangeDateTime		
LegalAgeGroupClassification		
Licenses		When this property is queried, another query runs about the user account's assignment status (`LicenseAssignmentStates`). This increases the runtime massively. Contains a list of objects with the `DisabledPlans`, `SkuId`, `AssignedByGroup`, `State`, and `Error` properties.
Mail		
MailNickname		
Manager		
MobilePhone		
OfficeLocation		
OnPremisesDistinguishedName		
OnPremisesDomainName		
OnPremisesImmutableId		
OnPremisesLastSyncDateTime		
OnPremisesSamAccountName		
OnPremisesSecurityIdentifier		
OnPremisesSyncEnabled		
OnPremisesUserPrincipalName		
PostalCode		

Property	Mandatory	Remark
PreferredLanguage		
ProxyAddresses		
State		
StreetAddress		
Surname		
UsageLocation		
UserDomain	x	
UserPrincipalName	x	
UserType	x	

Table 7: Unterstützte Eigenschaften für Gruppen (Schematyp: Group)

Property	Mandatory	Remark
Description		
DisplayName	x	
GroupTypes	x	
ID	x	
Licenses		Contains a list of objects with the `DisabledPlans` and `SkuId` properties.
Mail		
MailEnabled	x	
MailNickName	x	
Members		The property is not available in an initial query. The result contains the schema type and the ID.
OnPremisesSecurityIdentifier		
OnPremisesSyncEnabled		
Owners		The property is not available in an initial query. The result contains the schema type and the ID.
ProxyAddresses		
SecurityEnabled	x	

Table 8: Support properties for administration role (schema type: DirectoryRole)

Property	Mandatory	Remark
Description		
DisplayName	x	
ID	x	
Members		The property is not available in an initial query. The result contains the schema type and the ID.

Configuring the provisioning of memberships

Memberships, such as user accounts in groups, are saved in assignment tables in the One Identity Manager database. During provisioning of modified memberships, changes made in the target system may be overwritten. This behavior can occur under the following conditions:

- Memberships are saved as an object property in list form in the target system.

  Example: List of user accounts in the `Member` property of a Microsoft Entra ID group (`Group`)
- Memberships can be modified in either of the connected systems.
- A provisioning workflow and provisioning processes are set up.

If one membership in One Identity Manager changes, by default, the complete list of members is transferred to the target system. Therefore, memberships that were previously added to the target system are removed in the process and previously deleted memberships are added again.

To prevent this, provisioning can be configured such that only the modified membership is provisioned in the target system. The corresponding behavior is configured separately for each assignment table.

To allow separate provisioning of memberships

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Target system types** category.
2. In the result list, select the **Microsoft Entra ID** target system type.
3. Select the **Configure tables for publishing** task.
4. Select the assignment tables that you want to set up for single provisioning. Multi-select is possible.
5. Click **Merge mode**.

NOTE:

- This option can only be enabled for assignment tables that have a base table with a `XDateSubItem` column.
- Assignment tables that are grouped together in a virtual schema property in the mapping must be marked identically.

  Example: `AADUserInGroup` and `AADGroupInGroup`

6. Save the changes.

For each assignment table labeled like this, the changes made in One Identity Manager are saved in a separate table. Therefore, only newly added and deleted assignments are processed. During modification provisioning, the members list in the target system is compared to the entries in this table. This means that only modified memberships are provisioned and not the entire members list.

NOTE: The complete members list is updated by synchronization. During this process, objects with changes but incomplete provisioning are not handled. These objects are logged in the synchronization log.

You can restrict single provisioning of memberships with a condition. Once merge mode has been disabled for a table, the condition is deleted. Tables that have had the condition deleted or edited are marked with the following icon: . You can restore the original condition at any time.

To restore the original condition

1. Select the auxiliary table for which you want to restore the condition.
2. Right-click on the selected row and select the **Restore original values** context menu item.
3. Save the changes.

NOTE: To create the reference to the added or deleted assignments in the condition, use the `i` table alias.

Example of a condition on the `AADUserInGroup` assignment table:

```
exists (select top 1 1 from AADGroup g
     where g.UID_AADGroup = i.UID_AADGroup
     and <limiting condition>)
```

For more information about provisioning memberships, see the *One Identity Manager Target System Synchronization Reference Guide*.

Configuring single object synchronization

Changes made to individual objects in the target system can be immediately applied in the One Identity Manager database without having to start a full synchronization of the target system environment. Individual objects can only be synchronized if the object is already present in the One Identity Manager database. The changes are applied to the mapped object properties. If a membership list belongs to one of these properties, the entries in the

assignment table will also be updated. If the object is no longer present in the target system, then it is deleted from the One Identity Manager database.

Prerequisites

- A synchronization step exists that can import the changes to the changed object into One Identity Manager.
- The path to the base object of the synchronization is defined for the table that contains the changed object.

Single object synchronization is fully configured for synchronization projects created using the default project template. If you want to incorporate custom tables into this type of synchronization project, you must configure single object synchronization for these tables. For more information about this, see the *One Identity Manager Target System Synchronization Reference Guide*.

To define the path to the base object for synchronization for a custom table

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Target system types** category.
2. In the result list, select the **Microsoft Entra ID** target system type.
3. Select the **Assign synchronization tables** task.
4. In the **Add assignments** pane, assign the custom table for which you want to use single object synchronization.
5. Save the changes.
6. Select the **Configure tables for publishing** task.
7. Select the custom table and enter the **Root object path**.

   Enter the path to the base object in the ObjectWalker notation of the VI.DB.

   Example: `FK(UID_AADOrganization).XObjectKey`
8. Save the changes.

Related topics

- Synchronizing single objects on page 54
- Post-processing outstanding objects on page 56

Accelerating provisioning and single object synchronization

To smooth out spikes in data traffic, handling of processes for provisioning and single object synchronization can be distributed over several Job servers. This will also accelerate these processes.

NOTE: You should not implement load balancing for provisioning or single object synchronization on a permanent basis. Parallel processing of objects might result in dependencies not being resolved because referenced objects from another Job server have not been completely processed.

Once load balancing is no longer required, ensure that the synchronization server runs the provisioning processes and single object synchronization.

To configure load balancing

1. Configure the server and declare it as a Job server in One Identity Manager.
   - Job servers that share processing must have the **No process assignment** option enabled.
   - Assign the **Microsoft Entra ID connector** server function to the Job server.

   All Job servers must access the same Microsoft Entra ID tenant as the synchronization server for the respective base object.
2. In the Synchronization Editor, assign a custom server function to the base object.

   This server function is used to identify all the Job servers being used for load balancing.

   If there is no custom server function for the base object, create a new one.

   For more information about editing base objects, see the *One Identity Manager Target System Synchronization Reference Guide*.
3. In the Manager, assign this server function to all the Job servers that will be processing provisioning and single object synchronization for the base object.

   Only select those Job servers that have the same configuration as the base object's synchronization server.

Once all the processes have been handled, the synchronization server takes over provisioning and single object synchronization again.

To use the synchronization server without load balancing.

- In the Synchronization Editor, remove the server function from the base object.

For more information about load balancing, see the *One Identity Manager Target System Synchronization Reference Guide*.

Detailed information about this topic

- Job server for Microsoft Entra ID-specific process handling on page 275

Running synchronization

Synchronization is started using scheduled process plans. It is possible to start synchronization manually in the Synchronization Editor. You can simulate synchronization beforehand to estimate synchronization results and discover errors in the synchronization

configuration. If synchronization stopped unexpectedly, you must reset the start information to be able to restart synchronization.

If you want to specify the order in which target systems are synchronized, use the start up sequence to run synchronization. In a start up sequence, you can combine start up configurations from different synchronization projects and specify the order in which they are run. For more information about start up sequences, see the *One Identity Manager Target System Synchronization Reference Guide*.

Detailed information about this topic

- Starting synchronization on page 52
- Deactivating synchronization on page 53
- Displaying synchronization results on page 54
- Synchronizing single objects on page 54
- Pausing handling of target system specific processes (Offline mode) on page 60

Starting synchronization

When you set up the initial synchronization project using the Launchpad, a default schedule for regular synchronization is created and assigned. Activate this schedule to synchronize on a regular basis.

To synchronize on a regular basis

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > Start up configurations** category.
3. Select a start up configuration in the document view and click **Edit schedule**.
4. Edit the schedule properties.
5. To enable the schedule, click **Activate**.
6. Click **OK**.

You can also start synchronization manually if there is no active schedule.

To start initial synchronization manually

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > Start up configurations** category.
3. Select a start up configuration in the document view and click **Run**.
4. Confirm the security prompt with **Yes**.

IMPORTANT: As long as a synchronization process is running, you must not start another synchronization process for the same target system. This especially applies, if the same synchronization objects would be processed.

- If another synchronization process is started with the same start up configuration, the process is stopped and is assigned **Frozen** status. An error message is written to the One Identity Manager Service log file.
    - Ensure that start up configurations that are used in start up sequences are not started individually at the same time. Assign start up sequences and start up configurations different schedules.
- Starting another synchronization process with different start up configuration that addresses same target system may lead to synchronization errors or loss of data. Specify One Identity Manager behavior in this case, in the start up configuration.
    - Use the schedule to ensure that the start up configurations are run in sequence.
    - Group start up configurations with the same start up behavior.

Deactivating synchronization

Regular synchronization cannot be started until the synchronization project and the schedule are active.

To prevent regular synchronization

1. In the Synchronization Editor, open the synchronization project.
2. Select the start up configuration and deactivate the configured schedule.

   Now you can only start synchronization manually.

An activated synchronization project can only be edited to a limited extend. The schema in the synchronization project must be updated if schema modifications are required. The synchronization project is deactivated in this case and can be edited again.

Furthermore, the synchronization project must be deactivated if synchronization should not be started by any means (not even manually).

To deactivate the synchronization project

1. In the Synchronization Editor, open the synchronization project.
2. Select the **General** view on the home page.
3. Click **Deactivate project**.

Related topics

- Creating a synchronization project for initial synchronization of a Microsoft Entra ID tenant on page 28
- Pausing handling of target system specific processes (Offline mode) on page 60

Displaying synchronization results

Synchronization results are summarized in the synchronization log. You can specify the extent of the synchronization log for each system connection individually.
One Identity Manager provides several reports in which the synchronization results are organized under different criteria.

To display a synchronization log

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Logs** category.
3. Click ▶ in the navigation view toolbar.

   Logs for all completed synchronization runs are displayed in the navigation view.
4. Select a log by double-clicking it.

   An analysis of the synchronization is shown as a report. You can save the report.

To display a provisioning log

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Logs** category.
3. Click ⚡ in the navigation view toolbar.

   Logs for all completed provisioning processes are displayed in the navigation view.
4. Select a log by double-clicking it.

   An analysis of the provisioning is shown as a report. You can save the report.

The log is marked in color in the navigation view. This mark shows you the status of the synchronization/provisioning.

TIP: The logs are also displayed in the Manager under the **<target system> > synchronization log** category.

Related topics

- Configuring the synchronization log on page 34
- Troubleshooting on page 59

Synchronizing single objects

Individual objects can only be synchronized if the object is already present in the One Identity Manager database. The changes are applied to the mapped object properties. If a membership list belongs to one of these properties, the entries in the assignment table will also be updated.

NOTE: If the object is no longer present in the target system, then it is deleted from the One Identity Manager database.

To synchronize a single object

1. In the Manager, select the **Microsoft Entra ID** category.
2. Select the object type in the navigation view.
3. In the result list, select the object that you want to synchronize.
4. Select the **Synchronize this object** task.

   A process for reading this object is entered in the job queue.

Features of synchronizing memberships

If you synchronize changes in an object's member list, run single object synchronization on the assignment's root object, The base table of an assignment contains an `XDateSubItem` column containing information about the last change to the memberships.

Example: Base object for synchronizing member lists

Base object for assigning user accounts to groups is the group.

In the target system, a user account was assigned to a group. To synchronize this assignment, in the Manager, select the group that the user account was assigned to and run single object synchronization. In the process, all of the group's memberships are synchronized.

The user account must already exist as an object in the One Identity Manager database for the assignment to be made.

NOTE: To load changes to the assignment of subscriptions to user accounts, run single object synchronization on the user account.

Detailed information about this topic

- Configuring single object synchronization on page 49

Tasks following synchronization

After the synchronization of data from the target system into the One Identity Manager database, rework may be necessary. Check the following tasks:

- Post-processing outstanding objects on page 56
- Adding custom tables to the target system synchronization on page 58
- Managing Microsoft Entra ID user accounts through account definitions on page 58

Post-processing outstanding objects

Objects, which do not exist in the target system, can be marked as outstanding in One Identity Manager by synchronizing. This prevents objects being deleted because of an incorrect data situation or an incorrect synchronization configuration.

Outstanding objects:

- Cannot be edited in One Identity Manager.
- Are ignored by subsequent synchronizations.
- Are ignored by inheritance calculations.

This means, all memberships and assignments remain intact until the outstanding objects have been processed.

Start target system synchronization to do this.

To post-process outstanding objects

1. In the Manager, select the **Microsoft Entra ID > Target system synchronization: Microsoft Entra ID** category.

   The navigation view lists all the synchronization tables assigned to the `Microsoft Entra ID` target system type.

2. On the **Target system synchronization** form, in the **Table/object** column, open the node of the table for which you want to post-process outstanding objects.

   All objects that are marked as outstanding are shown. The **Last log entry** and **Last method run** columns display the time at which the last entry was made in the synchronization log and which processing method was run. The `No log available` entry can mean the following:

   - The synchronization log has already been deleted.

     - OR -

   - An assignment from a member list has been deleted from the target system.

     The base object of the assignment was updated during the synchronization. A corresponding entry appears in the synchronization log. The entry in the assignment table is marked as outstanding, but there is no entry in the synchronization log.

   - An object that contains a member list has been deleted from the target system.

     During synchronization, the object and all corresponding entries in the assignment tables are marked as outstanding. However, an entry in the synchronization log appears only for the deleted object.

TIP:

To display the properties of an outstanding object

1. Select the object on the target system synchronization form.
2. Open the context menu and click **Show object**.
3. For memberships, select the object whose properties you want to display.

3. Select the objects you want to rework. Multi-select is possible.
4. Click on one of the following icons in the form toolbar to run the respective method.

Table 9: Methods for handling outstanding objects

Icon	Method	Description
	Delete	The object is immediately deleted from the One Identity Manager database. Deferred deletion is not taken into account. Indirect memberships cannot be deleted.
	Publish	The object is added to the target system. The `Outstanding` label is removed from the object. This runs a target system specific process that triggers the provisioning process for the object. Prerequisites: • The table containing the object can be published. • The target system connector has write access to the target system.
	Reset	The `Outstanding` label is removed for the object.

TIP: If a method cannot be run due to certain restrictions, the respective icon is disabled.

- To display the constraint's details, click the **Show** button in the **Constraints** column.

5. Confirm the security prompt with **Yes**.

NOTE: By default, the selected objects are processed in parallel, which speeds up the selected method. If an error occurs during processing, the action is stopped and all changes are discarded.

Bulk processing of objects must be disabled if errors are to be localized, which means the objects are processed sequentially. Failed objects are named in the error message. All changes that were made up until the error occurred are saved.

To disable bulk processing

- Disable the icon in the form's toolbar.

NOTE: The target system connector must have write access to the target system in order to publish outstanding objects that are being post-processed. That means, the **Connection is read-only** option must not be set for the target system connection.

Adding custom tables to the target system synchronization

You must customize your target system synchronization to synchronize custom tables.

To add custom tables to target system synchronization

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Target system types** category.
2. In the result list, select the **Microsoft Entra ID** target system type.
3. Select the **Assign synchronization tables** task.
4. In the Add assignments pane, assign **custom** tables to the outstanding objects you want to handle.
5. Save the changes.
6. Select the **Configure tables for publishing** task.
7. Select the custom tables that contain the outstanding objects that can be published in the target system and set the **Publishable** option.
8. Save the changes.

Related topics

- Post-processing outstanding objects on page 56

Managing Microsoft Entra ID user accounts through account definitions

In the default installation, after synchronizing, identities are automatically created for the user accounts.If an account definition for the client is not known at the time of synchronization, user accounts are linked with identities. However, account definitions are not assigned. The user accounts are therefore in a **Linked** state.

To manage the user accounts using account definitions, assign an account definition and a manage level to these user accounts.

To manage user accounts through account definitions

1. Create an account definition.
2. Assign an account definition to the tenant.
3. Assign a user account in the **Linked** state to the account definition. The account definition's default manage level is applied to the user account.
   a. In the Manager, select the **Microsoft Entra ID > User accounts > Linked but not configured > <tenant>** category.
   b. Select the **Assign account definition to linked accounts** task.
   c. In the **Account definition** drop-down, select the account definition.
   d. Select the user accounts that contain the account definition.
   e. Save the changes.

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64

Troubleshooting

Synchronization Editor helps you to analyze and eliminate synchronization errors.

- Simulating synchronization

  The simulation allows you to estimate the result of synchronization. This means you can, for example, recognize potential errors in the synchronization configuration.
- Analyzing synchronization

  You can generate the synchronization analysis report for analyzing problems which occur during synchronization, for example, insufficient performance.
- Logging messages

  One Identity Manager offers different options for logging errors. These include the synchronization log, the log file for One Identity Manager Service, the logging of messages with NLOG, and similar.
- Reset start information

  If synchronization stopped unexpectedly, for example, because a server was not available, the start information must be reset manually. Only then can the synchronization be restarted.

For more information about these topics, see the *One Identity Manager Target System Synchronization Reference Guide*.

Related topics

- Displaying synchronization results on page 54

Ignoring data error in synchronization

By default, objects with incorrect data are not synchronized. These objects can be synchronized once the data has been corrected. In certain situations, however, it might be necessary to synchronize objects like these and ignore the data properties that have errors. This synchronization behavior can be configured in One Identity Manager.

To ignoring data errors during synchronization in One Identity Manager

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Configuration > One Identity Manager connection** category.
3. In the **General** view, click **Edit connection**.

   This starts the system connection wizard.
4. On the **Additional options** page, enable **Try to ignore data errors**.

   This option is only effective if **Continue on error** is set in the synchronization workflow.

   Default columns, such as primary keys, UID columns, or mandatory input columns cannot be ignored.
5. Save the changes.

IMPORTANT: If this option is set, One Identity Manager tries to ignore commit errors that could be related to data errors in a single column. This causes the data changed in the affected column to be discarded and the object is subsequently saved again. This effects performance and leads to loss of data.

Only set this option in the exceptional circumstance of not being able to correct the data before synchronization.

Pausing handling of target system specific processes (Offline mode)

If a target system connector is not able to reach the target system temporarily, you can enable offline mode for the target system. This stops target system specific processes from being frozen and having to be manually re-enabled later.

Whether offline mode is generally available for a target system connection is set in the base object of the respective synchronization project. Once a target system is truly unavailable, the target system connection can be switched offline and online again with the Launchpad.

In offline mode, all Job servers assigned to the base object are stopped. This includes the synchronization server and all Job servers involved in load balancing. If one of the Job servers also handles other tasks, these are not processed either.

Prerequisites

Offline mode can only be specified for a base object if certain prerequisites are fulfilled.

- The synchronization server is not used for any other base object as a synchronization server.
- If a server function is assigned to the base object, none of the Job servers with this server function may have any other server function (for example, update server).
- A dedicated synchronization server must be set up to exclusively process the Job queue for this base object. The same applies to all Job servers that are determined by the server function.

To allow offline mode for a base object

1. In the Synchronization Editor, open the synchronization project.
2. Select the **Base objects** category.
3. Select a base object in the document view and click .
4. Enable **Offline mode available**.
5. Click **OK**.
6. Save the changes.

IMPORTANT: To prevent data inconsistencies, the offline phase should be kept as short as possible.

The number of processes to handle depends on the extent of the changes in the One Identity Manager database and their effect on the target system during the offline phase. To establish data consistency between the One Identity Manager database and the target system, all pending processes must be handled before synchronization can start.

Only use offline mode, if possible, for short system downtimes such as maintenance windows.

To flag a target system as offline

1. Start the Launchpad and log in on the One Identity Manager database.
2. Select **Manage > System monitoring > Flag target systems as offline**.
3. Click **Run**.

   This opens the **Manage offline systems** dialog. The **Base objects** section displays the base objects of target system connections that can be switched to offline.
4. Select the base object whose target system connection is not available.
5. Click **Switch offline**.
6. Confirm the security prompt with **OK**.

   This stops all the Job servers assigned to the base object. No more synchronization or provisioning Jobs are performed. The Job Queue Info program shows when a Job server has been switched offline and the corresponding tasks are not being processed.

For more information about offline mode, see the *One Identity Manager Target System Synchronization Reference Guide*.

Related topics

- Deactivating synchronization on page 53

3

Managing Microsoft Entra ID user accounts and identities

The main feature of One Identity Manager is to map identities together with the main data and permissions available to them in different target systems. To achieve this, information about user accounts and permissions can be read from the target system into the One Identity Manager database and linked to identities. This provides an overview of the permissions for each identity in all of the connected target systems. One Identity Manager offers the option of managing user accounts and their permissions. You can provision modifications in the target systems. Identities are supplied with the necessary permissions in the connected target systems according to their function in the company. Regular synchronization keeps data consistent between target systems and the One Identity Manager database.

Because requirements vary between companies, One Identity Manager offers different methods for supplying user accounts to identities. One Identity Manager supports the following methods for linking identities and their user accounts:

- Identities can automatically obtain their account definitions using user account resources.

  If an identity does not yet have a user account in a Microsoft Entra ID tenant, a new user account is created. This is done by assigning account definitions to an identity using the integrated inheritance mechanisms and subsequent process handling.

  When you manage account definitions through user accounts, you can specify the way user accounts behave when identities are enabled or deleted.
- When user accounts are inserted, they can be automatically assigned to an existing identity or a new identity can be created if necessary. In the process, the identity main data is created on the basis of existing user account main data. This mechanism can be implemented if a new user account is created manually or by synchronization. However, this is not the One Identity Manager default method. You must define criteria for finding identities for automatic identity assignment.
- Identities and user accounts can be entered manually and assigned to each other.

For more information about identity handling and administration, see the *One Identity Manager Target System Base Module Administration Guide*.

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64
- Assigning identities automatically to Microsoft Entra ID user accounts on page 87
- Supported user account types on page 92
- Updating identities when Microsoft Entra ID user account are modified on page 98
- Specifying deferred deletion for Microsoft Entra ID user accounts on page 100
- Creating and editing Microsoft Entra ID user accounts on page 215

Account definitions for Microsoft Entra ID user accounts

One Identity Manager has account definitions for automatically allocating user accounts to identities. You can create account definitions for every target system. If an identity does not yet have a user account in a target system, a new user account is created. This is done by assigning account definitions to an identity.

The data for the user accounts in the respective target system comes from the basic identity main data. The identities must have a central user account. The assignment of the IT operating data to the identity's user account is controlled through the primary assignment of the identity to a location, a department, a cost center, or a business role. Processing is done through templates. There are predefined templates for determining the data required for user accounts included in the default installation. You can customize templates as required.

Specify the manage level for an account definition for managing user accounts. The user account's manage level specifies the extent of the identity's properties that are inherited by the user account. This allows an identity to have several user accounts in one target system, for example:

- Default user account that inherits all properties from the identity.
- Administrative user account that is associated to an identity but should not inherit the properties from the identity.

For more information about the principles of account definitions, manage levels, and determining the valid IT operating data, see the *One Identity Manager Target System Base Module Administration Guide*.

The following steps are required to implement an account definition:

- Creating account definitions
- Configuring manage levels
- Creating the formatting rules for IT operating data
- Collecting IT operating data
- Assigning account definitions to identities and target systems

Detailed information about this topic

- Creating account definitions on page 65
- Editing account definitions on page 66
- Main data for an account definition on page 66
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- Creating mapping rules for IT operating data on page 73
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- Assigning account definitions to identities on page 77
- Assigning account definitions to Microsoft Entra ID tenants on page 84
- Deleting account definitions on page 85

Creating account definitions

Create one or more account definitions for the target system.

To create a new account definition

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Click in the result list.
3. On the main data form, enter the main data of the account definition.
4. Save the changes.

Related topics

- Main data for an account definition on page 66
- Editing account definitions on page 66
- Assigning manage levels to account definitions on page 72

Editing account definitions

You can edit the main data of account definitions.

To edit an account definition

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Change main data** task.
4. Enter the account definition's main data.
5. Save the changes.

Related topics

- Main data for an account definition on page 66
- Creating account definitions on page 65
- Assigning manage levels to account definitions on page 72

Main data for an account definition

Enter the following data for an account definition:

Table 10: Main data for an account definition

Property	Description
Account definition	Account definition name.
User account table	Table in the One Identity Manager schema that maps user accounts. For Microsoft Entra ID user accounts, select `AADUser`.
Target system	Target system to which the account definition applies.
Required account definition	Specifies the required account definition. Define the dependencies between account definitions. When this account definition is requested or assigned, the required account definition is assigned automatically. Leave empty for Microsoft Entra ID tenants. In federations, you can enter the account definition of the Active Directory domain.
Description	Text field for additional explanation.

Property	Description
Manage level (initial)	Manage level to use by default when you add new user accounts.
Risk index	Value for evaluating the risk of assigning the account definition to identities. Set a value in the range **0** to **1**. This input field is only visible if the **QER | CalculateRiskIndex** configuration parameter is set. For more information, see the *One Identity Manager Risk Assessment Administration Guide*.
Service item	Service item through which you can request the account definition resource in the IT Shop. Assign an existing service item or add a new one.
IT Shop	Specifies whether the account definition can be requested through the IT Shop. This account definition can be requested through the Web Portal and allocated by defined approval processes. The resource can also be assigned directly to identities and roles outside the IT Shop.
Only for use in IT Shop	Specifies whether the account definition can only be requested through the IT Shop. This account definition can be requested through the Web Portal and allocated by defined approval processes. The account definition cannot be directly assigned to roles outside the IT Shop.
Automatic assignment to identities	Specifies whether the account definition is automatically assigned to all internal identities. To automatically assign the account definition to all internal identity, use the **Enable automatic assignment to identities** The account definition is assigned to every identity that is not marked as external. Once a new internal identity is created, they automatically obtain this account definition. To automatically remove the account definition assignment from all identities, use the **Disable automatic assignment to identities**. The account definition cannot be reassigned to identities from this point on. Existing account definition assignments remain intact.
Retain account definition if permanently disabled	Specifies the account definition assignment to permanently deactivated identities. Option set: The account definition assignment remains in effect. The user account remains intact. Option not set (default): The account definition assignment is not in effect. The associated user account is deleted.
Retain account definition	Specifies the account definition assignment to temporarily

Property	Description
if temporarily disabled	deactivated identities. Option set: The account definition assignment remains in effect. The user account remains intact. Option not set (default): The account definition assignment is not in effect. The associated user account is deleted.
Retain account definition on deferred deletion	Specifies the account definition assignment on deferred deletion of identities. Option set: The account definition assignment remains in effect. The user account remains intact. Option not set (default): The account definition assignment is not in effect. The associated user account is deleted.
Retain account definition on security risk	Specifies the account definition assignment to identities posing a security risk. Option set: The account definition assignment remains in effect. The user account remains intact. Option not set (default): The account definition assignment is not in effect. The associated user account is deleted.
Resource type	Resource type for grouping account definitions.
Spare field 01 - spare field 10	Additional company-specific information. Use the Designer to customize display names, formats, and templates for the input fields.
Groups can be inherited	Specifies whether the user account can inherit groups through the linked identity. If the option is set, the user account inherits groups through hierarchical roles, in which the identity is a member, or through IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned groups to this department, the user account inherits these groups. • If an identity has requested group membership in the IT Shop and the request is granted approval, the identity's user account only inherits the group if the option is set.
Subscriptions can be inherited	Specifies whether the user account can inherit Microsoft Entra ID subscriptions through the identity. If this option is set, the user account inherits Microsoft Entra ID subscriptions through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a

Property	Description
	department, for example, and you have assigned Microsoft Entra ID subscriptions to this department, the user account inherits these Microsoft Entra ID subscriptions. • If an identity has requested a Microsoft Entra ID subscription in the IT Shop and the request is granted approval, the identity's user account only inherits the Microsoft Entra ID subscription if the option is set.
Administrator roles can be inherited	Specifies whether the user account can inherit Microsoft Entra ID administrator roles through the identity If this option is set, the user account inherits administrator roles through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned administrator roles to this department, the user account inherits these administrator roles. • If an identity has requested an administrator role in the IT Shop and the request is granted approval, the identity's user account only inherits the administrator role if the option is set.
Disabled service plans can be inherited	Specifies whether the user account can inherit disabled Microsoft Entra ID service plans through the identity. If this option is set, the user account inherits disabled service plans through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned disabled service plans to this department, the user account inherits these disabled service plans. • If an identity has requested a disabled service plan in the IT Shop and the request is granted approval, the identity's user account only inherits the disabled service plan if the option is set.
Microsoft 365 groups can be inherited	NOTE: This property is only available if the Exchange Online Module is installed. Specifies whether the user account can inherit Microsoft 365 groups through the linked identity. If the option is set, the user account inherits Microsoft 365 groups through hierarchical roles, in which the identity is a member, or through IT Shop requests. • If you add an identity with a user account to a

Property	Description
	department, for example, and you have assigned Microsoft 365 groups to this department, the Microsoft Entra ID user account inherits these Microsoft 365 groups. • If an identity has requested group membership in the IT Shop and the request is granted approval, the identity's Microsoft Entra ID user account only inherits the Microsoft 365 group if the option is set. For more information about Microsoft 365 groups, see the *One Identity Manager Administration Guide for Connecting to Exchange Online*.

Editing manage levels

One Identity Manager supplies a default configuration for manage levels:

- **Unmanaged**: User accounts with the **Unmanaged** manage level are linked to the identity but they do no inherit any further properties. When a new user account is added with this manage level and an identity is assigned, some of the identity's properties are transferred initially. If the identity properties are changed at a later date, the changes are not passed onto the user account.
- **Full managed**: User accounts with the **Full managed** manage level inherit defined properties of the assigned identity. When a new user account is created with this manage level and an identity is assigned, the identity's properties are transferred in an initial state. If the identity properties are changed at a later date, the changes are passed onto the user account.

NOTE: The **Full managed** and **Unmanaged** manage levels are analyzed in templates. You can customize the supplied templates in the Designer.

You can define other manage levels depending on your requirements. You need to amend the templates to include manage level approaches.

Specify how an identity's temporary deactivation, permanent deactivation, deletion, and security risks affect its user accounts and group memberships at each manage level. For more information about manage levels, see the *One Identity Manager Target System Base Module Administration Guide*.

- Identity user accounts can be locked when they are disabled, deleted, or rated as a security risk so that permissions are immediately withdrawn. If the identity is reinstated at a later date, the user accounts are also reactivated.
- You can also define group membership inheritance. Inheritance can be discontinued if desired when, for example, the identity’s user accounts are disabled and therefore cannot be members in groups. During this time, no inheritance processes should be calculated for this identity. Existing group memberships are deleted.

To edit a manage level

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Manage levels** category.
2. Select the manage level in the result list.
3. Select the **Change main data** task.
4. Edit the manage level's main data.
5. Save the changes.

Related topics

- Main data for manage levels on page 72
- Creating manage levels on page 71
- Assigning manage levels to account definitions on page 72

Creating manage levels

One Identity Manager supplies a default configuration for the **Unmanaged** and **Full managed** manage levels. You can define other manage levels depending on your requirements.

IMPORTANT: In the Designer, extend the templates by adding the procedure for the additional manage levels. For more information about templates, see the *One Identity Manager Configuration Guide*

To create a manage level

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Manage levels** category.
2. Click in the result list.
3. On the main data form, edit the main data of the manage level.
4. Save the changes.

Related topics

- Main data for manage levels on page 72
- Editing account definitions on page 66
- Assigning manage levels to account definitions on page 72

Assigning manage levels to account definitions

IMPORTANT: The **Unmanaged** manage level is assigned automatically when you create an account definition and it cannot be removed.

To assign manage levels to an account definition

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Assign manage level** task.
4. In the **Add assignments** pane, assign the manage level.

   TIP: In the **Remove assignments** pane, you can remove assigned manage levels.

   To remove an assignment

   - Select the manage level and double-click ✓.
5. Save the changes.

Main data for manage levels

Enter the following data for a manage level.

Table 11: Main data for manage levels

Property	Description
Manage level	Name of the manage level.
Description	Text field for additional explanation.
IT operating data overwrites	Specifies whether user account data formatted from IT operating data is automatically updated. Permitted values are: • `Never`: Data is not updated. (Default) • `Always`: Data is always updated. • `Only initially`: Data is only determined at the start.
Retain groups if temporarily disabled	Specifies whether user accounts of temporarily deactivated retain their group memberships.
Lock user accounts if temporarily disabled	Specifies whether user accounts of temporarily deactivated identities are locked.
Retain groups if	Specifies whether user accounts of permanently deactivated

Property	Description
permanently disabled	identities retain group memberships.
Lock user accounts if permanently disabled	Specifies whether user accounts of permanently deactivated identities are locked.
Retain groups on deferred deletion	Specifies whether user accounts of identities marked for deletion retain their group memberships.
Lock user accounts if deletion is deferred	Specifies whether user accounts of identities marked for deletion are locked.
Retain groups on security risk	Specifies whether user accounts of identities posing a security risk retain their group memberships.
Lock user accounts if security is at risk	Specifies whether user accounts of identities posing a security risk are locked.
Retain groups if user account disabled	Specifies whether disabled user accounts retain their group memberships.

Creating mapping rules for IT operating data

An account definition specifies which rules are used to form the IT operating data and which default values will be used if no IT operating data can be found through the identity's primary roles.

The following IT operating data is used in the One Identity Manager default configuration for automatically creating user accounts for an identity in the target system and modifying them.

- Groups can be inherited
- Administrator roles can be inherited
- Subscriptions can be inherited
- Disabled service plans can be inherited
- Change password at next login
- Identity
- Privileged user account.

To create a mapping rule for IT operating data

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Edit IT operating data mapping** task.

4. Click **Add** and enter the following information:
   - **Column**: User account property for which the value is set. In the drop-down, you can select the columns that use the `TSB_ITDataFromOrg` script in their template. For more information about this, see the *One Identity Manager Target System Base Module Administration Guide*.
   - **Source**: Specifies which roles to use in order to find the user account properties. You have the following options:
     - Primary department
     - Primary location
     - Primary cost center
     - Primary business roles

       NOTE: The business role can only be used if the Business Roles Module is available.
     - Empty

       If you select a role, you must specify a default value and set the **Always use default value** option.
   - **Default value**: Default value of the property for an identity's user account if the value is not determined dynamically from the IT operating data.
   - **Always use default value**: Specifies whether the user account property is always set with the default value. IT operating data is not determined dynamically from a role.
   - **Notify when applying the default**: Specifies whether an email is sent to a specific mailbox when the default value is used. The **Identity - new user account with default properties created** mail template is used.

     To change the mail template, in the Designer, adjust the **TargetSystem | AzureAD | Accounts | MailTemplateDefaultValues** configuration parameter.

     To change the mail template, in the Designer, adjust the **TargetSystem | AzureAD | ExchangeOnline | Accounts | MailTemplateDefaultValues** configuration parameter.
5. Save the changes.

Related topics

- Entering IT operating data on page 74

Entering IT operating data

To create user accounts for an identity with the **Full managed** manage level, you need to know which IT operating data is required. The operating data required for each specific target system is defined with its business roles, departments, locations, or cost centers. An

identity is assigned a primary business role, primary location, primary department, or primary cost center. The necessary IT operating data is ascertained from these assignments and used in creating the user accounts. Default values are used if valid IT operating data cannot be found over the primary roles.

You can also specify IT operating data directly for a specific account definition.

Example: Mapping IT operating data

Normally, each identity in department A obtains a default user account in the tenant A. In addition, certain identities in department A obtain administrative user accounts in the tenant A.

Create an account definition A for the default user account of the tenant A and an account definition B for the administrative user account of tenant A.In the IT operating data mapping rule for the account definitions A and B, specify the `Department` property in order to determine the valid IT operating data.

Specify the effective IT operating data of department A for the tenant A. This IT operating data is used for standard user accounts. In addition, for department A, specify the effective IT operating data of account definition B. This IT operating data is used for administrative user accounts.

To define IT operating data

1. In the Manager, select the role in the **Organizations** or **Business roles** category.
2. Select the **Edit IT operating data** task.
3. Click **Add** and enter the following data.
   - **Effects on**: Specify an IT operating data application scope. The IT operating data can be used for a target system or a defined account definition.

     To specify an application scope

     a. Click → next to the field.

     b. Under **Table**, select the table that maps the target system for select the `TSBAccountDef` table or an account definition.

     c. Select the specific target system or account definition under **Effects on**.

     d. Click **OK**.
   - **Column**: Select the user account property for which the value is set.

     In the drop-down, you can select the columns that use the `TSB_ITDataFromOrg` script in their template. For more information about this, see the *One Identity Manager Target System Base Module Administration Guide*.
   - **Value**: Enter a fixed value to assign to the user account's property.
4. Save the changes.

Related topics

- Creating mapping rules for IT operating data on page 73

Modify IT operating data

If IT operating data changes, you must transfer the changes to the existing user accounts. To do this, templates must be rerun on the affected columns. Before you can run the templates, you can check what effect a change to the IT operating data has on the existing user accounts. You can decide whether the change is transferred to the One Identity Manager database in the case of each affected column in each affected database.

Prerequisites

- The IT operating data of a department, a cost center, a business role, or a location have been changed.

  - OR -

- The default values in the IT operating data template were modified for an account definition.

NOTE: If the assignment of an identity to a primary department, cost center, to a primary business role or to a primary location changes, the templates are automatically run.

To run the template

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Run templates** task.

   This displays a list of all user accounts that were created with the selected account definition and whose properties were changed by modifying the IT operating data. That means:

   - **Old value**: Value of the object property before changing the IT operating data.
   - **New value**: Value of the object property after changing the IT operating data.
   - **Selection**: Specifies whether the new value is copied to the user account.
4. Mark all the object properties in the **selection** column that will be given the new value.
5. Click **Apply**.

   The templates are applied to all selected user accounts and properties.

Assigning account definitions to identities

Account definitions are assigned to company identities.

Indirect assignment is the default method for assigning account definitions to identities. Account definitions are assigned to departments, cost centers, locations, or roles. The identities are categorized into these departments, cost centers, locations, or roles depending on their function in the company and thus obtain their account definitions. To react quickly to special requests, you can assign individual account definitions directly to identities.

You can automatically assign special account definitions to all company identities. It is possible to assign account definitions to the IT Shop as requestable products. Department managers can then request user accounts from the Web Portal for their staff. It is also possible to add account definitions to system roles. These system roles can be assigned to identities through hierarchical roles or added directly to the IT Shop as products.

In the One Identity Manager default installation, the processes are checked at the start to see if the identity already has a user account in the target system that has an account definition. If no user account exists, a new user account is created with the account definition's default manage level.

NOTE: If a user account already exists and is disabled, then it is re-enabled. In this case, you must change the user account manage level afterward.

As long as an account definition for an identity is valid, the identity retains the user account that was created by it. If the account definition assignment is removed, the user account that was created from this account definition, is deleted. User accounts marked as **Outstanding** are only deleted if the **QER | Person | User | DeleteOptions | DeleteOutstanding** configuration parameter is set.

Prerequisites for indirect assignment of account definitions to identities

- Assignment of identities and account definitions is permitted for role classes (departments, cost centers, locations, or business roles).

To configure assignments to roles of a role class

1. In the Manager, select role classes in the **Organizations > Basic configuration data > Role classes** category.

   - OR -

   In the Manager, select role classes in the **Business roles > Basic configuration data > Role classes** category.
2. Select the **Configure role assignments** task and configure the permitted assignments.
   - To generally allow an assignment, enable the **Assignments allowed** column.
   - To allow direct assignment, enable the **Direct assignments permitted** column.
3. Save the changes.

For more information about preparing role classes to be assigned, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Detailed information about this topic

- Assigning account definitions to departments, cost centers, and locations on page 78
- Assigning account definitions to business roles on page 79
- Assigning account definitions to all identities on page 79
- Assigning account definitions directly to identities on page 80
- Assigning account definitions to system roles on page 81
- Adding account definitions in the IT Shop on page 82

Assigning account definitions to departments, cost centers, and locations

Assign account definitions to departments, cost centers, and locations in order to assign identities to them through these organizations.

To add account definitions to hierarchical roles

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Assign organizations** task.
4. In the **Add assignments** pane, assign the organizations:
   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
5. Save the changes.

Related topics

- Assigning account definitions to business roles on page 79
- Assigning account definitions to all identities on page 79
- Assigning account definitions directly to identities on page 80

- Assigning account definitions to system roles on page 81
- Adding account definitions in the IT Shop on page 82

Assigning account definitions to business roles

NOTE: This function is only available if the Business Roles Module is installed.

You can assign account definitions to business roles in order to assign them to identities through business roles.

To add account definitions to hierarchical roles

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

Related topics

- Assigning account definitions to departments, cost centers, and locations on page 78
- Assigning account definitions to all identities on page 79
- Assigning account definitions directly to identities on page 80
- Assigning account definitions to system roles on page 81
- Adding account definitions in the IT Shop on page 82

Assigning account definitions to all identities

Use this task to assign the account definition to all internal identities. Identities that are marked as external do not obtain this account definition. Once a new internal identity is created, they automatically obtain this account definition. The assignment is calculated by the DBQueue Processor.

IMPORTANT: Only run this task if you can ensure that all current internal identities in the database and all pending newly added internal identities obtain a user account in this target system.

To assign an account definition to all identities

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Change main data** task.
4. Select the **Disable automatic assignment to identities** task.
5. Confirm the security prompt with **Yes**.
6. Save the changes.

NOTE: To automatically remove the account definition assignment from all identities, run the DISABLE AUTOMATIC ASSIGNMENT TO IDENTITIES task. The account definition cannot be reassigned to identities from this point on. Existing assignments remain intact.

Related topics

- Assigning account definitions to departments, cost centers, and locations on page 78
- Assigning account definitions to business roles on page 79
- Assigning account definitions directly to identities on page 80
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Assigning account definitions directly to identities

Account definitions can be assigned directly or indirectly to identities. Indirect assignment is carried out by allocating identities and account definitions in company structures, like departments, cost centers, locations, or business roles.

To react quickly to special requests, you can assign account definitions directly to identities.

To assign an account definition directly to identities

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Assign to identities** task.
4. In the **Add assignments** pane, add identities.

   TIP: In the **Remove assignments** pane, you can remove assigned identities.

   To remove an assignment

   - Select the identity and double-click ✓.
5. Save the changes.

Related topics

- Assigning account definitions to departments, cost centers, and locations on page 78
- Assigning account definitions to business roles on page 79
- Assigning account definitions to all identities on page 79
- Assigning account definitions to system roles on page 81
- Adding account definitions in the IT Shop on page 82

Assigning account definitions to system roles

NOTE: This function is only available if the System Roles Module is installed.

Use this task to add an account definition to system roles.

NOTE: Account definitions with the **Only use in IT Shop** option set can only be assigned to system roles that also have this option set.

To add account definitions to a system role

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Assigning account definitions to departments, cost centers, and locations on page 78
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Adding account definitions in the IT Shop

An account definition can be requested by shop customers when it is assigned to an IT Shop shelf. To ensure it can be requested, further prerequisites need to be guaranteed.

- The account definition must be labeled with the **IT Shop** option.
- The account definition must be assigned to a service item.

  TIP: In the Web Portal, all products that can be requested are grouped together by service category. To make the account definition easier to find in the Web Portal, assign a service category to the service item.
- If the account definition is only assigned to identities using IT Shop assignments, you must also set the **Only for use in IT Shop** option. Direct assignment to hierarchical roles may not be possible.

NOTE: IT Shop administrators can assign account definitions to IT Shop shelves if login is role-based. Target system administrators are not authorized to add account definitions in the IT Shop.

To add an account definition to the IT Shop (role-based login)

1. In the Manager, select the **Entitlements > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Add to IT Shop** task.
4. To assign the account definition to shelves, select the **IT Shop shelves** tab and, in the **Add assignments** section, select the shelves with a double-click.
5. To assign the account definition to IT Shop templates, select the **IT Shop templates** tab and, in the **Add assignments** section, select the template with a double-click.
6. Save the changes.

To add an account definition to the IT Shop (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Add to IT Shop** task.
4. To assign the account definition to shelves, select the **IT Shop shelves** tab and, in the **Add assignments** section, select the shelves with a double-click.
5. To assign the account definition to IT Shop templates, select the **IT Shop templates** tab and, in the **Add assignments** section, select the template with a double-click.
6. Save the changes.

To remove an account definition from individual IT Shop shelves (role-based login)

1. In the Manager, select the **Entitlements > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Add to IT Shop** task.
4. To remove the account definition from the shelves, select the **IT Shop shelves** tab and, in the **Remove assignments** section, double-click the shelves.
5. To remove the account definition from the IT Shop templates, select the **IT Shop templates** tab and, in the **Remove assignments** section, double-click the templates.
6. Save the changes.

To remove an account definition from individual IT Shop shelves (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Add to IT Shop** task.
4. To remove the account definition from the shelves, select the **IT Shop shelves** tab and, in the **Remove assignments** section, double-click the shelves.
5. To remove the account definition from the IT Shop templates, select the **IT Shop templates** tab and, in the **Remove assignments** section, double-click the templates.
6. Save the changes.

To remove an account definition from all IT Shop shelves (role-based login)

1. In the Manager, select the **Entitlements > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Remove from all shelves (IT Shop)** task.
4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The account definition is removed from all shelves by the One Identity Manager Service. At the same time, any requests and assignment requests with this account definition are canceled.

To remove an account definition from all IT Shop shelves (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
2. Select an account definition in the result list.
3. Select the **Remove from all shelves (IT Shop)** task.

4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The account definition is removed from all shelves by the One Identity Manager Service. At the same time, any requests and assignment requests with this account definition are canceled.

For more information about requesting company resources through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Related topics

- Main data for an account definition on page 66
- Assigning account definitions to departments, cost centers, and locations on page 78
- Assigning account definitions to business roles on page 79
- Assigning account definitions to all identities on page 79
- Assigning account definitions directly to identities on page 80
- Assigning account definitions to system roles on page 81

Assigning account definitions to Microsoft Entra ID tenants

The following prerequisites must be fulfilled if you implement automatic assignment of user accounts and identities resulting in administered user accounts (**Linked configured** state):

- The account definition is assigned to the target system.
- The account definition has the default manage level.

User accounts are only linked to the identity (**Linked** state) if no account definition is given. This is the case on initial synchronization, for example.

To assign the account definition to a target system

1. In the Manager, select the Microsoft Entra ID tenant in the **Microsoft Entra ID > Tenants** category.
2. Select the **Change main data** task.
3. From the **Account definition (initial)** drop-down, select the account definition for user accounts.
4. Save the changes.

Detailed information about this topic

- Assigning identities automatically to Microsoft Entra ID user accounts on page 87

Deleting account definitions

You can delete account definitions if they are not assigned to target systems, identities, hierarchical roles or any other account definitions.

To delete an account definition

1. Remove automatic assignments of the account definition from all identities.
    a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
    b. Select an account definition in the result list.
    c. Select the **Change main data** task.
    d. Select the **Disable automatic assignment to identities** task.
    e. Confirm the security prompt with **Yes**.
    f. Save the changes.
2. Remove direct assignments of the account definition to identities.
    a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
    b. Select an account definition in the result list.
    c. Select the **Assign to identities** task.
    d. In the **Remove assignments** pane, remove identities.
    e. Save the changes.
3. Remove the account definition's assignments to departments, cost centers, and locations.
    a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
    b. Select an account definition in the result list.
    c. Select the **Assign organizations** task.
    d. In the **Remove assignments** pane, remove the relevant departments, cost centers, and locations.
    e. Save the changes.
4. Remove the account definition's assignments to business roles.
    a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
    b. Select an account definition in the result list.
    c. Select the **Assign business roles** task.
    d. In the **Remove assignments** pane, remove the business roles.
    e. Save the changes.

5. If the account definition was requested through the IT Shop, it must be canceled and removed from all IT Shop shelves.

   For more information about unsubscribing requests, see the *One Identity Manager Web Portal User Guide*.

   To remove an account definition from all IT Shop shelves (role-based login)

   a. In the Manager, select the **Entitlements > Account definitions** category.
   b. Select an account definition in the result list.
   c. Select the **Remove from all shelves (IT Shop)** task.
   d. Confirm the security prompt with **Yes**.
   e. Click **OK**.

      The account definition is removed from all shelves by the One Identity Manager Service. At the same time, any requests and assignment requests with this account definition are canceled.

   To remove an account definition from all IT Shop shelves (non role-based login)

   a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
   b. Select an account definition in the result list.
   c. Select the **Remove from all shelves (IT Shop)** task.
   d. Confirm the security prompt with **Yes**.
   e. Click **OK**.

      The account definition is removed from all shelves by the One Identity Manager Service. At the same time, any requests and assignment requests with this account definition are canceled.

6. Remove the required account definition assignment. As long as the account definition is required for another account definition, it cannot be deleted. Check all the account definitions.

   a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
   b. Select an account definition in the result list.
   c. Select the **Change main data** task.
   d. From the **Required account definition** drop-down, remove the account definition.
   e. Save the changes.

7. Remove the account definition's assignments to target systems.

a. In the Manager, select the Microsoft Entra ID tenant in the **Microsoft Entra ID > Tenants** category.
b. Select the **Change main data** task.
c. On the **General** tab, remove the assigned account definitions.
d. Save the changes.

8. Delete the account definition.
   a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Account definitions > Account definitions** category.
   b. Select an account definition in the result list.
   c. Click to delete an account definition.

Assigning identities automatically to Microsoft Entra ID user accounts

When you add a user account, an existing identity can automatically be assigned to it. If necessary, a new identity can be created. The identity main data is created on the basis of existing user account main data. This mechanism can be triggered after a new user account is created either manually or through synchronization.

Define criteria for finding identities to apply to automatic identity assignment. If a user account is linked to an identity through the current mode, the user account is given, through an internal process, the default manage level of the account definition entered in the user account's target system. You can customize user account properties depending on how the behavior of the manage level is defined.

If you run this procedure during working hours, automatic assignment of identities to user accounts takes place from that moment onwards. If you disable the procedure again later, the changes only affect user accounts added or updated after this point in time. Existing identity assignments to user accounts remain intact.

NOTE: It is not recommended to assign identities using automatic identity assignment in the case of administrative user accounts. Use **Change main data** to assign identities to administrative user accounts for the respective user account.

For more information about assigning identities automatically, see the *One Identity Manager Target System Base Module Administration Guide*.

Run the following tasks to assign identities automatically:

- If you want identities to be assigned during the synchronization of user accounts, in the Designer, set the **TargetSystem | AzureAD | PersonAutoFullsync** configuration parameter and select the required mode.
- If you want identities to be assigned outside synchronization in the Designer, set the **TargetSystem | AzureAD | PersonAutoDefault** configuration parameter and select the required mode.

- In the **TargetSystem | AzureAD | PersonExcludeList** configuration parameter, define the user accounts for which no automatic assignment to identities is to take place.

  Example:

  `ADMINISTRATOR|GUEST`

  TIP: You can edit the value of the configuration parameter in the **Exclude list for automatic identity assignment** dialog.

  To edit the exclude list for automatic identity assignment

  1. In the Designer, edit the **PersonExcludeList** configuration parameter.
  2. Click **...** next to the **Value** field.

     This opens the **Exclude list for Microsoft Entra ID user accounts** dialog.
  3. To add a new entry, click **Add**.

     To edit an entry, select it and click **Edit**.
  4. Enter the name of the user account that does not allow identities to be assigned automatically.

     Each entry in the list is handled as part of a regular expression. You are allowed to use the usual special characters for regular expressions.
  5. To delete an entry, select it and click **Delete**.
  6. Click **OK**.
- Use the **TargetSystem | AzureAD | PersonAutoDisabledAccounts** configuration parameter to specify whether identities can be automatically assigned to disabled user accounts. User accounts do not obtain an account definition.
- Assign an account definition to the tenant. Ensure that the manage level to be used is entered as the default manage level.
- Define the search criteria for identity assignment in the tenant.

NOTE:

The following applies for synchronization:

- Automatic identity assignment takes effect if user accounts are added or updated.

The following applies outside synchronization:

- Automatic identity assignment takes effect if user accounts are added.

NOTE:

In the default installation, after synchronizing, identities are automatically created for the user accounts.If an account definition for the client is not known at the time of synchronization, user accounts are linked with identities. However, account definitions are not assigned. The user accounts are therefore in a **Linked** state.

To manage the user accounts using account definitions, assign an account definition and a manage level to these user accounts.

For more information, see Managing Microsoft Entra ID user accounts through account definitions on page 58.

Related topics

- Creating account definitions on page 65
- Assigning account definitions to Microsoft Entra ID tenants on page 84
- Changing manage levels for Microsoft Entra ID user accounts on page 92
- Editing search criteria for automatic identity assignment on page 89
- Finding identities and directly assigning them to user accounts on page 90

Editing search criteria for automatic identity assignment

NOTE: One Identity Manager supplies a default mapping for identity assignment. Only carry out the following steps when you want to customize the default mapping.

The criteria for identity assignments are defined for the tenant. You specify which user account properties must match the identity's properties such that the identity can be assigned to the user account. You can limit search criteria further by using format definitions.

The search criterion is written in XML notation to the **Search criteria for automatic identity assignment** column (`AccountToPersonMatchingRule`) in the `AADOrganization` table.

Search criteria are evaluated when identities are automatically assigned to user accounts. Furthermore, you can create a suggestion list for assignments of identities to user accounts based on the search criteria and make the assignment directly.

NOTE: Object definitions for user accounts that can have search criteria applied to them are predefined. For example, if you require other objects definitions that limit a preselection of user accounts, set up the respective custom object definitions in the Designer. For more information, see the *One Identity Manager Configuration Guide*.

To specify criteria for identity assignment

1. In the Manager, select the **Microsoft Entra ID > Tenants** category.
2. Select the tenant in the result list.
3. Select the **Define search criteria for identity assignment** task.
4. Specify which user account properties must match with which identity so that the

identity is linked to the user account.

Table 12: Default search criteria for user accounts and contacts

Apply to	Column for identity	Column for user account
Microsoft Entra ID user accounts	Central user account (`CentralAccount`)	Alias (`MailNickName`)

5. Save the changes.

For more information about defining search criteria, see the *One Identity Manager Target System Base Module Administration Guide*.

Related topics

- Assigning identities automatically to Microsoft Entra ID user accounts on page 87
- Finding identities and directly assigning them to user accounts on page 90

Finding identities and directly assigning them to user accounts

Based on the search criteria, you can create a suggestion list for the assignment of identities to user accounts and make the assignment directly. User accounts are grouped in different views for this.

- **Suggested assignments**: This view lists all user accounts to which One Identity Manager can assign an identity. All identities are shown that were found using the search criteria and can be assigned.
- **Assigned user accounts**: This view lists all user accounts to which an identity is assigned.
- **No identity assignment**: This view lists all user accounts to which no identity is assigned and for which no identity was found using the search criteria.

NOTE: To display disabled user accounts or deactivated identities in the view, enable the **Even locked accounts are mapped** option.

If you assign a deactivated identity to a user account, it might be locked or deleted depending on the configuration.

To apply search criteria to user accounts

1. In the Manager, select the **Microsoft Entra ID > Tenants** category.
2. Select the tenant in the result list.
3. Select the **Define search criteria for identity assignment** task.
4. At the bottom of the form, click **Reload**.

All possible assignments based on the search criteria are found in the target system for all user accounts. The three views are updated.

TIP: By double-clicking on an entry in the view, you can view the user account and identity main data.

The assignment of identities to user accounts creates connected user accounts (**Linked** state). To create managed user accounts (**Linked configured** state), you can assign an account definition at the same time.

To assign identities directly to user accounts

- Click **Suggested assignments**.
  1. Click the **Selection** box of all user accounts to which you want to assign the suggested identities. Multi-select is possible.
  2. (Optional) Select an account definition in the **Assign this account definition** drop-down, and select a manage level in the **Assign this account manage level** drop-down.
  3. Click **Assign selected**.
  4. Confirm the security prompt with **Yes**.

     The identities determined using the search criteria are assigned to the selected user accounts. If an account definition was selected, this is assigned to all selected user accounts.

  - OR -

- Click **No identity assignment**.
  1. Click **Select identity** for the user account to which you want to assign an identity. Select an identity from the drop-down.
  2. Click the **Selection** box of all user accounts to which you want to assign the selected identities. Multi-select is possible.
  3. (Optional) Select an account definition in the **Assign this account definition** drop-down, and select a manage level in the **Assign this account manage level** drop-down.
  4. Click **Assign selected**.
  5. Confirm the security prompt with **Yes**.

     The identities displayed in the **Identity** column are assigned to the selected user accounts. If an account definition was selected, this is assigned to all selected user accounts.

To remove assignments

- Click **Assigned user accounts**.
  1. Click the **Selection** box of all the user accounts with the identity assignment you want to delete. Multi-select is possible.
  2. Click **Remove selected**.

3. Confirm the security prompt with **Yes**.

   The assigned identities are removed from the selected user accounts.

Changing manage levels for Microsoft Entra ID user accounts

The default manage level is applied if you create user accounts using automatic identity assignment. You can change a user account manage level later.

To change the manage level for a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.
4. Select the manage level in the **Manage level** list on the **General** tab.
5. Save the changes.

Related topics

- Creating and editing Microsoft Entra ID user accounts on page 215

Supported user account types

Different types of user accounts, such as default user accounts, administrative user accounts, service accounts, or privileged user accounts, can be mapped in One Identity Manager.

The following properties are used for mapping different user account types.

- Identity type

  The **Identity type** property (`IdentityType` column) is used to describe the type of user account.

  Table 13: Identity types of user accounts

Identity type	Description	Value of the IdentityType column
Primary identity	Identity's default user account.	Primary

Identity type	Description	Value of the IdentityType column
Organizational identity	Secondary user account used for different roles in the company, for example for subcontracts with other functional areas.	Organizational
Personalized administrator identity	User account with administrative permissions, used by an identity.	Admin
Sponsored identity	User account used for a specific purpose. For example, for training purposes.	Sponsored
Shared identity	User account with administrative permissions, used by multiple identities.	Shared
Service identity	Service account.	Service

- Privileged user account

  Privileged user accounts are used to provide identities with additional privileges. This includes administrative user accounts or service accounts, for example. The user accounts are labeled with the **Privileged user account** property (`IsPrivilegedAccount` column).

Detailed information about this topic

- Default user accounts on page 93
- Administrative user accounts on page 94
- Providing administrative user accounts for one identity on page 95
- Providing administrative user accounts for several people on page 96
- Privileged user accounts on page 97

Default user accounts

Normally, each identity obtains a default user account, which has the permissions they require for their regular work. The user accounts are linked to the identity. The effect of the link and the scope of the identity's inherited properties on the user accounts can be configured through an account definition and its manage levels.

To create default user accounts through account definitions

1. Create an account definition and assign the **Unmanaged** and **Full managed** manage levels.

2. Specify how an identity's temporary deactivation, permanent deactivation, deletion, and security risks affect its user accounts and group memberships at each manage level.
3. Create a formatting rule for IT operating data.

   You use the mapping rule to define which rules are used to map IT operating data for user accounts and which default values are used if no IT operating data can be determined through an identity's primary roles.

   The type of IT operating data required depends on the target system. The following setting are recommended for default user accounts:

   - In the mapping rules for the `IsGroupAccount_Group`, `IsGroupAccount_SubSku`, `IsGroupAccount_DeniedService`, and `IsGroupAccount_DirectoryRole` columns, use the default value **`1`** and set the **Always use default value** option.
   - In the mapping rule for the `IdentityType` column, use the default value **`Primary`** and enable **Always use default value**.
4. Enter the effective IT operating data for the target system. Select the concrete target system under **Effects on**.

   Specify in the departments, cost centers, locations, or business roles that IT operating data should apply when you set up a user account.
5. Assign the account definition to identities.

   When the account definition is assigned to an identity, a new user account is created through the inheritance mechanism and subsequent processing.

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64

Administrative user accounts

An administrative user account must be used for certain administrative tasks. Administrative user accounts are usually predefined by the target system and have fixed names and login names, such as **Administrator**.

Administrative user accounts are imported into One Identity Manager during synchronization.

> NOTE: Some administrative user accounts can be automatically identified as privileged user accounts. To do this, in the Designer, enable the **Mark selected user accounts as privileged** schedule.

Related topics

- Providing administrative user accounts for one identity on page 95
- Providing administrative user accounts for several people on page 96

Providing administrative user accounts for one identity

Use this task to create an administrative user account that can be used by an identity.

Prerequisites

- The user account must be labeled as a personalized administrator identity.
- The identity that will be using the user account must be marked as a personalized administrator identity.
- The identity that will be using the user account must be linked to a main identity.

To prepare an administrative user account for an identity

1. Label the user account as a personalized administrator identity.
   a. In the Manager, select the **Microsoft Entra ID > User accounts** category.
   b. Select the user account in the result list.
   c. Select the **Change main data** task.
   d. On the **General** tab, in the **Identity** selection list, select `Personalized administrator identity`.
2. Link the user account to the identity that will be using this administrative user account.
   a. In the Manager, select the **Microsoft Entra ID > User accounts** category.
   b. Select the user account in the result list.
   c. Select the **Change main data** task.
   d. On the **General** tab, in the **Identity** selection list, select the identity that will be using this administrative user account.

      TIP: If you are the target system manager, you can selected to create a new identity.

Related topics

- Providing administrative user accounts for several people on page 96
- For more information about mapping identity types, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Providing administrative user accounts for several people

Use this task to create an administrative user account that can be used by more that one identity.

Prerequisites

- The user account must be labeled as a shared identity.
- There must be an identity with the type **`Shared identity`** available. The shared identity must have a manager.
- The identities who are permitted to use the user account must be labeled as a primary identity.

To prepare an administrative user account for multiple identities

1. Label the user account as a shared identity.
   a. In the Manager, select the **Microsoft Entra ID > User accounts** category.
   b. Select the user account in the result list.
   c. Select the **Change main data** task.
   d. On the **General** tab, in the **Identity** drop-down, select **`Shared identity`**.
2. Link the user account to an identity.
   a. In the Manager, select the **Microsoft Entra ID > User accounts** category.
   b. Select the user account in the result list.
   c. Select the **Change main data** task.
   d. On the **General** tab, in the **Identity** drop-down, select an identity the type **`Shared identity`**.

   > TIP: If you are the target system manager, you can use the button to create a new shared identity.
3. Assign the identities who will use this administrative user account to the user account.
   a. In the Manager, select the **Microsoft Entra ID > User accounts** category.
   b. Select the user account in the result list.
   c. Select the **Assign identities authorized to use** task.
   d. In the **Add assignments** pane, add identities.

   > TIP: In the **Remove assignments** pane, you can remove assigned identities.

To remove an assignment

- Select the identity and double-click ✓.

Related topics

- Providing administrative user accounts for one identity on page 95
- For more information about mapping identity types, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Privileged user accounts

Privileged user accounts are used to provide identities with additional privileges. This includes administrative user accounts or service accounts, for example. The user accounts are labeled with the **Privileged user account** property (`IsPrivilegedAccount` column).

NOTE: The criteria according to which user accounts are automatically identified as privileged are defined as extensions to the view definition (ViewAddOn) in the `TSBVAccountIsPrivDetectRule` table (which is a table of the **`Union`** type). The evaluation is done in the `TSB_SetIsPrivilegedAccount` script.

To create privileged users through account definitions

1. Create an account definition. Create a new manage level for privileged user accounts and assign this manage level to the account definition.
2. If you want to prevent the properties for privileged user accounts from being overwritten, set the **IT operating data overwrites** property for the manage level to **`Only initially`**. In this case, the properties are populated just once when the user accounts are created.
3. Specify how an identity's temporary deactivation, permanent deactivation, deletion, and security risks affect its user accounts and group memberships in the manage level.
4. Create a formatting rule for the IT operating data.

   You use the mapping rule to define which rules are used to map IT operating data for user accounts and which default values are used if no IT operating data can be determined through an identity's primary roles.

   The type of IT operating data required depends on the target system. The following settings are recommended for privileged user accounts:

   - In the mapping rule for the `IsPrivilegedAccount` column, use the default value **`1`** and set the **Always use default value** option.
   - You can also specify a mapping rule for the `IdentityType` column. The column owns different permitted values that represent user accounts.
   - To prevent privileged user accounts from inheriting the entitlements of the default user, define a mapping rule for the `IsGroupAccount_Group`,

`IsGroupAccount_SubSku`, and `IsGroupAccount_DeniedService` columns with a default value of `0` and set the **Always use default value** option.

5. Enter the effective IT operating data for the target system.

   Specify in the departments, cost centers, locations, or business roles which IT operating data should apply when you set up a user account.

6. Assign the account definition directly to identities who work with privileged user accounts.

   When the account definition is assigned to an identity, a new user account is created through the inheritance mechanism and subsequent processing.

TIP: If customization requires that the login names of privileged user accounts follow a defined naming convention, specify how the login names are formatted in the template.

- To use a prefix for the login name, in the Designer, set the **TargetSystem | AzureAD | Accounts | PrivilegedAccount | AccountName_Prefix** configuration parameter.
- To use a postfix for the login name, in the Designer, set the **TargetSystem | AzureAD | Accounts | PrivilegedAccount | AccountName_Postfix** configuration parameter.

These configuration parameters are evaluated in the default installation, if a user account is marked with the **Privileged user account** property (`IsPrivilegedAccount` column). The user account login names are renamed according to the formatting rules. This also occurs if the user accounts are labeled as privileged using the **Mark selected user accounts as privileged** schedule. If necessary, modify the schedule in the Designer.

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64

Updating identities when Microsoft Entra ID user account are modified

In One Identity Manager, modifications to identity properties are forwarded to the associated user accounts and subsequently provisioned in the target system. In certain circumstances, it may be necessary to forward user account modifications in the target system to identity properties in One Identity Manager.

Example: Changes to user accounts in the target system

During testing, user accounts from the target system are only read into One Identity Manager and identities created. User account administration (creating, modifying, and deleting) should be done later through One Identity Manager. During testing, user accounts are modified further in the target system, which can lead to drifts in user account properties and identity properties. Due to this, user account modifications loaded on resynchronization should be temporarily published to identities who are already created. This means data is not lost when user account administration is put into effect through One Identity Manager.

To update identities when user accounts are modified

- In the Designer, set the **TargetSystem | AzureAD | PersonUpdate** configuration parameter.

Modifications to user accounts are loaded into One Identity Manager during synchronization. These modifications are forwarded to the associated identities through subsequent scripting and processing.

NOTE:

- When making changes to user accounts, the identities are only updated for user accounts with the **Unmanaged** manage level and that are linked to an identity.
- Only the identity created by the modified user account is updated. The data source from which the identity was created is shown in the **Import data source** property. If other user accounts are assigned to the identity, changes to these user accounts do not cause the identity to be updated.
- For identities who do not yet have the **Import data source** set, the user account's target system is entered as the data source for the import during the first update of the connected user account.

User account properties are mapped to identity properties using the `AAD_PersonUpdate_AADUser` script. To make the mapping easier to customize, the script is overwritable.

To customize, create a copy of the script and start the script coding follows:

```
Public Overrides Function AAD_PersonUpdate_AADUser (ByVal UID_Account As String,
oldUserPrincipalName As String, ProcID As String)
```

This redefines the script and overwrites the original. The process does not have to be changed in this case.

Specifying deferred deletion for Microsoft Entra ID user accounts

You can use deferred deletion to specify how long the user accounts remain in the database after deletion is triggered before they are finally removed. By default, user accounts are finally deleted from the database after 30 days. First, the user accounts are disabled or locked. You can reenable the user accounts up until deferred deletion runs. After deferred deletion is run, the user accounts are deleted from the database and cannot be restored anymore.

You have the following options for configuring deferred deletion.

- Global deferred deletion: Deferred deletion applies to user accounts in all target system. The default value is `30` days.

  In the Designer, enter a different value for deferred deletion in the **Deferred deletion [days]** property of the `AADUser` table.

- Object-specific deferred deletion: Deferred deletion can be configured depending on certain properties of the accounts.

  To use object-specific deferred deletion, in the Designer, create a `Script (deferred deletion)` for the **AADUser** table.

Example: Script for object-specific deferred deletion

Deferred deletion of privileged user accounts is 10 days. The following **Script (deferred deletion)** is entered in the table.

```
If Not $IsPrivilegedAccount:Bool$ Then
     Value = 10
End If
```

For more information on editing table definitions and configuring deferred deletion in the Designer, see the *One Identity Manager Configuration Guide*.

Managing memberships in Microsoft Entra ID groups

Microsoft Entra ID user accounts can be grouped into Microsoft Entra ID groups that can be used to regulate access to resources.

In One Identity Manager, you can assign Microsoft Entra ID groups directly to user accounts or they can be inherited through departments, cost centers, locations, or business roles. Users can also request the groups through the Web Portal. To do this, groups are provided in the IT Shop.

NOTE: Assignments to Microsoft Entra ID groups that are synchronized with the local Active Directory are not allowed in One Identity Manager. These groups cannot be requested through the web portal. You can only manage these groups in your locally. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Detailed information about this topic

- Assigning Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 101
- Effectiveness of group memberships on page 112
- Microsoft Entra ID group inheritance based on categories on page 115
- Overview of all assignments on page 117

Assigning Microsoft Entra ID groups to Microsoft Entra ID user accounts

Microsoft Entra ID groups can be assigned directly or indirectly to Microsoft Entra ID user accounts.

In the case of indirect assignment, identities and Microsoft Entra ID groups are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. The Microsoft Entra ID groups assigned to an identity are calculated from the position in the hierarchy and the direction of inheritance. If you add an identity to roles and that identity

owns a Microsoft Entra ID user account, the Microsoft Entra ID user account is added to the Microsoft Entra ID group.

Furthermore, Microsoft Entra ID groups can be requested through the Web Portal. To do this, add identities to a shop as customers. All Microsoft Entra ID groups are assigned to this shop can be requested by the customers. Requested Microsoft Entra ID groups are assigned to the identities after approval is granted.

Through system roles, Microsoft Entra ID groups can be grouped together and assigned to identities and workdesks as a package. You can create system roles that contain only Microsoft Entra ID groups. You can also group any number of company resources into a system role.

To react quickly to special requests, you can assign Microsoft Entra ID groups directly to Microsoft Entra ID user accounts.

For more information see the following guides:

Topic	Guide
Basic principles for assigning and inheriting company resources	*One Identity Manager Identity Management Base Module Administration Guide* *One Identity Manager Business Roles Administration Guide*
Assigning company resources through IT Shop requests	*One Identity Manager IT Shop Administration Guide*
System roles	*One Identity Manager System Roles Administration Guide*

Detailed information about this topic

- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Adding Microsoft Entra ID groups automatically to the IT Shop on page 109
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
- Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts on page 112

Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts

In the case of indirect assignment, identities and Microsoft Entra ID groups are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. When assigning Microsoft Entra ID groups indirectly, check the following settings and modify them if necessary:

1. Assignment of identities and Microsoft Entra ID groups is permitted for role classes (departments, cost centers, locations, or business roles).

   For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.

   To configure assignments to roles of a role class

   1. In the Manager, select role classes in the **Organizations > Basic configuration data > Role classes** category.

      - OR -

      In the Manager, select role classes in the **Business roles > Basic configuration data > Role classes** category.
   2. Select the **Configure role assignments** task and configure the permitted assignments.
      - To generally allow an assignment, enable the **Assignments allowed** column.
      - To allow direct assignment, enable the **Direct assignments permitted** column.
   3. Save the changes.
2. Settings for assigning Microsoft Entra ID groups to Microsoft Entra ID user accounts.
   - The Microsoft Entra ID user account is linked to an identity.
   - The Microsoft Entra ID user account has the **Groups can be inherited** option set.

NOTE: There are other configuration settings that play a role when company resources are inherited through departments, cost centers, locations, and business roles. For example, role inheritance might be blocked or inheritance of identities not allowed. For more information about the basic principles for assigning company resources, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Related topics

- Creating and editing Microsoft Entra ID user accounts on page 215
- General main data of Microsoft Entra ID user accounts on page 216

Assigning Microsoft Entra ID groups to departments, cost centers and locations

Assign groups to departments, cost centers, or locations so that the group can be assigned to user accounts through these organizations.

This task is not available for dynamic groups.

To assign a group to departments, cost centers, or locations (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign organizations** task.
4. In the **Add assignments** pane, assign the organizations:
   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
5. Save the changes.

To assign groups to a department, a cost center, or a location (non role-based login or role-based login)

1. In the Manager, select the **Organizations > Departments** category.

   - OR -

   In the Manager, select the **Organizations > Cost centers** category.

   - OR -

   In the Manager, select the **Organizations > Locations** category.
2. Select the department, cost center, or location in the result list.
3. Select the **Assign Microsoft Entra ID groups** task.
4. In the **Add assignments** pane, assign groups.

   TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

   To remove an assignment

   - Select the group and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
- Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts on page 112
- One Identity Manager users for managing Microsoft Entra ID on page 13

Assigning Microsoft Entra ID groups to business roles

NOTE: This function is only available if the Business Roles Module is installed.

Assign the group to business roles so that the group is assigned to user accounts through these business roles.

This task is not available for dynamic groups.

To assign a group to a business role (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

To assign groups to a business role (non role-based login or role-based login)

1. In the Manager, select the **Business roles > <role class>** category.
2. Select the business role in the result list.
3. Select the **Assign Microsoft Entra ID groups** task.
4. In the **Add assignments** pane, assign the groups.

TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

To remove an assignment

- Select the group and double-click ✓.

5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Adding Microsoft Entra ID groups to system roles on page 106
- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
- Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts on page 112
- One Identity Manager users for managing Microsoft Entra ID on page 13

Adding Microsoft Entra ID groups to system roles

NOTE: This function is only available if the System Roles Module is installed.

Use this task to add a group to system roles.

If you assign a system role to identities, all Microsoft Entra ID user accounts owned by these identities inherit the group.

This task is not available for dynamic groups.

NOTE: Groups with **Only use in IT Shop** set can only be assigned to system roles that also have this option set. For more information, see the *One Identity Manager System Roles Administration Guide*.

To assign a group to system roles

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign system roles** task.

4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.

5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
- Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts on page 112

Adding Microsoft Entra ID groups to the IT Shop

When you assign a group to an IT Shop shelf, it can be requested by the shop customers. To ensure it can be requested, further prerequisites need to be guaranteed:

- The group is not a dynamic group.
- The group must be labeled with the **IT Shop** option.
- The group must be assigned a service item.

  TIP: In the Web Portal, all products that can be requested are grouped together by service category. To make the group easier to find in the Web Portal, assign a service category to the service item.

- If you only want the group to be assigned to identities through IT Shop requests, the group must also be labeled with the **Use only in IT Shop** option. Direct assignment to hierarchical roles or user accounts is no longer permitted.

NOTE: With role-based login, the IT Shop administrators can assign groups to IT Shop shelves. Target system administrators are not authorized to add groups to IT Shop.

To add a group to the IT Shop.

1. In the Manager, select the **Microsoft Entra ID > Groups** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID groups** (role-based login) category.
2. In the result list, select the group.
3. Select the **Add to IT Shop** task.
4. To assign the group to shelves, select the **IT Shop shelves** tab and, in the **Add assignments** section, double-click on the shelves.
5. To assign the group to IT Shop templates, select the **IT Shoptemplates** tab and, in the **Add assignments** section, double-click on the templates.
6. Save the changes.

To remove a group from individual shelves of the IT Shop

1. In the Manager, select the **Microsoft Entra ID > Groups** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID groups** (role-based login) category.
2. In the result list, select the group.
3. Select the **Add to IT Shop** task.
4. To remove the group from the shelves, select the **IT Shop shelves** tab and, in the **Remove assignments** section, double-click the shelves.
5. To remove the group from the IT Shop templates, select the **IT Shop templates** tab and, in the **Remove assignments** section, double-click the templates.
6. Save the changes.

To remove a group from all shelves of the IT Shop

1. In the Manager, select the **Microsoft Entra ID > Groups** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID groups** (role-based login) category.
2. In the result list, select the group.
3. Select the **Remove from all shelves (IT Shop)** task.
4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The group is removed from all shelves by the One Identity Manager Service. All requests and assignment requests with this group are canceled.

For more information about requesting company resources through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Related topics

- General main data for Microsoft Entra ID groups on page 236
- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Adding Microsoft Entra ID groups automatically to the IT Shop on page 109
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
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Adding Microsoft Entra ID groups automatically to the IT Shop

The following steps can be used to automatically add Microsoft Entra ID groups to the IT Shop. Synchronization ensures that the Microsoft Entra ID groups are added to the IT Shop. If necessary, you can manually start synchronization with the Synchronization Editor. New Microsoft Entra ID groups created in One Identity Manager also are added automatically to the IT Shop.

To add Microsoft Entra ID groups automatically to the IT Shop

1. In the Designer, set the **QER | ITShop | AutoPublish | AADGroup** configuration parameter.
2. In order not to add Microsoft Entra ID groups to the IT Shop automatically, in the Designer, set the **QER | ITShop | AutoPublish | AADGroup | ExcludeList** configuration parameter.

   This configuration parameter contains a listing of all Microsoft Entra ID groups that should not be allocated to the IT Shop automatically. You can extend this list if required. To do this, enter the names of the groups in the configuration parameter. Names are listed in a pipe (|) delimited list. Regular expressions are supported.
3. Compile the database.

The Microsoft Entra ID groups are added automatically to the IT Shop from now on.

The following steps are run to add a Microsoft Entra ID group to the IT Shop.

1. A service item is determined for the Microsoft Entra ID group.

   The service item is tested for each Microsoft Entra ID group and modified if necessary. The name of the service item corresponds to the name of the Microsoft Entra ID group.

   - The service item is modified for Microsoft Entra ID groups with service items.
   - Microsoft Entra ID groups without service items are allocated new service items.

2. The service item is assigned to either the **Microsoft Entra ID groups | Security groups** default service category or the **Microsoft Entra ID groups | Distribution groups** default service category.

3. An application role for product owners is determined and assigned to the service item.

   Product owners can approve requests for membership in these Microsoft Entra ID groups. The default product owner is the Microsoft Entra ID group's owner.

   NOTE: The application role for the product owner must be added under the **Request & Fulfillment | IT Shop | Product owner** application role.

   - If the owner of the Microsoft Entra ID group is already a member of an application role for product owners, this application role is assigned to the service item. Therefore, all members of this application role become product owners of the Microsoft Entra ID group.
   - If the owner of the Microsoft Entra ID group is not yet a member of an application role for product owners, a new application role is created. The name of the application corresponds to the name of the owner.
     - If the owner is a user account, the user account's identity is added to the application role.
     - If it is a group of owners, the identities of all this group's user accounts are added to the application role.

4. The Microsoft Entra ID group is labeled with the **IT Shop** option and assigned to the **IT Shop groups** Microsoft Entra ID shelf in the **Identity & Access Lifecycle** shop.

Then the shop customers can use the Microsoft Entra ID to request memberships in Web Portal groups.

NOTE: If a Microsoft Entra ID group is irrevocably deleted from the One Identity Manager database, the associated service item is also deleted.

For more information about configuring the One Identity Manager IT Shop Administration Guide, see the *IT Shop*. For more information about requesting access requests in the Web Portal, see the *One Identity Manager Web Portal User Guide*.

Related topics

- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104

- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111
- Adding Microsoft Entra ID groups to Microsoft Entra ID groups on page 238

Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups

To react quickly to special requests, you can assign groups directly to user accounts. You cannot directly assign groups that have the **Only use in IT Shop** option.

NOTE: User accounts cannot be manually added to dynamic groups.

To assign user accounts directly to a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign user accounts** task.
4. In the **Add assignments** pane, assign the user accounts.

   TIP: In the **Remove assignments** pane, you can remove assigned user accounts.

   To remove an assignment

   - Select the user account and double-click ✓.
5. Save the changes.

Related topics

- Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts on page 112
- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Adding Microsoft Entra ID groups to the IT Shop on page 107

Assigning Microsoft Entra ID groups directly to Microsoft Entra ID user accounts

To react quickly to special requests, you can assign groups directly to user accounts. You cannot directly assign groups that have the **Only use in IT Shop** option.

NOTE: User accounts cannot be manually added to dynamic groups.

To assign groups directly to user accounts

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign groups** task.
4. In the **Add assignments** pane, assign the groups.

   TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

   To remove an assignment

   - Select the group and double-click ✓.
5. Save the changes.

Related topics

- Assigning Microsoft Entra ID groups to departments, cost centers and locations on page 104
- Assigning Microsoft Entra ID groups to business roles on page 105
- Adding Microsoft Entra ID groups to system roles on page 106
- Adding Microsoft Entra ID groups to the IT Shop on page 107
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID groups on page 111

Effectiveness of group memberships

When groups are assigned to user accounts an identity may obtain two or more groups, which are not permitted in this combination. To prevent this, you can declare mutually exclusive groups. To do this, you specify which of the two groups should apply to the user accounts if both are assigned.

It is possible to assign an excluded group at any time either directly, indirectly, or with an IT Shop request. One Identity Manager determines whether the assignment is effective.

NOTE:

- You cannot define a pair of mutually exclusive groups. That means, the definition "Group A excludes group B" `AND` "Group B excludes groups A" is not permitted.
- You must declare each group to be excluded from a group separately. Exclusion definitions cannot be inherited.
- One Identity Manager does not check if membership of an excluded group is permitted in another group (table).

The effectiveness of the assignments is mapped in the `AADUserInGroup` and `AADBaseTreeHasGroup` tables by the `XIsInEffect` column.

Example: The effect of group memberships

- Group A is defined with permissions for triggering requests in a tenant. A group B is authorized to make payments. A group C is authorized to check invoices.
- Group A is assigned through the "Marketing" department, group B through "Finance", and group C through the "Control group" business role.

Jo User1 has a user account in this tenant. They primarily belong to the "Marketing" department. The "Control group" business role and the "Finance" department are assigned to them secondarily. Without an exclusion definition, the user account obtains all the permissions of groups A, B, and C.

By using suitable controls, you want to prevent an identity from being able to trigger a request and to pay invoices. That means, groups A, B, and C are mutually exclusive. An identity that checks invoices may not be able to make invoice payments as well. That means, groups B and C are mutually exclusive.

Table 14: Specifying excluded groups (`AADGroupExclusion` table)

Effective group	Excluded group
Group A	
Group B	Group A
Group C	Group B

Table 15: Effective assignments

Identity	Member in role	Effective group
Pat Identity1	Marketing	Group A
Jan User3	Marketing, finance	Group B
Jo User1	Marketing, finance, control group	Group C
Chris User2	Marketing, control group	Group A, Group C

Only the group C assignment is in effect for Jo User1. It is published in the target system. If Jo User1 leaves the "control group" business role at a later date, group B also takes effect.

The groups A and C are in effect for Chris User2 because the groups are not defined as mutually exclusive. That means that the identity is authorized to trigger requests and to check invoices. If this should not be allowed, define further exclusion for group C.

Table 16: Excluded groups and effective assignments

Identity	Member in role	Assigned group	Excluded group	Effective group
Chris User2	Marketing	Group A		Group C
	Control group	Group C	Group B Group A	

Prerequisites

- The **QER | Structures | Inherite | GroupExclusion** configuration parameter is set.

  In the Designer, set the configuration parameter and compile the database.

  NOTE: If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.

- Mutually exclusive groups belong to the same tenant.

To exclude a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select a group in the result list.
3. Select the **Exclude groups** task.
4. In the **Add assignments** pane, assign the groups that are mutually exclusive to the selected group.

   - OR -

   In the **Remove assignments** pane, remove the groups that are no longer mutually exclusive.
5. Save the changes.

Microsoft Entra ID group inheritance based on categories

In One Identity Manager, user accounts can selectively inherit groups, administrator roles, subscriptions, and disabled service plans. To do this, the groups (administrator roles, subscriptions, and disabled service plans) and the user accounts are divided into categories. The categories can be freely selected and are specified using a mapping rule. Each category is given a specific position within the template. The mapping rule contains different tables. Use the user account table to specify categories for target system dependent user accounts. In the other tables, enter your categories for the groups, administrator roles, subscriptions, and disabled service plans. Each table contains the category positions **position 1** to **position 63**.

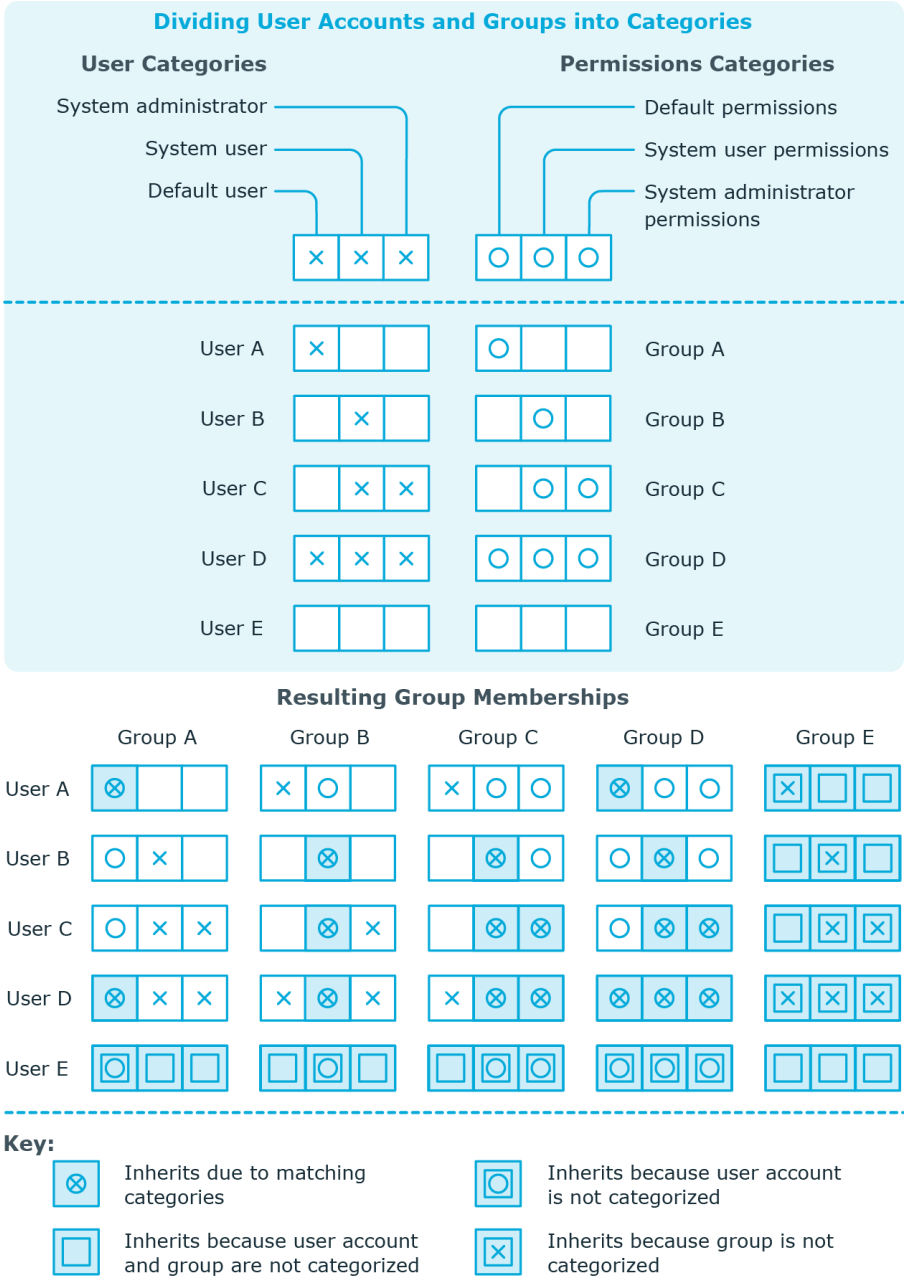
Every user account can be assigned to one or more categories. Each group can also be assigned to one or more categories. The group is inherited by the user account when at least one user account category items matches an assigned group. The group is also inherited by the user account if the group or the user account is not put into categories.

NOTE: Inheritance through categories is only taken into account when groups are assigned indirectly through hierarchical roles. Categories are not taken into account when groups are directly assigned to user accounts.

Table 17: Category examples

Category item	Categories for user accounts	Categories for groups
1	Default user	Default permissions
2	System users	System user permissions
3	System administrator	System administrator permissions

Figure 2: Example of inheriting through categories.



To use inheritance through categories

1. In the Manager, define the categories in the Microsoft Entra ID tenant.
2. In the Manager, assign categories to user accounts through their main data.
3. In the Manager, assign categories to groups through their main data.

Related topics

- Defining categories for the inheritance of entitlements on page 209
- General main data of Microsoft Entra ID user accounts on page 216
- General main data for Microsoft Entra ID groups on page 236
- Editing Microsoft Entra ID subscription main data on page 249

Overview of all assignments

The **Overview of all assignments** report is displayed for some objects, such as authorizations, compliance rules, or roles. The report finds all the roles, for example, departments, cost centers, locations, business roles, and IT Shop structures in which there are identities who own the selected base object. In this case, direct as well as indirect base object assignments are included.

Example: Assignment overview

- If the report is created for a resource, all roles are determined in which there are identities with this resource.
- If the report is created for a group or another system entitlement, all roles are determined in which there are identities with this group or system entitlement.
- If the report is created for a compliance rule, all roles are determined in which there are identities who violate this compliance rule.
- If the report is created for a department, all roles are determined in which identities of the selected department are also members.
- If the report is created for a business role, all roles are determined in which identities of the selected business role are also members.

To display detailed information about assignments

- To display the report, select the base object from the navigation or the result list and select the **Overview of all assignments** report.
- Click the **Used by** button in the report toolbar to select the role class for which you want to determine whether roles exist that contain identities with the selected base object.

  All the roles of the selected role class are shown. The color coding of elements identifies the role in which there are identities with the selected base object. The meaning of the report control elements is explained in a separate legend. To access the legend, click the icon in the report's toolbar.
- Double-click a control to show all child roles belonging to the selected role.

- By clicking the button in a role's control, you display all identities in the role with the base object.
- Use the small arrow next to to start a wizard that allows you to bookmark this list of identities for tracking. This creates a new business role to which the identities are assigned.

Figure 3: Toolbar of the Overview of all assignments report.

Table 18: Meaning of icons in the report toolbar

Icon	Meaning
	Show the legend with the meaning of the report control elements
	Saves the current report view as a graphic.
	Selects the role class used to generate the report.
	Displays all roles or only the affected roles.

Managing Microsoft Entra ID administrator roles assignments

In One Identity Manager, you can assign the Microsoft Entra ID administrator roles directly to user accounts or they can be inherited through departments, cost centers, locations, or business roles. Users can also request the administrator roles through the Web Portal. To do this, administrator roles are provided in the IT Shop.

Detailed information about this topic

- Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID user accounts on page 119
- Microsoft Entra ID administrator role inheritance based on categories on page 130
- Overview of all assignments on page 117

Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID user accounts

Microsoft Entra ID administrator roles can be assigned directly or indirectly to Microsoft Entra ID user accounts.

In the case of indirect assignment, identities and Microsoft Entra ID administrator roles are assigned to hierarchical roles, such as, departments, cost centers, locations, or business roles. The Microsoft Entra ID administrator roles assigned to an identity are calculated from the position in the hierarchy and the direction of inheritance. If you add an identity to roles and that identity owns a Microsoft Entra ID user account, the Microsoft Entra ID user account is added to the Microsoft Entra ID administrator roles.

You can also request Microsoft Entra ID administration roles in the Web Portal. To do this, add identities to a shop as customers. All Microsoft Entra ID administrator roles assigned as products to this shop, can be requested by the customers. Requested Microsoft Entra ID administrator roles are assigned to the identities after approval is granted.

Through system roles, Microsoft Entra ID administrator roles can be grouped together and assigned to identities as a package. You can create system roles that contain only Microsoft Entra ID administrator roles. You can also group any number of company resources into a system role.

To react quickly to special requests, you can assign Microsoft Entra ID administrator roles directly to Microsoft Entra ID user accounts.

For more information see the following guides:

Topic	Guide
Basic principles for assigning and inheriting company resources	*One Identity Manager Identity Management Base Module Administration Guide* *One Identity Manager Business Roles Administration Guide*
Assigning company resources through IT Shop requests	*One Identity Manager IT Shop Administration Guide*
System roles	*One Identity Manager System Roles Administration Guide*

Detailed information about this topic

- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Adding Microsoft Entra ID administrator roles to system roles on page 124
- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128
- Assigning Microsoft Entra ID administrator roles directly to Microsoft Entra ID user accounts on page 129

Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts

In the case of indirect assignment, identities and Microsoft Entra ID administrator roles are assigned to hierarchical roles, such as, departments, cost centers, locations, or business roles. When assigning Microsoft Entra ID administrator roles indirectly, check the following settings and modify them if necessary.

1. Assignment of identities and Microsoft Entra ID administrator roles is permitted for role classes (departments, cost centers, locations, or business roles).

   For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.

   To configure assignments to roles of a role class

   1. In the Manager, select role classes in the **Organizations > Basic configuration data > Role classes** category.

      - OR -

      In the Manager, select role classes in the **Business roles > Basic configuration data > Role classes** category.
   2. Select the **Configure role assignments** task and configure the permitted assignments.
      - To generally allow an assignment, enable the **Assignments allowed** column.
      - To allow direct assignment, enable the **Direct assignments permitted** column.
   3. Save the changes.
2. Settings for assigning Microsoft Entra ID administrator roles to Microsoft Entra ID user accounts.
   - The Microsoft Entra ID user account is linked to an identity.
   - The Microsoft Entra ID user account has the **Administrator roles can be inherited** option set.

NOTE: There are other configuration settings that play a role when company resources are inherited through departments, cost centers, locations, and business roles. For example, role inheritance might be blocked or inheritance of identities not allowed. For more information about the basic principles for assigning company resources, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Related topics

- Creating and editing Microsoft Entra ID user accounts on page 215
- General main data of Microsoft Entra ID user accounts on page 216

Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations

By assigning administrator roles to departments, cost centers, or locations, you enable the group to be assigned to user accounts through these organizations.

To assign an administrator role to departments, cost centers, or locations (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Assign organizations** task.
4. In the **Add assignments** pane, assign the organizations:
   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
5. Save the changes.

To assign administrator roles to departments, cost centers or locations (role-based login)

1. In the Manager, select the **Organizations > Departments** category.

   - OR -

   In the Manager, select the **Organizations > Cost centers** category.

   - OR -

   In the Manager, select the **Organizations > Locations** category.
2. Select the department, cost center or location in the result list.
3. Select the **Assign Microsoft Entra ID administrator roles** task.
4. In the **Add assignments** pane, assign administrator roles.

   TIP: In the **Remove assignments** pane, you can remove assigned administrator roles.

   To remove an assignment

   - Select the administrator role and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128
- Adding Microsoft Entra ID administrator roles to system roles on page 124

- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125
- One Identity Manager users for managing Microsoft Entra ID on page 13

Assigning Microsoft Entra ID administrator roles to business roles

NOTE: This function is only available if the Business Roles Module is installed.

By assigning administrator roles to business roles, the administrator role can be assigned to user accounts through these business roles.

To assign an administrator role to business roles (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

To assign administrator roles to a business role (non role-based login)

1. In the Manager, select the **Business roles > <role class>** category.
2. Select the business role in the result list.
3. Select the **Assign Microsoft Entra ID administrator roles** task.
4. In the **Add assignments** pane, assign administrator roles.

   TIP: In the **Remove assignments** pane, you can remove assigned administrator roles.

   To remove an assignment

   - Select the administrator role and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121

- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128
- Adding Microsoft Entra ID administrator roles to system roles on page 124
- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125
- One Identity Manager users for managing Microsoft Entra ID on page 13

Adding Microsoft Entra ID administrator roles to system roles

NOTE: This function is only available if the System Roles Module is installed.

Use this task to add an administrator role to system roles. If you assign a system role to identities, all Microsoft Entra ID user accounts owned by these identities inherit the administrator role.

NOTE: Applications in which the **Only use in IT Shop** option is set can only be assigned to system roles that also have this option set. For more information, see the *One Identity Manager System Roles Administration Guide*.

To assign an administrator role to system roles

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128
- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125

Adding Microsoft Entra ID administrator roles in the IT Shop

Once an administrator role has been assigned to an IT Shop shelf, it can be requested by the shop customers. To ensure it can be requested, further prerequisites need to be guaranteed:

- The administrator role must be labeled with the **IT Shop** option.
- The administrator role must be assigned to a service item.
- If the administrator role can only be assigned to identities using IT Shop requests, the administrator role must be also labeled with the **Only use in IT Shop** option. Direct assignment to hierarchical roles may not be possible.

NOTE: IT Shop administrators can assign administrator roles to IT Shop shelves in the case of role-based login. Target system administrators are not authorized to add administrator roles in the IT Shop.

To add an administrator role in the IT Shop

1. In the Manager, select the **Microsoft Entra ID > administrator roles** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID administrator roles** (role-based login) category.
2. Select the administrator role in the result list.
3. Select the **Add to IT Shop** task.
4. In the **Add assignments** pane, assign the administrator role to the IT Shop shelves.
5. Save the changes.

To remove an administrator role from individual IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > administrator roles** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID administrator roles** (role-based login) category.
2. Select the administrator role in the result list.
3. Select the **Add to IT Shop** task.
4. In the **Remove assignments** pane, remove the administrator role from the IT Shop shelves.
5. Save the changes.

To remove an administrator role from all IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > administrator roles** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID administrator roles** (role-based login) category.
2. Select the administrator role in the result list.
3. Select the **Remove from all shelves (IT Shop)** task.
4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The administrator role is removed from all shelves by the One Identity Manager Service. All requests and assignment requests with this administrator role are canceled at the same time.

For more information about requesting company resources through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Related topics

- Editing main data of Microsoft Entra ID administrator roles on page 243
- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128
- Adding Microsoft Entra ID administrator roles to system roles on page 124

Adding Microsoft Entra ID administrator roles to the IT Shop

The following steps can be used to automatically add Microsoft Entra ID administration roles to the IT Shop. Synchronization ensures that the Microsoft Entra ID administration roles are added to the IT Shop. If necessary, you can manually start synchronization with the Synchronization Editor. New Microsoft Entra ID administration roles created in One Identity Manager also are added automatically to the IT Shop.

To automatically include Microsoft Entra ID administrator roles to the IT Shop

1. In the Designer, set the **QER | ITShop | AutoPublish | AADDirectoryRole** configuration parameter.
2. In order not to add Microsoft Entra ID administration roles to the IT Shop automatically, in the Designer, set the **QER | ITShop | AutoPublish | AADDirectoryRole | ExcludeList** configuration parameter.

   This configuration parameter contains a listing of all Microsoft Entra ID administration roles that should not be allocated to the IT Shop automatically. You can extend this list if required. To do this, enter the name of the administration roles in the configuration parameter Names are listed in a pipe (|) delimited list. Regular expressions are supported.
3. Compile the database.

The Microsoft Entra ID administration roles are added automatically to the IT Shop from now on.

The following steps are run to add a Microsoft Entra ID administration role to the IT Shop.

1. A service item is determined for the Microsoft Entra ID administration role.

   The service item is tested for each Microsoft Entra ID administration role and modified if necessary. The name of the service item corresponds to the name of the Microsoft Entra ID administration role.

   - The service item is modified for Microsoft Entra ID administration roles with service items.
   - Microsoft Entra ID administration roles without service items are allocated new service items.
2. The service item is assigned to the **Microsoft Entra ID administrator roles** default service category.
3. An application role for product owners is determined and assigned to the service item.

   Product owners can approve requests for membership in these Microsoft Entra ID groups. The default product owner is the Microsoft Entra ID group's owner.

   NOTE: The application role for the product owner must be added under the **Request & Fulfillment | IT Shop | Product owner** application role.

   - If the owner of the Microsoft Entra ID group is already a member of an application role for product owners, this application role is assigned to the service item. Therefore, all members of this application role become product owners of the Microsoft Entra ID group.
   - If the owner of the Microsoft Entra ID group is not yet a member of an application role for product owners, a new application role is created. The name of the application corresponds to the name of the owner.
     - If the owner is a user account, the user account's identity is added to the application role.

- If it is a group of owners, the identities of all this group's user accounts are added to the application role.

4. The Microsoft Entra ID administrator role is labeled with the **IT Shop** option and assigned to the **Microsoft Entra ID** administrator roles IT Shop shelf in the **Identity & Access Lifecycle** shop.

Then the shop customers can use the Microsoft Entra ID to request memberships in Web Portal groups.

NOTE: If a Microsoft Entra ID group is irrevocably deleted from the One Identity Manager database, the associated service item is also deleted.

For more information about configuring the One Identity Manager IT Shop Administration Guide, see the *IT Shop*. For more information about requesting access requests in the Web Portal, see the *One Identity Manager Web Portal User Guide*.

Related topics

- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Adding Microsoft Entra ID administrator roles to system roles on page 124
- Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles on page 128

Assigning Microsoft Entra ID user accounts directly to Microsoft Entra ID administrator roles

To react quickly to special requests, you can assign administrator roles directly to user accounts. You cannot directly assign administration roles that have the **Only use in IT Shop** option set.

To assign a user account directly to an administrator role.

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Assign user accounts** task.
4. In the **Add assignments** pane, assign the user accounts.

   TIP: In the **Remove assignments** pane, you can remove assigned user accounts.

To remove an assignment

- Select the user account and double-click ✓.

5. Save the changes.

Related topics

- Assigning Microsoft Entra ID administrator roles directly to Microsoft Entra ID user accounts on page 129
- Assigning Microsoft Entra ID administrator roles to departments, cost centers, and locations on page 121
- Assigning Microsoft Entra ID administrator roles to business roles on page 123
- Adding Microsoft Entra ID administrator roles to system roles on page 124
- Adding Microsoft Entra ID administrator roles in the IT Shop on page 125

Assigning Microsoft Entra ID administrator roles directly to Microsoft Entra ID user accounts

To react quickly to special requests, you can assign administrator roles directly to the user account. You cannot directly assign administration roles that have the **Only use in IT Shop** option set.

To assign administrator roles directly to user accounts

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign administrator roles** task.
4. In the **Add assignments** pane, assign administrator roles.

   TIP: In the **Remove assignments** pane, you can remove assigned administrator roles.

   To remove an assignment

   - Select the administrator role and double-click ✓.

5. Save the changes.

Related topics

- Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID user accounts on page 119

Microsoft Entra ID administrator role inheritance based on categories

The procedure described under Microsoft Entra ID group inheritance based on categories on page 115 can also be applied for administrator roles.

To use inheritance through categories

1. In the Manager, define the categories in the Microsoft Entra ID tenant.
2. In the Manager, assign categories to user accounts through their main data.
3. In the Manager, assign categories to administrator roles through their main data.

Related topics

- Defining categories for the inheritance of entitlements on page 209
- General main data of Microsoft Entra ID user accounts on page 216
- Editing main data of Microsoft Entra ID administrator roles on page 243

Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments

The user requires a Microsoft Entra ID subscription to access the service plans in Microsoft Entra ID. An Microsoft Entra ID subscription defines the scope of service plans that the user can access. Use of individual service plans by the user can be permitted or not.

Example:

The Microsoft Entra ID subscription A contains service plan 1, service plan 2, and the service plan 3.

- The subscription A is assigned to the user.
- The user is not permitted service plan 2.

Therefore, the user can use service plans 1 and 3.

In Microsoft Entra ID, Microsoft Entra ID subscriptions can be assigned to users and groups. Service plans can be permitted or not depending on the assignment method. The user obtains all the permitted service plans.

Example:

The Microsoft Entra ID subscription A contains service plan 1, service plan 2, and the service plan 3.

- The Microsoft Entra ID subscription A is assigned directly to the user.
- The user is not permitted service plan 2.

The Microsoft Entra ID subscription B contains the service plan 4, service plan 5, and the service plan 6.

- The Microsoft Entra ID subscription B is assigned to group A.
- The group A is not permitted to use service plan 6.
- The user is in group A.

Therefore, the user can use service plans 1, 3, 4, and 5.

It is possible that a user obtains the same Microsoft Entra ID subscription directly as well as through one or more groups. If a service plan is permitted by one assignment method and not by another, the user is given the service plan.

Example:

The Microsoft Entra ID subscription A contains service plan 1, service plan 2, and the service plan 3.

- The Microsoft Entra ID subscription A is assigned directly to the user.
- The user is not permitted to use service plan 2.
- The Microsoft Entra ID subscription A is assigned to group A.
- All service plans for the group A are allowed.
- The user is in group A.

Therefore, the user can use service plans 1, 2, and 3.

In One Identity Manager, Microsoft Entra ID subscriptions and service plans and their assignments to Microsoft Entra ID user accounts and Microsoft Entra ID groups are mapped as follows.

Table 19: Microsoft Entra ID subscription and service plans in the One Identity Manager schema map

Table	Description
AADSubSku	This table contains all Microsoft Entra ID subscriptions. The information about Microsoft Entra ID subscriptions within a Microsoft Entra ID tenant is loaded into One Identity Manager by synchronization. You cannot create new Microsoft Entra ID subscription in One Identity Manager.
AADServicePlan	This table contains the service plans. The information about service plans within a Microsoft Entra ID tenant is

Table	Description
	loaded into One Identity Manager by synchronization. You cannot create new service plans in One Identity Manager.
AADServicePlanInSubSku	This table contains service plan assignments to Microsoft Entra ID subscriptions. The assignments are loaded into One Identity Manager by synchronization. You cannot edit the assignments in One Identity Manager.
AADDeniedServicePlan	This table contains service plan assignments to Microsoft Entra ID subscriptions for mapping service plans that are not permitted. The entries are created automatically in One Identity Manager after synchronizing Microsoft Entra ID subscriptions. Service plans that are not permitted are called "disabled service plans" in One Identity Manager. By assigning a disabled service plans to a Microsoft Entra ID user account in One Identity Manager, you disable the use of this service plan in Microsoft Entra ID.
AADUserHasSubSku	This table contains Microsoft Entra ID subscription assignments to Microsoft Entra ID user accounts. It maps direct assignments of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts and assignments that a Microsoft Entra ID user accounts obtains through its Microsoft Entra ID groups. The assignments are loaded by synchronization. You can assign Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts in One Identity Manager either directly, though IT Shop requests, or through departments, cost centers, locations and business roles. You cannot edit assignments by Microsoft Entra ID groups in One Identity Manager. The Microsoft Entra ID group that an assignment results from, is mapped in the **Microsoft Entra ID source group** column (`AADUserHasSubSku.UID_AADGroupSource`). If the column is empty, this assignment of the Microsoft Entra ID subscription to the Microsoft Entra ID user account is created either directly, though IT Shop requests, or through departments, cost centers, locations, and business roles. Assignments through Microsoft Entra ID groups are marked in the **Origin** column with the **`Assignment by group`** value (`AADUserHasSubSku.XOrigin=16`). The table also contains a list of disabled service plans

Table	Description
	from its Microsoft Entra ID subscriptions for each Microsoft Entra ID user account (`AADUserHasSubSku.DenyList`).
AADUserHasDeniedService	This table contains disabled service plan assignments to Microsoft Entra ID user accounts. The entries are taken from the list of disabled service plans (`AADUserHasSubSku.DenyList`). You can assign disabled service plans to Microsoft Entra ID user accounts in One Identity Manager either directly, though IT Shop requests, or through departments, cost centers, locations and business roles. By assigning a disabled service plan, you disable the use of this service plan in Microsoft Entra ID. NOTE: A disabled service plan that is assigned to a user account can be permitted if the user accounts obtains the service plan additional through a group and the service plan for the group is allowed. The assignment by group is not mapped in this table.
AADUserHasServicePlan	This table contains the service plan assignments for Microsoft Entra ID user accounts that are in effect. The assignments are calculated in One Identity Manager from the entries in the `AADUserHasSubSku`, `AADUserHasDeniedService`, `AADGroupHasSubSku`, and `AADGroupHasDeniedService` tables.
AADGroupHasSubSku	This table contains Microsoft Entra ID subscription assignments to Microsoft Entra ID groups. This table also contains a list of disabled service plans from its Microsoft Entra ID subscriptions for each Microsoft Entra ID group (`AADUserHasSubSku.DenyList`). The assignments are loaded into One Identity Manager by synchronization. You cannot edit the assignments in One Identity Manager.
AADGroupHasDeniedService	This table contains disabled service plan assignments to Microsoft Entra ID groups. The entries are taken from the list of disabled service plans (`AADGroupHasSubSku.DenyList`). You cannot edit the assignments in One Identity Manager.

Detailed information about this topic

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

- Assigning Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 137
- Inheriting Microsoft Entra ID subscriptions based on categories on page 160
- Assigning disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 149
- Inheritance of disabled Microsoft Entra ID service plans based on categories on page 160
- Overview of all assignments on page 117

Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups

A Microsoft Entra ID user account can obtain Microsoft Entra ID subscriptions and Microsoft Entra ID service plans directly or through its Microsoft Entra ID groups.

NOTE: It is possible that a Microsoft Entra ID user obtains the same Microsoft Entra ID subscription directly as well as through one or more Microsoft Entra ID groups. If a service plan is permitted by one assignment method and not by another, the user is given the service plan.

This means:

A disabled service plan that is assigned to a user account can be permitted if the user accounts obtains the service plan additional through a group and the service plan for the group is allowed.

For more information, see Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131.

To display information about Microsoft Entra ID subscriptions and service plans for a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Microsoft Entra ID user account overview** task.

   The following information about Microsoft Entra ID subscriptions and service plans for a user account is displayed on the overview form.

   - **Microsoft Entra ID subscriptions (owned)**: Microsoft Entra ID subscriptions assigned to the user account either directly, though IT Shop requests, or through departments, cost centers, locations and business roles.

- **Microsoft Entra ID subscriptions (inherited)**: Microsoft Entra ID subscriptions that the user account has obtained through its Microsoft Entra ID groups.
- **Microsoft Entra ID service plans**: Microsoft Entra ID service plans permitted for this user account.
- **Disabled Microsoft Entra ID service plans from owned subscriptions**: Disabled Microsoft Entra ID service plans assigned to this user account either directly, through the IT Shop or through departments, cost centers, locations and business roles.

4. Select the **License overview** report.

   The report contains a summary of assigned and effective subscriptions and service plans for a Microsoft Entra ID user account.

To display information about Microsoft Entra ID subscriptions and service plans for a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Microsoft Entra ID group overview** task.

   The following information about Microsoft Entra ID subscriptions and service plans for a group is displayed on the overview form.

   - **Microsoft Entra ID subscriptions**: Microsoft Entra ID subscriptions assigned to Microsoft Entra ID groups.
   - **Enabled Microsoft Entra ID service plans**: Microsoft Entra ID service plans permitted for this group.
   - **Disabled Microsoft Entra ID service plans**: Microsoft Entra ID service plans not permitted for this group.
   - **Microsoft Entra ID user accounts**: Microsoft Entra ID user accounts assigned to group and therefore contain subscriptions and service plans.

Related topics

- Assigning Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 137
- Assigning disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 149

Assigning Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts

Microsoft Entra ID subscriptions can be assigned directly or indirectly to Microsoft Entra ID user accounts.

In the case of indirect assignment, identities and Microsoft Entra ID subscriptions are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. Microsoft Entra ID subscriptions assigned to an identity are calculated from the position in the hierarchy and the direction of inheritance. If the identity has a user account in Microsoft Entra ID, Microsoft Entra ID role subscriptions are inherited by this Microsoft Entra ID user account.

Furthermore, Microsoft Entra ID subscriptions can be assigned to identities through IT Shop requests. Add identities to a shop as customers so that Microsoft Entra ID subscriptions can be assigned through IT Shop requests. All Microsoft Entra ID subscriptions assigned to this shop can be requested by the customers. Requested Microsoft Entra ID subscriptions are assigned to the identities after approval is granted.

> TIP: You can combine the account definition for creating the Microsoft Entra ID user account and the Microsoft Entra ID subscription that will be used into one system role. In this way, the identity automatically obtains a user account and a Microsoft Entra ID subscription.
>
> An identity can obtain this system role directly through departments, cost centers, locations, or business roles, or an IT Shop request.

To react quickly to special requests, you can assign Microsoft Entra ID subscriptions directly to Microsoft Entra ID user accounts.

> NOTE: A Microsoft Entra ID user account can also obtain Microsoft Entra ID subscriptions through its Microsoft Entra ID groups. You cannot edit assignments by Microsoft Entra ID groups in One Identity Manager.
>
> The `AADUserhasSubSku` contains the assignments of Microsoft Entra ID subscription to Microsoft Entra ID user accounts with their origin. For more information, see Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131.

For more information see the following guides:

Topic	Guide
Basic principles for assigning and inheriting company resources	*One Identity Manager Identity Management Base Module Administration Guide* *One Identity Manager Business Roles Administration Guide*
Assigning company resources through	*One Identity Manager IT Shop*

Topic	Guide
IT Shop requests	*Administration Guide*
System roles	*One Identity Manager System Roles Administration Guide*

Detailed information about this topic

- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143
- Adding Microsoft Entra ID subscriptions automatically to the IT Shop on page 144
- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146
- Assigning Microsoft Entra ID subscriptions directly to Microsoft Entra ID user accounts on page 147

Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts

In the case of indirect assignment, identities and Microsoft Entra ID subscriptions are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. When assigning Microsoft Entra ID subscriptions indirectly, check the following settings and modify them if necessary:

1. Assignment of identities and Microsoft Entra ID subscriptions is permitted for role classes (departments, cost centers, locations, or business roles).

   For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.

To configure assignments to roles of a role class

1. In the Manager, select role classes in the **Organizations > Basic configuration data > Role classes** category.

   - OR -

   In the Manager, select role classes in the **Business roles > Basic configuration data > Role classes** category.

2. Select the **Configure role assignments** task and configure the permitted assignments.
   - To generally allow an assignment, enable the **Assignments allowed** column.
   - To allow direct assignment, enable the **Direct assignments permitted** column.
3. Save the changes.

2. Settings for assigning Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts.
   - The Microsoft Entra ID user account is linked to an identity.
   - The Microsoft Entra ID user account has a location.
   - The Microsoft Entra ID user account has the **Subscriptions can be inherited** option set.

NOTE: There are other configuration settings that play a role when company resources are inherited through departments, cost centers, locations, and business roles. For example, role inheritance might be blocked or inheritance of identities not allowed. For more information about the basic principles for assigning company resources, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Related topics

- Creating and editing Microsoft Entra ID user accounts on page 215
- General main data of Microsoft Entra ID user accounts on page 216

Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations

Assign Microsoft Entra ID subscriptions to departments, cost centers, and locations so that user accounts are assigned to them through these organizations.

To assign a Microsoft Entra ID subscription to departments, cost centers, or locations (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.

3. Select the **Assign organizations** task.
4. In the **Add assignments** pane, assign the organizations:
   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
5. Save the changes.

To assign Microsoft Entra ID subscriptions to a department, cost center, or location (role-based login)

1. In the Manager, select the **Organizations > Departments** category.

   - OR -

   In the Manager, select the **Organizations > Cost centers** category.

   - OR -

   In the Manager, select the **Organizations > Locations** category.
2. Select the department, cost center or location in the result list.
3. Select the **Assign Microsoft Entra ID subscriptions** task.
4. In the **Add assignments** pane, select the Microsoft Entra ID tenant and assign the Microsoft Entra ID subscriptions.

   TIP: In the **Remove assignments** pane, you can remove assigned Microsoft Entra ID subscriptions.

   To remove an assignment

   - Select the ✓ subscription and double-click Microsoft Entra ID.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146
- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143
- One Identity Manager users for managing Microsoft Entra ID on page 13

Assigning Microsoft Entra ID subscriptions to business roles

NOTE: This function is only available if the Business Roles Module is installed.

Assign Microsoft Entra ID subscriptions to business roles so that they are assigned to user accounts through these business roles.

To assign a Microsoft Entra ID subscription to business roles (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

To assign Microsoft Entra ID subscriptions to a business role (role-based login)

1. In the Manager, select the **Business roles > <role class>** category.
2. Select the business role in the result list.
3. Select the **Assign Microsoft Entra ID subscriptions** task.
4. In the **Add assignments** pane, select the Microsoft Entra ID tenant and assign the Microsoft Entra ID subscriptions.

   TIP: In the **Remove assignments** pane, you can remove assigned Microsoft Entra ID subscriptions.

   To remove an assignment

   - Select the ✓ subscription and double-click Microsoft Entra ID.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146

- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143
- One Identity Manager users for managing Microsoft Entra ID on page 13

Adding Microsoft Entra ID subscriptions to system roles

NOTE: This function is only available if the System Roles Module is installed.

Use this task to add a Microsoft Entra ID subscription to system roles. If you assign a system role to identities, all Microsoft Entra ID user accounts owned by these identities inherit the Microsoft Entra ID subscription.

NOTE: Microsoft Entra ID subscriptions in which the **Only use in IT Shop** option is set can only be assigned to system roles that also have this option set. For more information, see the *One Identity Manager System Roles Administration Guide*.

TIP: You can combine the account definition for creating the Microsoft Entra ID user account and the Microsoft Entra ID subscription that will be used into one system role. In this way, the identity automatically obtains a user account and a Microsoft Entra ID subscription.

An identity can obtain this system role directly through departments, cost centers, locations, or business roles, or an IT Shop request.

To assign a Microsoft Entra ID subscription to a system role

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141

- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143

Adding Microsoft Entra ID subscriptions to the IT Shop

Once a Microsoft Entra ID subscription is assigned to an IT Shop shelf, it can be requested by customers. To ensure it can be requested, further prerequisites need to be guaranteed.

- The Microsoft Entra ID subscription must be labeled with the **IT Shop** option.
- The Microsoft Entra ID subscription must be assigned to a service item.
- If the Microsoft Entra ID subscription is only supposed to be available to identities through IT Shop requests, the Microsoft Entra ID subscription must also be labeled with the **Only for use in IT Shop** option. Direct assignment to hierarchical roles may not be possible.

NOTE: IT Shop administrators can assign Microsoft Entra ID subscriptions to IT Shop shelves in the case of role-based login. Target system administrators are not authorized to add Microsoft Entra ID subscriptions in the IT Shop.

To add a Microsoft Entra ID subscription to the IT Shop

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID subscriptions** (role-based login) category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select **Add to IT Shop**.
4. In the **Add assignments** pane, assign the Microsoft Entra ID subscription to the IT Shop shelves.
5. Save the changes.

To remove a Microsoft Entra ID subscription from individual IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID subscriptions** (role-based login) category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select **Add to IT Shop**.

4. In the **Remove assignments** pane, remove the Microsoft Entra ID subscription from the IT Shop shelves.
5. Save the changes.

To remove a Microsoft Entra ID subscription from all IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Microsoft Entra ID subscriptions** (role-based login) category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select the **Remove from all shelves (IT Shop)** task.
4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The Microsoft Entra ID subscription is removed from all shelves by the One Identity Manager Service. All request and assignment requests for this Microsoft Entra ID subscription are canceled in the process.

For more information about requesting company resources through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Related topics

- Editing Microsoft Entra ID subscription main data on page 249
- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Adding Microsoft Entra ID subscriptions automatically to the IT Shop on page 144
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146

Adding Microsoft Entra ID subscriptions automatically to the IT Shop

The following steps can be used to automatically add Microsoft Entra ID subscriptions to the IT Shop. Synchronization ensures that the Microsoft Entra ID subscriptions are added to the IT Shop. If necessary, you can manually start synchronization with the

Synchronization Editor. New Microsoft Entra ID subscriptions created in One Identity Manager also are added automatically to the IT Shop.

To add Microsoft Entra ID subscriptions automatically to the IT Shop

1. In the Designer, set the **QER | ITShop | AutoPublish | AADSubSku** configuration parameter.
2. In order not to add Microsoft Entra ID subscriptions to the IT Shop automatically, in the Designer, set the **QER | ITShop | AutoPublish | AADSubSku | ExcludeList** configuration parameter.

   This configuration parameter contains a listing of all Microsoft Entra ID subscriptions that should not be allocated to the IT Shop automatically. You can extend this list if required. To do this, enter the name of the subscription in the configuration parameter. Names are listed in a pipe (|) delimited list. Regular expressions are supported.
3. Compile the database.

The Microsoft Entra ID subscriptions are added automatically to the IT Shop from now on.

The following steps are run to add a Microsoft Entra ID subscription to the IT Shop.

1. A service item is determined for the Microsoft Entra ID subscription.

   The service item is tested for each Microsoft Entra ID subscription and modified if necessary. The name of the service item corresponds to the name of the Microsoft Entra ID subscription.

   - The service item is modified for Microsoft Entra ID subscriptions with service items.
   - Microsoft Entra ID subscriptions without service items are allocated new service items.
2. The service item is assigned to the **Microsoft Entra ID subscriptions** default service category.
3. An application role for product owners is determined and assigned to the service item.

   Product owners can approve requests for these Microsoft Entra ID subscriptions. The default product owner is the Microsoft Entra ID subscription's owner.

   NOTE: The application role for the product owner must be added under the **Request & Fulfillment | IT Shop | Product owner** application role.

   - If the owner of the Microsoft Entra ID subscription is already a member of an application role for product owners, this application role is assigned to the service item. Therefore, all members of this application role become product owners of the Microsoft Entra ID subscription.
   - If the owner of the Microsoft Entra ID subscription is not yet a member of an application role for product owners, a new application role is created. The name of the application corresponds to the name of the owner.

- If the owner is a user account, the user account's identity is added to the application role.
- If it is a group of owners, the identities of all this group's user accounts are added to the application role.

4. The Microsoft Entra ID subscription is labeled with the **IT Shop** option and assigned to the **Microsoft Entra ID subscriptions** IT Shop shelf in the **Identity & Access Lifecycle** shop.

Then the shop customers use the Web Portal to request the Microsoft Entra ID subscription.

NOTE: If a Microsoft Entra ID subscription is irrevocably deleted from the One Identity Manager database, the associated service item is also deleted.

For more information about configuring the One Identity Manager IT Shop Administration Guide, see the *IT Shop*. For more information about requesting access requests in the Web Portal, see the *One Identity Manager Web Portal User Guide*.

Related topics

- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions on page 146
- Adding Microsoft Entra ID subscriptions to system roles on page 142

Assigning Microsoft Entra ID user account directly to Microsoft Entra ID subscriptions

To react quickly to special requests, you can assign subscriptions directly to Microsoft Entra ID user accounts. You cannot directly assign Microsoft Entra ID subscriptions that have the **Only use in IT Shop** option set.

Special on the assignment form

On the form, assignments of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts are shown with their origin. This means:

- **Microsoft Entra ID source group**: Microsoft Entra ID group resulting from an assignment. If the column is empty, this assignment of the Microsoft Entra ID subscription to the Microsoft Entra ID user account is created either directly, though IT Shop requests, or through departments, cost centers, locations, and business roles.

- **Origin**: Type of assignment. Assignments through Microsoft Entra ID groups are marked with the **`Assigned by group`** value (`AADUserHasSubSku.XOrigin=16`).

NOTE: You cannot delete assignments that are not derived from a Microsoft Entra ID group.

To assign a Microsoft Entra ID subscription directly to user accounts

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select in the **Assign user accounts** task.
4. Click **Add** and select the user account in the **Microsoft Entra ID user account** drop-down.
5. Save the changes.

To remove direct assignments of Microsoft Entra ID subscriptions

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select in the **Assign user accounts** task.
4. Select the assignment and click **Remove**.
5. Save the changes.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Assigning Microsoft Entra ID subscriptions directly to Microsoft Entra ID user accounts on page 147
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143

Assigning Microsoft Entra ID subscriptions directly to Microsoft Entra ID user accounts

To react quickly to special requests, you can assign subscriptions directly to Microsoft Entra ID user accounts. You cannot directly assign Microsoft Entra ID subscriptions that have the **Only use in IT Shop** option set.

Special on the assignment form

On the form, assignments of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts are shown with their origin. This means:

- **Microsoft Entra ID source group**: Microsoft Entra ID group resulting from an assignment. If the column is empty, this assignment of the Microsoft Entra ID subscription to the Microsoft Entra ID user account is created either directly, though IT Shop requests, or through departments, cost centers, locations, and business roles.
- **Origin**: Type of assignment. Assignments through Microsoft Entra ID groups are marked with the **`Assigned by group`** value (`AADUserHasSubSku.XOrigin=16`).

NOTE: You cannot delete assignments that are not derived from a Microsoft Entra ID group.

To assign subscriptions directly to Microsoft Entra ID user accounts

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Directly assign subscriptions** task.
4. Click **Add** and in the **Microsoft Entra ID subscription** drop-down, select a Microsoft Entra ID subscription.
5. Save the changes.

To remove direct assignments of Microsoft Entra ID subscriptions

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Directly assign subscriptions** task.
4. Select the assignment and click **Remove**.
5. Save the changes.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Assigning Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 137
- Assigning Microsoft Entra ID subscriptions to departments, cost centers, and locations on page 139
- Assigning Microsoft Entra ID subscriptions to business roles on page 141
- Adding Microsoft Entra ID subscriptions to system roles on page 142
- Adding Microsoft Entra ID subscriptions to the IT Shop on page 143

Assigning disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts

Disabled Microsoft Entra ID service plans can be assigned directly or indirectly to Microsoft Entra ID user accounts.

In the case of indirect assignment, identities and disabled service plans are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. The disabled Microsoft Entra ID service plans assigned to an identity are calculated from the position in the hierarchy and the direction of inheritance. If the identity has a user account in Microsoft Entra ID, disabled service plans belonging to roles are inherited by this user account.

Furthermore, disabled service plans can be assigned to identities through IT Shop requests. Add identities to a shop as customers so that disabled service plans can be assigned through IT Shop requests. All disabled service plans are assigned to this shop can be requested by the customers. Requested disabled service plans are assigned to the identities after approval is granted.

To react quickly to special requests, you can assign disabled service plans directly to Microsoft Entra ID user accounts.

NOTE: It is possible that a Microsoft Entra ID user obtains the same Microsoft Entra ID subscription directly as well as through one or more Microsoft Entra ID groups. If a service plan is permitted by one assignment method and not by another, the user is given the service plan.

This means:

A disabled service plan that is assigned to a user account can be permitted if the user accounts obtains the service plan additional through a group and the service plan for the group is allowed.

For more information, see Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131 and Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135.

For more information see the following guides:

Topic	Guide
Basic principles for assigning and inheriting company resources	*One Identity Manager Identity Management Base Module Administration Guide* *One Identity Manager Business Roles Administration Guide*
Assigning company resources through IT Shop requests	*One Identity Manager IT Shop Administration Guide*

Topic	Guide
System roles	*One Identity Manager System Roles Administration Guide*

Detailed information about this topic

- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Adding disabled Microsoft Entra ID service plans to system roles on page 154
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155
- Adding disabled Microsoft Entra ID service plans automatically to the IT Shop on page 156
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Assigning disabled Microsoft Entra ID service plans directly to Microsoft Entra ID user accounts on page 159

Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts

In the case of indirect assignment, identities and disabled service plans are assigned to hierarchical roles, such as departments, cost centers, locations, or business roles. When assigning disabled Microsoft Entra ID service plans indirectly, check the following settings and modify them if necessary:

1. Assignment of identities and disabled Microsoft Entra ID service plans is permitted for role classes (departments, cost centers, locations, or business roles).

   For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.

   To configure assignments to roles of a role class

   1. In the Manager, select role classes in the **Organizations > Basic configuration data > Role classes** category.

      - OR -

In the Manager, select role classes in the **Business roles > Basic configuration data > Role classes** category.

2. Select the **Configure role assignments** task and configure the permitted assignments.
   - To generally allow an assignment, enable the **Assignments allowed** column.
   - To allow direct assignment, enable the **Direct assignments permitted** column.
3. Save the changes.

2. Settings for assigning disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts.
   - The Microsoft Entra ID user account is linked to an identity.
   - The Microsoft Entra ID user account has the **Disabled service plans can be inherited** option set.

Related topics

- Creating and editing Microsoft Entra ID user accounts on page 215
- General main data of Microsoft Entra ID user accounts on page 216

Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations

Assign disabled Microsoft Entra ID service plans to departments, cost centers, and locations in order to assign user accounts to them through these organizations.

To assign a disabled service plan to departments, cost centers, or locations (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the service plan in the result list.
3. Select the **Assign organizations** task.
4. In the **Add assignments** pane, assign the organizations:
   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

To remove an assignment

- Select the organization and double-click ✓.

5. Save the changes.

To assign disabled service plans to a department, cost center, or location (role-based login)

1. In the Manager, select the **Organizations > Departments** category.

   - OR -

   In the Manager, select the **Organizations > Cost centers** category.

   - OR -

   In the Manager, select the **Organizations > Locations** category.
2. Select the department, cost center or location in the result list.
3. Select **Assigning disabled Microsoft Entra ID service plans**.
4. In the **Add assignments** pane, select the Microsoft Entra ID subscription and assign the disabled service plans.

   TIP: In the **Remove assignments** pane, you can remove assigned service plans.

   To remove an assignment

   - Select the service plan and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Adding disabled Microsoft Entra ID service plans to system roles on page 154
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155
- One Identity Manager users for managing Microsoft Entra ID on page 13

Assigning disabled Microsoft Entra ID service plans to business roles

NOTE: This function is only available if the Business Roles Module is installed.

Assign disabled Microsoft Entra ID service plans to business roles so that they can be assigned to user accounts through these business roles.

To assign a disabled service plan to a business role (non role-based login)

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the service plan in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.

1. Save the changes.

To assign disabled service plans to a business role (non role-based login)

1. In the Manager, select the **Business roles > <role class>** category.
2. Select the business role in the result list.
3. Select the **Assigning disabled Microsoft Entra ID service plans** task.
4. In the **Add assignments** pane, select the Microsoft Entra ID subscription and assign the disabled service plans.

   TIP: In the **Remove assignments** pane, you can remove assigned service plans.

   To remove an assignment

   - Select the service plan and double-click ✓.

5. Save the changes.

Related topics

- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Adding disabled Microsoft Entra ID service plans to system roles on page 154
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155
- One Identity Manager users for managing Microsoft Entra ID on page 13

Adding disabled Microsoft Entra ID service plans to system roles

NOTE: This function is only available if the System Roles Module is installed.

Use this task to add disabled Microsoft Entra ID service plans to system roles. If you assign a system role to identities, all Microsoft Entra ID user accounts owned by these identities inherit the disabled service plan.

NOTE: Disabled Microsoft Entra ID service plans in which the **Only use in IT Shop** option is set can only be assigned to system roles that also have this option set. For more information, see the *One Identity Manager System Roles Administration Guide*.

To assign a disabled service plan to system roles

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the service plan in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155

Adding disabled Microsoft Entra ID service plans to the IT Shop

A disabled Microsoft Entra IDservice plan can be requested by shop customers when it is assigned to an IT Shop shelf. To ensure it can be requested, further prerequisites need to be guaranteed.

- The disabled service plan must be labeled with the **IT Shop** option.
- The disabled service plan must be assigned to a service item.
- If the disabled service plan is only assigned to identities using IT Shop requests, you must also set the **Only for use in IT Shop** option. Direct assignment to hierarchical roles may not be possible.

NOTE: IT Shop administrators can assign disabled service plans to IT Shop shelves in the case of role-based login. Target system administrators are not authorized to add disabled service plans in the IT Shop.

To add a disabled service plan in the IT Shop

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Disabled Microsoft Entra ID service plans** (role-based login) category.
2. Select the service plan in the result list.
3. Select the **Add to IT Shop** task.
4. In the **Add assignments** pane, assign the disabled service plan to the IT Shop shelves.
5. Save the changes.

To remove a disabled service plan from individual IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Disabled Microsoft Entra ID service plans** (role-based login) category.
2. Select the service plan in the result list.
3. Select the **Add to IT Shop** task.
4. In the **Remove assignments** pane, remove the disabled service plan from the IT Shop shelves.
5. Save the changes.

To remove a disabled service plan from all IT Shop shelves

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** (non role-based login) category.

   - OR -

   In the Manager, select the **Entitlements > Disabled Microsoft Entra ID service plans** (role-based login) category.
2. Select the service plan in the result list.
3. Select the **Remove from all shelves (IT Shop)** task.
4. Confirm the security prompt with **Yes**.
5. Click **OK**.

   The disabled service plan is removed from all shelves by the One Identity Manager Service. All requests and assignment requests with this disabled service plan are canceled at the same time.

For more information about requesting company resources through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Related topics

- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150
- Adding disabled Microsoft Entra ID service plans automatically to the IT Shop on page 156
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Adding disabled Microsoft Entra ID service plans to system roles on page 154

Adding disabled Microsoft Entra ID service plans automatically to the IT Shop

The following steps can be used to automatically add disabled Microsoft Entra ID service plans to the IT Shop. Synchronization ensures that the disabled service plans are added to the IT Shop. If necessary, you can manually start synchronization with the Synchronization Editor. New disabled service plans created in One Identity Manager also are added automatically to the IT Shop.

To add disabled service plans automatically to the IT Shop

1. In the Designer, set the **QER | ITShop | AutoPublish | AADDeniedServicePlan** configuration parameter.
2. In order not to add disabled service plans to the IT Shop automatically, in the Designer, set the **QER | ITShop | AutoPublish | AADDeniedServicePlan | ExcludeList** configuration parameter.

   This configuration parameter contains a listing of all disabled service plans that should not be allocated to the IT Shop automatically. You can extend this list if required. To do this, enter the name of the subscription in the configuration parameter. Names are listed in a pipe (|) delimited list. Regular expressions are supported.
3. Compile the database.

The disabled service plans are added automatically to the IT Shop from now on.

The following steps are run to add a disabled service plan to the IT Shop.

1. A service item is determined for the disabled service plan.

   The service item is tested for each disabled service plan and modified if necessary. The name of the service item corresponds to the name of the disabled service plan.

   - The service item is modified for disabled service plans with service items.
   - Disabled service plans without service items are allocated new service items.
2. The service item is assigned to the **Disabled Microsoft Entra ID service plans** default service category.
3. An application role for product owners is determined and assigned to the service item.

   Product owners can approve requests for these disabled service plans. The default product owner is the disabled service plan's owner.

   NOTE: The application role for the product owner must be added under the **Request & Fulfillment | IT Shop | Product owner** application role.

   - If the owner of the disabled service plan is already a member of an application role for product owners, this application role is assigned to the service item. Therefore, all members of this application role become product owners of the disabled service plan.
   - If the owner of the disabled service plan is not yet a member of an application role for product owners, a new application role is created. The name of the application corresponds to the name of the owner.
     - If the owner is a user account, the user account's identity is added to the application role.
     - If it is a group of owners, the identities of all this group's user accounts are added to the application role.
4. The disabled service plan is labeled with the **IT Shop** option and assigned to the **Disabled Microsoft Entra ID service plans** IT Shop shelf in the **Identity & Access Lifecycle** shop.

Then the shop customers use the Web Portal to request the disabled service plan.

NOTE: If a disabled service plan is irrevocably deleted from the One Identity Manager database, the associated service item is also deleted.

For more information about configuring the One Identity Manager IT Shop Administration Guide, see the *IT Shop*. For more information about requesting access requests in the Web Portal, see the *One Identity Manager Web Portal User Guide*.

Related topics

- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans on page 158
- Adding disabled Microsoft Entra ID service plans to system roles on page 154

Assigning Microsoft Entra ID user accounts directly to disabled Microsoft Entra ID service plans

To react quickly to special requests, you can assign disabled service plans directly to a user account. You cannot directly assign disabled service plans with the **Only use in IT Shop** option set.

To assign a disabled service plan directly to a user account

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the service plan in the result list.
3. Select the **Assign user accounts** task.
4. In the **Add assignments** pane, assign the user accounts.

   TIP: In the **Remove assignments** pane, you can remove assigned user accounts.

   To remove an assignment

   - Select the user account and double-click ✓.
5. Save the changes.

Related topics

- Assigning Microsoft Entra ID subscriptions directly to Microsoft Entra ID user accounts on page 147
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Adding disabled Microsoft Entra ID service plans to system roles on page 154
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155

Assigning disabled Microsoft Entra ID service plans directly to Microsoft Entra ID user accounts

To react quickly to special requests, you can assign disabled service plans directly to a user account. You cannot directly assign disabled service plans with the **Only use in IT Shop** option set.

To assign disabled service plans directly to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign disabled service plans** task.
4. In the **Add assignments** pane, assign disabled service plans.

   TIP: In the **Remove assignments** pane, you can remove assigned disabled service plans.

   To remove an assignment

   - Select a disabled service plan and click ✓.
5. Save the changes.

Related topics

- Assigning disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 149
- Assigning disabled Microsoft Entra ID service plans directly to departments, cost centers, and locations on page 151
- Assigning disabled Microsoft Entra ID service plans to business roles on page 152
- Adding disabled Microsoft Entra ID service plans to system roles on page 154
- Adding disabled Microsoft Entra ID service plans to the IT Shop on page 155

Inheriting Microsoft Entra ID subscriptions based on categories

The procedure described under Microsoft Entra ID group inheritance based on categories on page 115 can also be used for Microsoft Entra ID subscriptions.

To use inheritance through categories

1. In the Manager, define the categories in the Microsoft Entra ID tenant.
2. In the Manager, assign categories to user accounts through their main data.
3. In the Manager, assign categories to Microsoft Entra ID subscriptions through their main data.

Related topics

- Defining categories for the inheritance of entitlements on page 209
- General main data of Microsoft Entra ID user accounts on page 216
- Editing Microsoft Entra ID subscription main data on page 249

Inheritance of disabled Microsoft Entra ID service plans based on categories

The procedure described under Microsoft Entra ID group inheritance based on categories on page 115 can also be used for disabled service plans.

To use inheritance through categories

1. In the Manager, define the categories in the Microsoft Entra ID tenant.
2. In the Manager, assign categories to user accounts through their main data.
3. In the Manager, assign categories to disabled service plans through their main data.

Related topics

- Defining categories for the inheritance of entitlements on page 209
- General main data of Microsoft Entra ID user accounts on page 216
- Editing main data of disabled Microsoft Entra ID service plans on page 252

Login credentials for Microsoft Entra ID user accounts

When new user accounts are created in One Identity Manager, the passwords needed to log in to the target system are created immediately also. Various options are available for assigning the initial password. Predefined password policies are applied to the passwords, and you can adjust these policies to suit your individual requirements if necessary. You can set up email notifications to distribute the login credentials generated to users.

Detailed information about this topic

- Password policies for Microsoft Entra ID user accounts on page 161
- Initial password for new Microsoft Entra ID user accounts on page 173
- Email notifications about login data on page 173

Password policies for Microsoft Entra ID user accounts

One Identity Manager provides you with support for creating complex password policies, for example, for system user passwords, the identities' central password as well as passwords for individual target systems. Password polices apply not only when the user enters a password but also when random passwords are generated.

Predefined password policies are supplied with the default installation that you can use or customize if required. You can also define your own password policies.

Detailed information about this topic

- Predefined password policies on page 162
- Using password policies on page 163
- Creating password policies on page 164
- Editing password policies on page 165

- Custom scripts for password requirements on page 169
- Password exclusion list on page 172
- Checking passwords on page 172
- Testing password generation on page 173

Predefined password policies

You can customize predefined password policies to meet your own requirements if necessary.

Password for logging in to One Identity Manager

The **One Identity Manager password policy** is applied for logging in to One Identity Manager. This password policy defines the settings for the system user passwords (`DialogUser.Password` and `Person.DialogUserPassword`) as well as the passcode for a one time log in on the Web Portal (`Person.Passcode`).

> NOTE: The **One Identity Manager password policy** is marked as the default policy. This password policy is applied if no other password policy can be found for identities, user accounts, or system users.

For more information about password policies for identities, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Password policy for forming identities' central passwords

An identity's central password is formed from the target system specific user accounts by respective configuration. The **Identity central password policy** defines the settings for the (`Person.CentralPassword`) central password. Members of the **Identity Management | Identities | Administrators** application role can adjust this password policy.

> IMPORTANT: Ensure that the **Identity central password policy** does not violate the target system-specific requirements for passwords.

For more information about password policies for identities, see the *One Identity Manager Identity Management Base Module Administration Guide*.

Password policies for user accounts

Predefined password policies are provided, which you can apply to the user account password columns of the user accounts.

> IMPORTANT: If you do not use password policies that are specific to the target system, the **One Identity Manager password policy** default policy applies. In this case, ensure that the default policy does not violate the target systems requirements.

The **Microsoft Entra ID password policy** is predefined for Microsoft Entra ID. You can apply this password policy to Microsoft Entra ID user accounts (`AADUser.Password`) of a Microsoft Entra ID tenant.

If the tenants' password requirements differ, it is recommended that you set up your own password policies for each tenant.

Furthermore, you can apply password policies based on the account definition of the user accounts or based on the manage level of the user accounts.

Using password policies

The **Microsoft Entra ID password policy** is predefined for Microsoft Entra ID. You can apply this password policy to Microsoft Entra ID user accounts (`AADUser.Password`) of a Microsoft Entra ID tenant.

If the tenants' password requirements differ, it is recommended that you set up your own password policies for each tenant.

Furthermore, you can apply password policies based on the account definition of the user accounts or based on the manage level of the user accounts.

The password policy that is to be used for a user account is determined in the following sequence:

1. Password policy of the user account's account definition.
2. Password policy of the user account's manage level.
3. Password policy of the user account's tenant.
4. The **One Identity Manager password policy** (default policy).

IMPORTANT: If you do not use password policies that are specific to the target system, the **One Identity Manager password policy** default policy applies. In this case, ensure that the default policy does not violate the target systems requirements.

To reassign a password policy

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. Select the password policy in the result list.
3. Select **Assign objects**.
4. Click **Add** in the **Assignments** section and enter the following data.
   - **Apply to**: Application scope of the password policy.

     To specify an application scope

     1. Click → next to the field.
     2. Select one of the following references under **Table**:

- The table that contains the base objects of synchronization.
- To apply the password policy based on the account definition, select the `TSBAccountDef` table.
- To apply the password policy based on the manage level, select the `TSBBehavior` table.

3. Under **Apply to**, select the table that contains the base objects.
   - If you have selected the table containing the base objects of synchronization, next select the specific target system.
   - If you have selected the `TSBAccountDef` table, next select the specific account definition.
   - If you have selected the `TSBBehavior` table, next select the specific manage level.
4. Click **OK**.

- **Password column**: Name of the password column.
- **Password policy**: Name of the password policy to use.

5. Save the changes.

To change a password policy's assignment

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. Select the password policy in the result list.
3. Select the **Assign objects** task.
4. In the **Assignments** pane, select the assignment you want to change.
5. From the **Password Policies** drop-down, select the new password policy you want to apply.
6. Save the changes.

Creating password policies

Predefined password policies are supplied with the default installation that you can use or customize if required. You can also define your own password policies.

To create a password policy

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. Click in the result list.
3. On the main data form, enter the main data of the password policy.
4. Save the changes.

Detailed information about this topic

- General main data of password policies on page 165
- Policy settings on page 166
- Character classes for passwords on page 167
- Custom scripts for password requirements on page 169
- Editing password policies on page 165

Editing password policies

Predefined password policies are supplied with the default installation that you can use or customize if required.

To edit a password policy

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. In the result list, select the password policy.
3. Select the **Change main data** task.
4. Edit the password policy's main data.
5. Save the changes.

Detailed information about this topic

- General main data of password policies on page 165
- Policy settings on page 166
- Character classes for passwords on page 167
- Custom scripts for password requirements on page 169
- Creating password policies on page 164

General main data of password policies

Enter the following main data of a password policy.

Table 20: main data for a password policy

Property	Meaning
Display name	Password policy name. Translate the given text using the button.

Property	Meaning
Description	Text field for additional explanation. Translate the given text using the button.
Error Message	Custom error message generated if the policy is not fulfilled. Translate the given text using the button.
Owner (Application Role)	Application roles whose members can configure the password policies.
Default policy	Mark as default policy for passwords. This option cannot be changed. NOTE: The **One Identity Manager password policy** is marked as the default policy. This password policy is applied if no other password policy can be found for identities, user accounts, or system users.

Policy settings

Define the following settings for a password policy on the **Password** tab.

Table 21: Policy settings

Property	Meaning
Initial password	Initial password for newly created user accounts. The initial password is used if a password is not entered when you create a user account or if a random password is not generated.
Password confirmation	Reconfirm password.
Minimum Length	Minimum length of the password. Specify the number of characters a password must have. If the value is **0**, no password is required.
Max. length	Maximum length of the password. Specify the number of characters a password can have. The maximum permitted value is **256**.
Max. errors	Maximum number of errors. Set the number of invalid passwords attempts. The number of failed logins is only taken into account when logging in to One Identity Manager. If the value is **0**, the number of failed logins is not taken into account. This data is only taken into account if the One Identity Manager login was through a system user or identity based authentication module. If a user has exceeded

Property	Meaning
	the maximum number of failed logins, the identity or system user will not be able to log in to One Identity Manager. You can use the Password Reset Portal to reset the passwords of identities and system users who have been locked. For more information, see the *One Identity Manager Web Portal User Guide*.
Validity period	Maximum age of the password. Enter the length of time a password can be used before it expires. If the value is **0**, then the password does not expire.
Password history	Enter the number of passwords to be saved. If, for example, a value of **5** is entered, the user's last five passwords are stored. If the value is **0**, then no passwords are stored in the password history.
Minimum password strength	Specifies how secure the password must be. The higher the password strength, the more secure it is. The value **0** means that the password strength is not tested. The values **1**, **2**, **3** and **4** specify the required complexity of the password. The value **1** represents the lowest requirements in terms of password strength. The value **4** requires the highest level of complexity.
Name properties denied	Specifies whether name properties are permitted in the password. If this option is set, name properties are not permitted in passwords. The values of these columns are taken into account if the **Contains name properties for password check** option is set. In the Designer, adjust this option in the column definition. For more information, see the *One Identity Manager Configuration Guide*.

Character classes for passwords

Use the **Character classes** tab to specify which characters are permitted for a password.

Table 22: Character classes for passwords

Property	Meaning
Required number of character classes	Number of rules for character classes that must be fulfilled so that a password adheres to the password policy. The following rules are taken into account for **Min. number letters**, **Min. number lowercase**, **Min. number uppercase**, **Min. number digits**, and **Min. number special characters**. That means:

Property	Meaning
	• Value **0**: All character class rules must be fulfilled. • Value **>0**: Minimum number of character class rules that must be fulfilled. At most, the value can be the number of rules with a value **>0**. NOTE: Generated passwords are not tested for this.
Min. number letters	Specifies the minimum number of alphabetical characters the password must contain.
Min. number lowercase	Specifies the minimum number of lowercase letters the password must contain.
Min. number uppercase	Specifies the minimum number of uppercase letters the password must contain.
Min. number digits	Specifies the minimum number of digits the password must contain.
Min. number special characters	Specifies the minimum number of special characters the password must contain.
Permitted special characters	List of permitted special characters.
Max. identical characters in total	Specifies the maximum number of identical characters that can be present in the password in total.
Max. identical characters in succession	Specifies the maximum number of identical character that can be repeated after each other.
Denied special characters	List of special characters that are not permitted.
Do not generate lowercase letters	Specifies whether a generated password can contain lowercase letters. This setting only applies when passwords are generated.
Do not generate uppercase	Specifies whether a generated password can contain uppercase letters. This setting only applies when passwords are generated.

Property	Meaning
letters	
Do not generate digits	Specifies whether a generated password can contain digits. This setting only applies when passwords are generated.
Do not generate special characters	Specifies whether a generated password can contain special characters. If this option is set, only letters, numbers, and spaces are allowed in passwords. This setting only applies when passwords are generated.

Custom scripts for password requirements

You can implement custom scripts for testing and generating passwords if the password requirements cannot be mapped with the existing settings options. Scripts are applied in addition to the other settings.

Detailed information about this topic

- Checking passwords with a script on page 169
- Generating passwords with a script on page 171

Checking passwords with a script

You can implement a script if additional policies need to be used for checking a password that cannot be mapped with the available settings.

Syntax of check scripts

```
Public Sub CCC_CustomPwdValidate( policy As VI.DB.Passwords.PasswordPolicy, spwd
As System.Security.SecureString)
```

With parameters:

`policy` = password policy object

`spwd` = password to check

TIP: To use a base object, take the `Entity` property of the `PasswordPolicy` class.

Example: Script that checks a password

A password cannot start with **?** or **!** . The password cannot start with three identical characters. The script checks a given password for validity.

```
Public Sub CCC_PwdValidate( policy As VI.DB.Passwords.PasswordPolicy, spwd
As System.Security.SecureString)
     Dim pwd = spwd.ToInsecureArray()
     If pwd.Length>0
          If pwd(0)="?" Or pwd(0)="!"
               Throw New Exception(#LD("Password can't start with '?' or
               '!'")#)
          End If
     End If
     If pwd.Length>2
          If pwd(0) = pwd(1) AndAlso pwd(1) = pwd(2)
               Throw New Exception(#LD("Invalid character sequence in
               password")#)
          End If
     End If
End Sub
```

To use a custom script for checking a password

1. In the Designer, create your script in the **Script Library** category.
2. Edit the password policy.
   a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
   b. In the result list, select the password policy.
   c. Select the **Change main data** task.
   d. On the **Scripts** tab, enter the name of the script to be used to check a password in the **Check script** field.
   e. (Optional) On the **Scripts** tab, in the **Additional requirements** field, enter the description of the additional requirements for the password that are checked by the script.
   f. Save the changes.

Related topics

- Generating passwords with a script on page 171

Generating passwords with a script

You can implement a generating script if additional policies need to be used for generating a random password, which cannot be mapped with the available settings.

Syntax for generating script

```
Public Sub CCC_PwdGenerate( policy As VI.DB.Passwords.PasswordPolicy, spwd As System.Security.SecureString)
```

With parameters:

`policy` = password policy object

`spwd` = generated password

TIP: To use a base object, take the `Entity` property of the `PasswordPolicy` class.

Example: Script that generates a password

In random passwords, this script replaces the invalid characters **?** and **!** at the beginning of a password with _.

```
Public Sub CCC_PwdGenerate( policy As VI.DB.Passwords.PasswordPolicy, spwd As System.Security.SecureString)
    Dim pwd = spwd.ToInsecureArray()
    ' replace invalid characters at first position
    If pwd.Length>0
        If pwd(0)="?" Or pwd(0)="!"
            spwd.SetAt(0, CChar("_"))
        End If
    End If
End Sub
```

To use a custom script for generating a password

1. In the Designer, create your script in the **Script Library** category.
2. Edit the password policy.
   a. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
   b. In the result list, select the password policy.
   c. Select the **Change main data** task.

d. On the **Scripts** tab, enter the name of the script to be used to generate a password in the **Generating script** field.

e. Save the changes.

Related topics

- Checking passwords with a script on page 169

Password exclusion list

You can add words to a list of restricted terms to prohibit them from being used in passwords.

NOTE: The restricted list applies globally to all password policies.

To add a term to the restricted list

1. In the Designer, select the **Base data > Security settings > Password policies** category.
2. Create a new entry with the **Object > New** menu item and enter the term you want to exclude from the list.
3. Save the changes.

Checking passwords

When you verify a password, all the password policy settings, custom scripts, and the restricted passwords are taken into account.

To verify if a password conforms to the password policy

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. In the result list, select the password policy.
3. Select the **Change main data** task.
4. Select the **Test** tab.
5. Select the table and object to be tested in **Base object for test**.
6. Enter a password in **Enter password to test**.

   A display next to the password shows whether it is valid or not.

Testing password generation

When you generate a password, all the password policy settings, custom scripts and the restricted passwords are taken into account.

To generate a password that conforms to the password policy

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Password policies** category.
2. In the result list, select the password policy.
3. Select the **Change main data** task.
4. Select the **Test** tab.
5. Click **Generate**.

   This generates and displays a password.

Initial password for new Microsoft Entra ID user accounts

You can issue an initial password for a new Microsoft Entra ID user account in the following ways:

- When you create the user account, enter a password in the main data.
- Assign a randomly generated initial password to enter when you create user accounts.
  - In the Designer, set the **TargetSystem | AzureAD | Accounts | InitialRandomPassword** configuration parameter.
  - Apply target system specific password policies and define the character sets that the password must contain.
  - Specify which identity will receive the initial password by email.

Related topics

- Password policies for Microsoft Entra ID user accounts on page 161
- Email notifications about login data on page 173

Email notifications about login data

You can configure the login credentials for new user accounts to be sent by email to a specified person. In this case, two messages are sent with the user name and the initial

password. Mail templates are used to generate the messages. The mail text in a mail template is defined in several languages. This means the recipient's language can be taken into account when the email is generated. Mail templates are supplied in the default installation with which you can configure the notification procedure.

The following prerequisites must be fulfilled in order to use notifications:

1. Ensure that the email notification system is configured in One Identity Manager. For more information, see the *One Identity Manager Installation Guide*.
2. In the Designer, set the **Common | MailNotification | DefaultSender** configuration parameter and enter the sender address for sending the email notifications.
3. Ensure that all identities have a default email address. Notifications are sent to this address. For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.
4. Ensure that a language can be determined for all identities. Only then can they receive email notifications in their own language. For more information, see the *One Identity Manager Identity Management Base Module Administration Guide*.

When a randomly generated password is issued for the new user account, the initial login data for a user account is sent by email to a previously specified identity.

To send initial login data by email

1. In the Designer, set the **TargetSystem | AzureAD | Accounts | InitialRandomPassword** configuration parameter.
2. In the Designer, set the **TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo** configuration parameter and enter the message recipient as a value.
3. In the Designer, set the **TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo | MailTemplateAccountName** configuration parameter.

   By default, the message sent uses the mail template **Identity - new user account created**. The message contains the name of the user account.
4. In the Designer, set the **TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo | MailTemplatePassword** configuration parameter.

   By default, the message sent uses the mail template **Identity - initial password for new user account**. The message contains the initial password for the user account.

TIP: To use custom mail templates for emails of this type, change the value of the configuration parameter.

Microsoft Entra ID role management

Role management describes extended role management functionality of role-based access control in Microsoft 365. This allows the user to manage roles and their members, as well as limiting role assignments to partial scopes in Microsoft Entra ID.

Microsoft Entra ID roles are read into One Identity Manager by synchronization. You can edit individual main data of the Microsoft Entra ID roles.

Related topics

- Microsoft Entra ID role management tenants on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198
- Microsoft Entra ID role main data on page 179
- Managing Microsoft Entra ID scoped role assignments on page 184
- Managing Microsoft Entra ID scoped role eligibilities on page 195

Microsoft Entra ID role management tenants

Microsoft Entra ID role management offers you a range of role management features. The scope of these features depends on the level of the Microsoft Entra ID license selected by the user, which is provided by the corresponding tenants.

Microsoft Entra ID "Free"

This license includes basic role management functionality. Integrated roles can be used without restrictions. These roles have predefined role definitions. With this license, it is

possible to add individual users to integrated roles and remove them. You can create groups.

IMPORTANT: Not included in the basic functionality are maintenance of directory roles in One Identity Manager and use of custom roles. This feature requires the Microsoft Entra ID P1 license or P2 license.

IMPORTANT: Directory roles must be maintained via the Microsoft Azure management portal.

IMPORTANT: This license enables role assignment to individual users. Assigning roles to groups is only possible with the Microsoft Entra ID P1 license and P2 licenses.

Microsoft Entra ID Premium P1 - Role Based Access Control (RBAC)

Role-based access control is provided by the Microsoft Entra ID Premium P1 license. In addition to the basic features, it includes access to role definitions and role assignments. Roles can be assigned to an entire group. This allows consistent role eligibilities within a group.

There are two different types of partial scopes to which role-based access control can be applied.

- Directory object limitation: Role assignments can be limited to specific objects, such as a registered application or a user, within the Microsoft Entra ID directory. Restricting elements of a defined administrative unit is also possible.
- Restricting custom elements of a service: Customized roles cannot be created in One Identity Manager, only through synchronization.

IMPORTANT: This license does not include the functionality of Microsoft Entra ID Privileged Identity Management.

Microsoft Entra ID Premium P2 - Privileged Identity Management (PIM)

In addition to the existing limitations of role-based access control, this license provides additional functionality to restrict and control role assignments. Privileged Identity Management distinguishes between active role assignments and assignment eligibilities.

Role assignment: A principal is assigned a role.

Role eligibility: A principal has no active role assignment, but can enable a temporary role assignment if required.

Configuration of role policies, such as time limits, is possible for both assignment types. Furthermore, it is possible to create attestations for roles.

NOTE: It is not possible to create role assignments for which multi-factor authentication is mandatory.

NOTE: Due to the constraints of Microsoft GraphAPI, the role management feature in One Identity Manager in PIM mode only supports the global directory scope for active role assignments.

Detailed information about this topic

- Overview of role-based access control in Microsoft Entra ID
- Basic information about roles in Microsoft Entra ID

Related topics

- Microsoft Entra ID role management on page 175
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Enabling new Microsoft Entra ID role management features

The introduction of the Microsoft 365 role management makes extended features available for managing roles and their members and for limiting role assignments in Microsoft Entra ID parts of One Identity Manager.

New and existing synchronization projects automatically obtain the basic mode (equivalent to the Entra ID Free license from Microsoft 365) with the introduction of Microsoft Entra ID role management. The basic mode includes all the current features of One Identity Manager. The new role management features can be accessed by activating RBAC mode (Entra ID P1 license) and PIM mode (Entra ID P2 license). This activation is necessary for existing synchronization projects, and also when creating a new synchronization project.

> NOTE: All existing Microsoft Entra ID features remain available in basic mode. It is only necessary to activate RBAC mode or PIM mode if you want to use extended role management features.

To enable extended role management features for RBAC

1. In the Synchronization Editor, select the synchronization project.
2. Select **Workflows**.
3. Select the **Initial Synchronization** workflow and click the **Enable/disable synchronization step** button.
4. Disable the **DirectoryRole** synchronization step.
5. Enable the following synchronization steps.
   a. **RBAC DirectoryRole**
   b. **RBAC DirectoryRole Assignments**
6. Save the changes.

7. Select the **Provisioning** workflow and click the **Enable/disable synchronization step** button.
8. Disable the **DirectoryRole** synchronization step.
9. Enable the **RBAC DirectoryRole Assignments** synchronization step.
10. Save the changes.
11. In the Object Browser, select the **AADOrganization** table.
12. Set the **RoleBehavior** value to RBAC.
13. Save the changes.

To enable extended role management features for PIM

1. In the Synchronization Editor, select the synchronization project.
2. Select **Workflows**.
3. Select the **Initial Synchronization** workflow and click the **Enable/disable synchronization step** button.
4. Disable the **DirectoryRole** synchronization step.
5. Enable the following synchronization steps.
   a. **RBAC DirectoryRole**
   b. **PIM DirectoryRole Assignments**
   c. **PIM DirectoryRole Eligibility**
   d. **PIM DirectoryRole Policies**
6. Save the changes.
7. Select the **Provisioning** workflow and click the **Enable/disable synchronization step** button.
8. Disable the **DirectoryRole** synchronization step.
9. Enable the following synchronization steps.
   a. **PIM DirectoryRole Assignments**
   b. **PIM DirectoryRole Eligibility**
10. Save the changes.
11. In the Object Browser, select the **AADOrganization** table.
12. Set the **RoleBehavior** value to PIM.
13. Save the changes.

Detailed information about this topic

- Overview of role-based access control in Microsoft Entra ID
- Basic information about roles in Microsoft Entra ID

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Microsoft Entra ID role main data

You are provided with the following general main data of a role.

Table 23: General main data

Property	Description
Display name	Name for displaying the role in the user interface of One Identity Manager tools.
Tenant	The role's Microsoft Entra ID tenant.
Owner (application role)	Application whose members can configure role assignment and role eligibilities.
Provider	Interface responsible for managing the role.
Version	Specifies the version of the role definition.
Description	Text field for additional explanation.
Built-in	Specifies whether the role definition is part of the Microsoft Entra ID basic settings or a customized definition.
Enabled	Specifies whether the role is available for assignment.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID business roles to scopes though role eligibilities on page 203

Main data of Microsoft Entra ID role settings

You are provided with the following general main data of a role setting.

NOTE: The main data is set by processes, you cannot set it manually.

Table 24: General main data

Property	Description
No permanent activation	Specifies whether a role is permanently enabled.
Max. activation period (hours)	Maximum time period for the role to be enabled.
Activation rules	Specifies the activation rules that apply.
Require approval to activate	Specifies whether role activation must be approved.
Require reason on activation	Specifies whether a reason must be given for activating a role.
Require reason on approval	Specifies whether a reason must be given for approving a role.
Approval mode	Specifies the approval mode.
Approval stages	Specifies the number of approval stages.
Recipient type	Type of recipient.
Authentication on activation	Specifies whether authentication is required for activating the role.
Authentication context	Context for authentication.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Microsoft Entra ID role main data on page 179
- Main data of Microsoft Entra ID scoped role assignments on page 183
- Main data of Microsoft Entra ID scoped role eligibilities on page 194

Displaying Microsoft Entra ID role settings main data

Navigate to the Microsoft Entra ID role settings main data via the overview form of the corresponding role.

| NOTE: The main data is set by processes, you cannot set it manually.

To display Microsoft Entra ID role settings main data

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. In the overview, select the **Microsoft Entra ID role setting** form element.
4. Select the form element **Microsoft Entra ID role setting** again.

Related topics

- Main data of Microsoft Entra ID role settings on page 180

Assigning temporary access passes to Microsoft Entra ID user accounts

Temporary access passes allow you to assign temporary access passes to new user accounts using the Manager. Only one temporary access pass can be assigned per user account.

To add a temporary access pass to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. In the result list, select the user account.
3. Select the **Assign temporary access pass** task.

   | NOTE: This feature is not available in B2C Tenants.
4. Click **Add** and enter the following information.
   - **Start time**: The time at which the temporary access pass becomes valid.

     | NOTE: If access is to start immediately, do not enter a start time.
   - **Valid for (minutes)**: Specify the validity period of the temporary access pass.
   - Specify whether this is a **One-time user** temporary access pass.
5. Save the changes.

To remove a temporary access pass to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign temporary access pass** task.
4. Select the temporary access pass and click **Remove**.
5. Save the changes.

Displaying Microsoft Entra ID scoped role assignments

Scoped role assignments for a Microsoft Entra ID role are displayed on the role's overview form.

To obtain an overview of a scoped role assignment for a role

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Microsoft Entra ID role overview** task.

   The **Microsoft Entra ID role assignments** form element shows the role assignments of the role.

Related topics

- Displaying scoped role eligibilities for Microsoft Entra ID roles on page 182

Displaying scoped role eligibilities for Microsoft Entra ID roles

The scoped role eligibilities for a Microsoft Entra ID role are displayed on the role's overview form.

To obtain an overview of a scoped role eligibilities for a role

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Microsoft Entra ID role overview** task.

   The **Microsoft Entra ID role eligibilities** form element shows the role eligibilities of the role.

Related topics

- Displaying Microsoft Entra ID scoped role assignments on page 182

Overview of Microsoft Entra ID scoped role assignments

Use this task to obtain an overview of the most important information about a scoped role assignment.

To obtain an overview of a scoped role assignment

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role assignment in the result list.
3. Select the **Microsoft Entra ID scoped role assignment overview** task.

Related topics

- Overview of Microsoft Entra ID scoped role eligibilities on page 194

Main data of Microsoft Entra ID scoped role assignments

You are provided with the following general main data of an assignment role.

Table 25: General main data

Property	Description
Microsoft Entra ID role	The role to which the scoped role assignment is assigned.
App scope	The application scope to which the principal should be given access.
Directory scope	The directory scope to which the principal should be given access.
Permitted scope	Specifies whether the scope for the role can be selected in Microsoft Entra ID.
Risk index	The risk index of the scoped role assignment.
Category	Category used to selectively inherit system entitlements, such as default user account, system administrator, system account.

Related topics

- Microsoft Entra ID role management on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Assigning Microsoft Entra ID scoped role assignments on page 187

Managing Microsoft Entra ID scoped role assignments

To edit a role assignment

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Assign role assignments** task.
4. Select the principal.
5. Edit the relevant main data.
6. Save the changes.

Related topics

- Managing Microsoft Entra ID scoped role eligibilities on page 195
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Adding Microsoft Entra ID scoped role assignments

Role management allows you to make additional role assignments for roles in Microsoft Entra ID partial scopes.

To assign a role assignment to a role

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Assign role assignments** task.

4. Click **Add** and enter the following information.
   - **Principal**: Specify the group, service principal, or user.
     a. Click → next to the field.
     b. Under **Table**, select one of the following tables:
        - To authorize a group, select `AADGroup`.
        - To authorize a service principal, select `AADServicePrincipal`.
        - To authorize a user account, select `AADUser`.
        - Under **Principal**, select the group, service principal, or user account.
     c. Click **OK**.
   - **Application scope**: Specify the organization to which the service principal is given access.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.
     c. Under **Application scope**, select the tenant.
     d. Click **OK**.

   - OR -
   - **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
     a. Click → next to the field.
     b. Under **Table**, select one of the following tables:
        - To authorize an administrative unit, select `AADAdministrativeUnit`.
        - To authorize an application, select `AADApplication`.
        - To authorize an organization, select `AADOrganization`.
        - To authorize a service principal, select `AADServicePrincipal`.
     c. Under **Directory scope**, select the tenant.
   - Specify whether this assignment is a **Direct assignment**.

     NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.
   - **Request procedure**: References the request procedure that results in the assignment.

     NOTE: The request procedure is set by processes and cannot be set manually.

Related topics

- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Managing Microsoft Entra ID scoped role assignments on page 184 Managing Microsoft Entra ID scoped role assignments
- Managing Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Editing Microsoft Entra ID scoped role assignments

To edit the main data of a role assignment

1. In the Manager, select the **Microsoft Entra ID > Role assignments** category.
2. Select the role assignment in the result list.
3. Select the **Change main data** task.
4. Edit the main data of the role assignment.
5. Save the changes.

Deleting Microsoft Entra ID scoped role assignments

To delete a role assignment

1. In the Manager, select the **Microsoft Entra ID > Role assignments** category.
2. Select the role assignment in the result list.
3. Click in the result list.
4. Confirm the security prompt with **Yes**.
5. Save the changes.

Assigning Microsoft Entra ID scoped role assignments

In Microsoft Entra ID, active role assignments for groups can be assigned in specified partial scopes.

To assign a system role to scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

To assign a business role to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

To assign an organization to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.
3. Select the **Assign organizations** task.

In the **Add assignments** pane, assign the organizations:

- On the **Departments** tab, assign departments.
- On the **Locations** tab, assign locations.
- On the **Cost centers** tab, assign cost centers.

TIP: In the **Remove assignments** pane, you can remove assigned organizations.

To remove an assignment

- Select the organization and double-click ✓.

4. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Assigning Microsoft Entra ID scoped role assignments to Microsoft Entra ID user accounts

To add a scoped role assignment to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign role assignments** task.
4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.
     c. Under **Application scope**, select the tenant.
     d. Click **OK**.

   - OR -

- **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
  a. Click → next to the field.
  b. Under **Table**, select one of the following tables:
     - To authorize an administrative unit, select `AADAdministrativeUnit`.
     - To authorize an application, select `AADApplication`.
     - To authorize an organization, select `AADOrganization`.
     - To authorize a service principal, select `AADServicePrincipal`.
  c. Under **Directory scope**, select the tenant.
  d. Click **OK**.
- Specify whether this assignment is a **Direct assignment**.

  NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.
- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is determined by processes and cannot be set manually.

5. Save the changes.

Assigning Microsoft Entra ID scoped role assignments to Microsoft Entra ID groups

This task only available for groups with the **Assignable to administrator roles** option enabled.

To add a scoped role assignment to a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign role assignments** task.
4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.

c. Under **Application scope**, select the tenant.

d. Click **OK**.

- OR -

- **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.

  a. Click → next to the field.

  b. Under **Table**, select one of the following tables:

     - To authorize an administrative unit, select `AADAdministrativeUnit`.
     - To authorize an application, select `AADApplication`.
     - To authorize an organization, select `AADOrganization`.
     - To authorize a service principal, select `AADServicePrincipal`.

  c. Under **Directory scope**, select the tenant.

  d. Click **OK**.

- Enter **Permanent** if it is a permanent assignment.
- **Start time**: The time from which the role eligibility is assigned.
- **End time**: The time at which the role eligibility expires.

  NOTE: Select **Permanent**, disables the **End time** entry.

- **Valid from**: Time at which the role eligibility becomes valid.
- Specify whether this assignment is a **Direct assignment**.

  NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.

- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is determined by processes and cannot be set manually.

5. Save the changes.

Assigning Microsoft Entra ID scoped role assignments to Microsoft Entra ID service principals

To add a scoped role assignment to a service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the service principal.
3. Select the **Assign role assignments** task.

4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.
     c. Under **Application scope**, select the tenant.
     d. Click **OK**.

   - OR -

   - **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
     a. Click → next to the field.
     b. Under **Table**, select one of the following tables:
        - To authorize an administrative unit, select `AADAdministrativeUnit`.
        - To authorize an application, select `AADApplication`.
        - To authorize an organization, select `AADOrganization`.
        - To authorize a service principal, select `AADServicePrincipal`.
     c. Under **Directory scope**, select the tenant.
     d. Click **OK**.
   - Specify whether this assignment is a **Direct assignment**.

     NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.
   - **Request procedure**: References the request procedure that results in the assignment.

     NOTE: The request procedure is determined by processes and cannot be set manually.
5. Save the changes.

Assigning Microsoft Entra ID system roles to scopes through role assignments

To assign a system role to scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.

3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Assigning Microsoft Entra ID business roles to scopes though role assignments

To assign a business role to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Assigning Microsoft Entra ID organizations to scopes through role assignments

To assign an organization to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role assignments** category.
2. Select the role in the result list.
3. Select the **Assign organizations** task.

   In the **Add assignments** pane, assign the organizations:

   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
4. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Overview of Microsoft Entra ID scoped role eligibilities

Use this task to obtain an overview of the most important information about a scoped eligibility.

To obtain an overview of a scoped role eligibility

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role eligibility in the result list.
3. Select the **Microsoft Entra ID scoped role eligibility overview** task.

Related topics

- Overview of Microsoft Entra ID scoped role assignments on page 183

Main data of Microsoft Entra ID scoped role eligibilities

You are provided with the following general main data for a role eligibility.

Table 26: General main data

Property	Description
Microsoft Entra ID role	The role to which the scoped role eligibility is assigned.
App scope	The application scope to which the principal should be given access.
Directory scope	The directory scope to which the principal should be given access.
Permitted scope	Specifies whether the scope for the role can be selected in Microsoft Entra ID.
Risk index	The risk index of the scoped role eligibility.
Category	Category used to selectively inherit system entitlements, such as default user account, system administrator, system account.

Related topics

- Microsoft Entra ID role management on page 175
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role eligibilities

Managing Microsoft Entra ID scoped role eligibilities

To edit a role eligibility

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Assign role eligibilities** task.
4. Select the principal.
5. Edit the relevant main data.
6. Save the changes.

Related topics

- Managing Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role assignments on page 184
- Adding Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Adding Microsoft Entra ID scoped role eligibilities

Role management allows you to assign additional role eligibilities to roles in Microsoft Entra ID partial scopes. These role eligibilities can be enabled by the assigned principal as required.

To assign a role eligibility to a role

1. In the Manager, select the **Microsoft Entra ID > Roles** category.
2. Select the role in the result list.
3. Select the **Assign role eligibilities** task.

NOTE: The **Assign role eligibilities** task is not available for role in RBAC mode (Microsoft Entra ID P1 license).

4. Click **Add** and enter the following information.
   - **Principal**: Specify the group, service principal, or user.
     a. Click → next to the field.
     b. Under **Table**, select one of the following tables:
        - To authorize a group, select `AADGroup`.
        - To authorize a service principal, select `AADServicePrincipal`.
        - To authorize a user account, select `AADUser`.
     c. Under **Principal**, select the group, service principal, or user account.
     d. Click **OK**.
   - **Application scope**: Specify the organization to which the service principal is given access.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.
     c. Under **Application scope**, select the tenant.
     d. Click **OK**.

     - OR -
   - **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
     a. Click → next to the field.
     b. Under **Table**, select one of the following tables:
        - To authorize an administrative unit, select `AADAdministrativeUnit`.
        - To authorize an application, select `AADApplication`.
        - To authorize an organization, select `AADOrganization`.
        - To authorize a service principal, select `AADServicePrincipal`.
     c. Under **Directory scope**, select the tenant.
   - Enter **Permanent** if it is a permanent assignment.
   - **Start time**: The time at which the role eligibility is assigned.
   - **End time**: The time at which the role eligibility expires.

     NOTE: Select **Permanent**, disables the **End time** entry.
   - **Valid from**: The role eligibility is valid from this point on.
   - **Valid until**: The role eligibility is expired from at this point on.
   - Specify whether this assignment is a **Direct assignment**.

     NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.

- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is set by processes and cannot be set manually.

5. Save the changes.

Related topics

- Adding Microsoft Entra ID scoped role assignments on page 184
- Managing Microsoft Entra ID scoped role assignments on page 184 Managing Microsoft Entra ID scoped role assignments
- Managing Microsoft Entra ID scoped role eligibilities on page 195
- Assigning Microsoft Entra ID scoped role assignments on page 187
- Assigning Microsoft Entra ID scoped role eligibilities on page 198

Editing Microsoft Entra ID scoped role eligibilities

To edit the main data of a role eligibility

1. In the Manager, select the **Microsoft Entra ID > Role eligibilities** category.
2. Select the role eligibility in the result list.
3. Select the **Change main data** task.
4. Edit the main data of role eligibility.
5. Save the changes.

Deleting Microsoft Entra ID scoped role eligibilities

To delete a role eligibility

1. In the Manager, select the **Microsoft Entra ID > Role eligibilities** category.
2. Select the role eligibility in the result list.
3. Click in the result list.
4. Confirm the security prompt with **Yes**.
5. Save the changes.

Assigning Microsoft Entra ID scoped role eligibilities

In Microsoft Entra ID, role eligibilities for groups can be assigned in specified partial scopes. This does not mean that a principal has an active role assignment, but can activate it at any time if necessary.

To assign a system role to scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

To assign a business role to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

To assign an organization to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign organizations** task.

In the **Add assignments** pane, assign the organizations:

- On the **Departments** tab, assign departments.
- On the **Locations** tab, assign locations.
- On the **Cost centers** tab, assign cost centers.

TIP: In the **Remove assignments** pane, you can remove assigned organizations.

To remove an assignment

- Select the organization and double-click ✓.

4. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role assignments on page 187

Assigning Microsoft Entra ID scoped role eligibilities to Microsoft Entra ID user accounts

To add a scoped role eligibility to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Assign role eligibilities** task.
4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.
     c. Under **Application scope**, select the tenant.
     d. Click **OK**.

   - OR -

- **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
  a. Click → next to the field.
  b. Under **Table**, select one of the following tables:
     - To authorize an administrative unit, select `AADAdministrativeUnit`.
     - To authorize an application, select `AADApplication`.
     - To authorize an organization, select `AADOrganization`.
     - To authorize a service principal, select `AADServicePrincipal`.
  c. Under **Directory scope**, select the tenant.
  d. Click **OK**.
- Specify whether this assignment is a **Direct assignment**.

  NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.
- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is determined by processes and cannot be set manually.

5. Save the changes.

Assigning Microsoft Entra ID scoped role eligibilities to Microsoft Entra ID groups

This task only available for groups with the **Assignable to administrator roles** option enabled.

To add a scoped role eligibility to a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign role eligibilities** task.
4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.
     a. Click → next to the field.
     b. Under **Table**, select the `AADOrganization` table.

c. Under **Application scope**, select the tenant.

d. Click **OK**.

- OR -

- **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.

  a. Click → next to the field.

  b. Under **Table**, select one of the following tables:

     - To authorize an administrative unit, select `AADAdministrativeUnit`.
     - To authorize an application, select `AADApplication`.
     - To authorize an organization, select `AADOrganization`.
     - To authorize a service principal, select `AADServicePrincipal`.

  c. Under **Directory scope**, select the tenant.

  d. Click **OK**.

- Specify whether this assignment is a **Direct assignment**.

  NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.

- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is determined by processes and cannot be set manually.

5. Save the changes.

Assigning Microsoft Entra ID scoped role eligibilities to Microsoft Entra ID service principals

To add a scoped role eligibility to a service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the service principal.
3. Select the **Assign role eligibilities** task.
4. Click **Add** and enter the following information.
   - **Microsoft Entra ID role**: Specify the role for authorization.
   - **Application scope**: Specify the organization for authorization.

a. Click → next to the field.
b. Under **Table**, select the `AADOrganization` table.
c. Under **Application scope**, select the tenant.
d. Click **OK**.

- OR -

- **Directory scope**: Specify the administrative unit, application, organization, or service principal for authorization.
  a. Click → next to the field.
  b. Under **Table**, select one of the following tables:
     - To authorize an administrative unit, select `AADAdministrativeUnit`.
     - To authorize an application, select `AADApplication`.
     - To authorize an organization, select `AADOrganization`.
     - To authorize a service principal, select `AADServicePrincipal`.
  c. Under **Directory scope**, select the tenant.
  d. Click **OK**.
- Enter **Permanent** if it is a permanent assignment.
- **Start time**: The time from which the role eligibility is assigned.
- **End time**: The time at which the role eligibility expires.

  NOTE: Select **Permanent**, disables the **End time** entry.
- **Valid from**: Time at which the role eligibility becomes valid.
- Specify whether this assignment is a **Direct assignment**.

  NOTE: The assignment specifications **Indirect assignment** and **Assignment request** are determined by processes and cannot be set manually.
- **Request procedure**: References the request procedure that results in the assignment.

  NOTE: The request procedure is determined by processes and cannot be set manually.

5. Save the changes.

Assigning Microsoft Entra ID system roles to scopes through role eligibilities

To assign a system role to scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign system roles** task.
4. In the **Add assignments** pane, assign system roles.

   TIP: In the **Remove assignments** pane, you can remove the system role assignment.

   To remove an assignment

   - Select the system role and double-click ✓.
5. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role assignments on page 187

Assigning Microsoft Entra ID business roles to scopes though role eligibilities

To assign a business role to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign business roles** task.
4. In the **Add assignments** pane, select the role class and assign business roles.

   TIP: In the **Remove assignments** pane, you can remove assigned business roles.

   To remove an assignment

   - Select the business role and double-click ✓.
5. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175

- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role assignments on page 187

Assigning Microsoft Entra ID organizations to scopes through role eligibilities

To assign an organization to a scope

1. In the Manager, select the **Microsoft Entra ID > Scoped role eligibilities** category.
2. Select the role in the result list.
3. Select the **Assign organizations** task.

   In the **Add assignments** pane, assign the organizations:

   - On the **Departments** tab, assign departments.
   - On the **Locations** tab, assign locations.
   - On the **Cost centers** tab, assign cost centers.

   TIP: In the **Remove assignments** pane, you can remove assigned organizations.

   To remove an assignment

   - Select the organization and double-click ✓.
4. Save the changes.

Related topics

- Microsoft Entra ID role management on page 175
- Microsoft Entra ID role management tenants on page 175
- Microsoft Entra ID role main data on page 179
- Assigning Microsoft Entra ID scoped role assignments on page 187

Mapping Microsoft Entra ID objects in One Identity Manager

In One Identity Manager, you can map user accounts, groups, administrator roles, subscriptions, service plans, applications, service principals, and app roles of a Microsoft Entra ID tenant. These objects are imported into the One Identity Manager database during synchronization. You cannot display or edit their properties in the Manager.

Detailed information about this topic

- Microsoft Entra ID core directories on page 205
- Microsoft Entra ID tenant on page 206
- Microsoft Entra ID domains on page 210
- Microsoft Entra ID user accounts on page 214
- Microsoft Entra ID user identities on page 230
- Microsoft Entra ID groups on page 233
- Microsoft Entra ID administrator roles on page 242
- Microsoft Entra ID administrative units on page 245
- Microsoft Entra ID subscriptions and Microsoft Entra ID service principals on page 248
- Disabled Microsoft Entra ID service plans on page 251
- Microsoft Entra ID app registrations and Microsoft Entra ID service principals on page 254
- Reports about Microsoft Entra ID objects on page 263

Microsoft Entra ID core directories

For more information about the Microsoft Entra ID structure, see the *Microsoft Entra ID documentation* from Microsoft.

You must provide details about your organization the first time you register for a Microsoft cloud service. This detailed information is used to make a new Microsoft Entra ID directory partition. The organization represents one Microsoft Entra ID tenant. In One Identity Manager, you can edit the main data of each tenant. However, you cannot create new tenants in One Identity Manager.

A base domain is linked to the core directory in the cloud. You can also add other user-defined domains in Microsoft Entra ID, which you can then allocate to Microsoft cloud services. One Identity Manager only loads verified domain data into the database. It is not possible to edit data in One Identity Manager.

Detailed information about this topic

- Microsoft Entra ID tenant on page 206
- Microsoft Entra ID domains on page 210
- Microsoft Entra ID policies for activity-based timeouts on page 211
- Microsoft Entra ID policies for home realm discovery on page 212
- Microsoft Entra ID policies for issuing tokens on page 212
- Microsoft Entra ID policies for token lifetime on page 213

Microsoft Entra ID tenant

You must provide details about your organization the first time you register for a Microsoft cloud service. This detailed information is used to make a new Microsoft Entra ID directory partition. The organization represents one Microsoft Entra ID tenant. In One Identity Manager, you can edit the main data of each Microsoft Entra ID tenant. However, you cannot create new Microsoft Entra ID tenants in One Identity Manager.

To edit Microsoft Entra ID tenant main data

1. In the Manager, select the **Microsoft Entra ID > Tenants** category.
2. In the result list, select the Microsoft Entra ID tenant.
3. Select the **Change main data** task.
4. Edit the Microsoft Entra ID tenant's main data.
5. Save the changes.

Detailed information about this topic

- General main data of Microsoft Entra ID tenants on page 207
- Information about local Active Directory on page 208
- Defining categories for the inheritance of entitlements on page 209
- Synchronizing single objects on page 54

General main data of Microsoft Entra ID tenants

Enter the following general main data.

Table 27: Microsoft Entra ID tenant main data

Property	Description
Display name	The Microsoft Entra ID tenant's display name.
Tenant type	The Microsoft Entra ID tenant type. Permitted values are: • **`AAD`**: Microsoft Entra ID tenant • **`AAD B2C`**: Microsoft Entra ID B2C tenants This property is loaded by synchronization and cannot be edited.
Account definition (initial)	Initial account definition for creating Microsoft Entra ID user accounts. This account definition is used if automatic assignment of identities to user accounts is used for this Microsoft Entra ID tenant and user accounts should be created which are already managed (**`Linked configured`** state). The account definition's default manage level is applied. User accounts are only linked to the identity (**`Linked`**) if no account definition is given. This is the case on initial synchronization, for example.
Target system managers	Application role, in which target system managers are specified for the Microsoft Entra ID tenant. Target system managers only edit the objects from Microsoft Entra ID tenants to which they are assigned. Each Microsoft Entra ID tenant can have a different target system manager assigned to it. Select the One Identity Manager application role whose members are responsible for administration of this Microsoft Entra ID tenant. Use the button to add a new application role.
Location	The Microsoft Entra ID tenant's location.
Street	Street or road.
City	City.
Zip code	Zip code.
Country	Country.
Synchronized by	Type of synchronization through which the data is synchronized between the Microsoft Entra ID tenant and One Identity Manager. You can no longer change the synchronization type once objects for this Microsoft Entra ID tenant are present in One Identity Manager.

Property	Description
	If you create a Microsoft Entra ID tenant with the Synchronization Editor, `One Identity Manager` is used. **Table 28: Permitted values** | Value | Synchronization by | Provisioned by | |---|---|---| | One Identity Manager | Microsoft Entra ID connector | Microsoft Entra ID connector | | No synchronization | none | none | NOTE: If you select `No synchronization`, you can define custom processes to exchange data between One Identity Manager and the target system.
Recipients (marketing notifications)	List of recipients of marketing notifications.
Recipient (technical notifications)	List of recipients of technical notifications.
Recipients (security notifications)	List of recipients of security notifications.
Phone numbers (security notifications)	Phone numbers for security notifications.

Related topics

- Assigning identities automatically to Microsoft Entra ID user accounts on page 87
- Target system managers for Microsoft Entra ID on page 273

Information about local Active Directory

The **Linked** tab shows information about the local Active Directory, which is linked to the Microsoft Entra ID tenant.

Table 29: Local Active Directory user account data

Property	Description
Synchronization with local Active Directory enabled	Specifies whether synchronization with a local Active Directory is enabled.
Last synchronization	Time of the last Microsoft Entra ID tenant synchronization with the local Active Directory.

Defining categories for the inheritance of entitlements

In One Identity Manager, user accounts can selectively inherit groups, administrator roles, subscriptions, and disabled service plans. To do this, the groups (administrator roles, subscriptions, and disabled service plans) and the user accounts are divided into categories. The categories can be freely selected and are specified using a mapping rule. Each category is given a specific position within the template. The mapping rule contains different tables. Use the user account table to specify categories for target system dependent user accounts. In the other tables, enter your categories for the groups, administrator roles, subscriptions, and disabled service plans. Each table contains the category positions **position 1** to **position 63**.

To define a category

1. In the Manager, select the Microsoft Entra ID tenant in the **Microsoft Entra ID > Tenants** category.
2. Select the **Change main data** task.
3. Switch to the **Mapping rule category** tab.
4. Extend the relevant roots of a table.
5. To enable the category, double-click .
6. Enter a category name of your choice for user accounts and groups (administrator roles, subscriptions, disabled service plans) in the login language that you use.
7. Save the changes.

Related topics

- Microsoft Entra ID group inheritance based on categories on page 115
- Microsoft Entra ID administrator role inheritance based on categories on page 130
- Inheriting Microsoft Entra ID subscriptions based on categories on page 160
- Inheritance of disabled Microsoft Entra ID service plans based on categories on page 160

Editing the synchronization project for a Microsoft Entra ID tenant

Synchronization projects in which a Microsoft Entra ID tenant is already used as a base object can also be opened in the Manager. You can, for example, check the configuration or view the synchronization log in this mode. The Synchronization Editor is not started with its full functionality. You cannot run certain functions, such as, running synchronization or simulation, starting the target system browser and others.

NOTE: The Manager is locked for editing throughout. To edit objects in the Manager, close the Synchronization Editor.

To open an existing synchronization project in the Synchronization Editor

1. In the Manager, select the **Microsoft Entra ID > Tenants** category.
2. In the result list, select the Microsoft Entra ID tenant.
3. Select the **Change main data** task.
4. Select the **Edit synchronization project** task.

Related topics

- Adjusting the synchronization configuration for Microsoft Entra ID environments on page 36

Microsoft Entra ID domains

A base domain is linked to the core directory in the cloud. You can also add other user-defined domains in Microsoft Entra ID, which you can then allocate to Microsoft cloud services. One Identity Manager only loads verified domain data into the database. It is not possible to edit data in One Identity Manager.

To obtain an overview of a domain

1. In the Manager, select the **Microsoft Entra ID > Verified domains** category.
2. Select the domain in the result list.
3. Select the **Microsoft Entra ID domain overview** task.

Table 30: Domain main data

Property	Description
Domain name	Full domain name.
Tenant	Microsoft Entra ID tenant entered for this domain.

Property	Description
Type	Type of domain.
Primary domain	Specifies whether this is the primary domain, for example, for creating new Microsoft Entra ID user accounts.
Initial domain	Specifies whether this is the initial domain. The initial domain is created when a tenant is registered in Microsoft Entra ID.
Available services	List of the services available in this domain.

Related topics

- Synchronizing single objects on page 54

Microsoft Entra ID policies for activity-based timeouts

You can use Microsoft Entra ID activity-based timeout policies to specify the idle time of web sessions for applications. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Microsoft Entra ID activity-based timeout policies are loaded into One Identity Manager during synchronization and cannot be changed.

To display information about a Microsoft Entra ID policy

1. In the Manager, select the **Microsoft Entra ID > Tenants > <your tenant> > Policies > Activity-based timeout policies** category.
2. In the result list, select the Microsoft Entra ID policy.
3. Select one of the following tasks:
    - **Activity-based timeout policy overview**: This shows you an overview of the Microsoft Entra ID policy and its dependencies.
    - **Change main data**: Shows the Microsoft Entra ID policy's main data. You cannot edit the main data.
        - **Display name**: The Microsoft Entra ID policy's display name.
        - **Description**: Description of the Microsoft Entra ID policy.
        - **Definition**: Definition of the Microsoft Entra ID in JSON format.
        - **Tenant**: Microsoft Entra ID tenant that owns the policy.
        - **Default policy**: Specifies whether this is the Microsoft Entra ID tenant's default policy.

Microsoft Entra ID policies for home realm discovery

You can use Microsoft Entra ID home realm discovery policies to accelerate logging users into federated domains. To provide an Microsoft Entra ID home realm discovery policy for a Microsoft Entra ID application, you assign the policy to the Microsoft Entra ID service principal. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Microsoft Entra ID home realm discovery policies are loaded into One Identity Manager during synchronization and cannot be changed.

To display information about a Microsoft Entra ID policy

1. In the Manager, select the **Microsoft Entra ID > Tenants > <your tenant> > Policies > Home realm discovery policies** category.
2. In the result list, select the Microsoft Entra ID policy.
3. Select one of the following tasks:
    - **Home realm discovery policy overview**: This shows you an overview of the Microsoft Entra ID policy and its dependencies.
    - **Change main data**: Shows the Microsoft Entra ID policy's main data. You cannot edit the main data.
        - **Display name**: The Microsoft Entra ID policy's display name.
        - **Description**: Description of the Microsoft Entra ID policy.
        - **Definition**: Definition of the Microsoft Entra ID in JSON format.
        - **Tenant**: Microsoft Entra ID tenant that owns the policy.
        - **Default policy**: Specifies whether this is the Microsoft Entra ID tenant's default policy.

Related topics

- Displaying Microsoft Entra ID service principal main data on page 261

Microsoft Entra ID policies for issuing tokens

You can use Microsoft Entra ID token issuance policies to specify SAML token properties for logging in. To provide a Microsoft Entra ID token issuance policy for a Microsoft Entra ID application, you assign the policy to the Microsoft Entra ID application. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Microsoft Entra ID token issuance policies are loaded into One Identity Manager during synchronization and cannot be changed.

To display information about a Microsoft Entra ID policy

1. In the Manager, select the **Microsoft Entra ID > Tenants > <your tenant> > Policies > Token issuance policies** category.
2. In the result list, select the Microsoft Entra ID policy.
3. Select one of the following tasks:
    - **Token issuance policy overview**: This shows you an overview of the Microsoft Entra ID policy and its dependencies.
    - **Change main data**: Shows the Microsoft Entra ID policy's main data. You cannot edit the main data.
        - **Display name**: The Microsoft Entra ID policy's display name.
        - **Description**: Description of the Microsoft Entra ID policy.
        - **Definition**: Definition of the Microsoft Entra ID in JSON format.
        - **Tenant**: Microsoft Entra ID tenant that owns the policy.
        - **Default policy**: Specifies whether this is the Microsoft Entra ID tenant's default policy.

Related topics

- Displaying Microsoft Entra ID app registration main data on page 256

Microsoft Entra ID policies for token lifetime

You can use Microsoft Entra ID token lifetime policies to specify the validity of token for logging in. To provide an Microsoft Entra ID token lifetime policy for a Microsoft Entra ID application, you assign the policy to the Microsoft Entra ID application. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Microsoft Entra ID token lifetime policies are loaded into One Identity Manager during synchronization and cannot be changed.

To display information about a Microsoft Entra ID policy

1. In the Manager, select the **Microsoft Entra ID > Tenants > <your tenant> > Policies > Token lifetime policies** category.
2. In the result list, select the Microsoft Entra ID policy.
3. Select one of the following tasks:
    - **Token lifetime policy overview**: This shows you an overview of the Microsoft Entra ID policy and its dependencies.
    - **Change main data**: Shows the Microsoft Entra ID policy's main data. You cannot edit the main data.

- **Display name**: The Microsoft Entra ID policy's display name.
- **Description**: Description of the Microsoft Entra ID policy.
- **Definition**: Definition of the Microsoft Entra ID in JSON format.
- **Tenant**: Microsoft Entra ID tenant that owns the policy.
- **Default policy**: Specifies whether this is the Microsoft Entra ID tenant's default policy.

Related topics

- Displaying Microsoft Entra ID app registration main data on page 256

Microsoft Entra ID user accounts

You use One Identity Manager to manage user accounts in Microsoft Entra ID. The user requires a subscription to access the service plans in Microsoft Entra ID. Microsoft Entra ID user accounts obtain the required access permissions to the resources through membership in groups.

Related topics

- Managing Microsoft Entra ID user accounts and identities on page 63
- Managing memberships in Microsoft Entra ID groups on page 101
- Managing Microsoft Entra ID administrator roles assignments on page 119
- Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131
- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Login credentials for Microsoft Entra ID user accounts on page 161
- Creating and editing Microsoft Entra ID user accounts on page 215
- Deploying user identities for Microsoft Entra ID user accounts on page 230
- Assigning extended properties to Microsoft Entra ID user accounts on page 226
- Disabling Microsoft Entra ID user accounts on page 227
- Deleting and restoring Microsoft Entra ID user accounts on page 228
- Displaying the Microsoft Entra ID user account overview on page 229
- Displaying Active Directory user accounts for Microsoft Entra ID user accounts on page 230
- Synchronizing single objects on page 54

Creating and editing Microsoft Entra ID user accounts

A user account can be linked to an identity in One Identity Manager. You can also manage user accounts separately from identities.

NOTE: It is recommended to use account definitions to set up user accounts for company identities. In this case, some of the main data described in the following is mapped through templates from identity main data.

NOTE: If identities are to obtain their user accounts through account definitions, the identities must own a central user account and obtain their IT operating data through assignment to a primary department, a primary location, or a primary cost center.

TIP: You can combine the account definition for creating the user account and the subscription that will be used into one system role. In this way, the identity automatically obtains a user account and a subscription.

An identity can obtain this system role directly through departments, cost centers, locations, or business roles, or an IT Shop request.

To create a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Click in the result list.
3. On the main data form, edit the main data of the user account.
4. Save the changes.

To edit main data of a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.
4. Edit the user account's resource data.
5. Save the changes.

To manually assign a user account for an identity

1. In the Manager, select the **Identities > Identities** category.
2. Select the identity in the result list.
3. Select the **Assign Microsoft Entra ID user accounts** task.
4. Assign a user account.
5. Save the changes.

Detailed information about this topic

- General main data of Microsoft Entra ID user accounts on page 216
- Contact data for Microsoft Entra ID user accounts on page 222
- Information about the user profile for Microsoft Entra ID user accounts on page 223
- Organizational data for Microsoft Entra ID user accounts on page 224
- Information about the local Active Directory user account on page 225

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64
- Supported user account types on page 92
- Login credentials for Microsoft Entra ID user accounts on page 161
- Managing Microsoft Entra ID user accounts and identities on page 63
- Managing memberships in Microsoft Entra ID groups on page 101
- Managing Microsoft Entra ID administrator roles assignments on page 119
- Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131

General main data of Microsoft Entra ID user accounts

Enter the following general main data.

Table 31: Additional main data of a user account

Property	Description
Identity	Identity that uses this user account. • An identity is already entered if the user account was generated by an account definition. • If you are using automatic identity assignment, an associated identity is found and added to the user account when you save the user account. • If you create the user account manually, you can select an identity in the drop-down. The drop-down displays activated and deactivated identities by default. If you do not want to see any deactivated identities, set the **QER | Person| HideDeactivatedIdentities** configuration parameter.

Property	Description
	NOTE: If you assign a deactivated identity to a user account, it might be locked or deleted depending on the configuration. You can create a new identity for a user account with an identity of type **`Organizational identity`**, **`Personalized administrator identity`**, **`Sponsored identity`**, **`Shared identity`**, or **`Service identity`**. To do this, click next next to the input field and enter the required identity main data. Which login data is required depends on the selected identity type.
No link to an identity required	Specifies whether the user account is intentionally not assigned an identity. The option is automatically set if a user account is included in the exclusion list for automatic identity assignment or a corresponding attestation is carried out. You can set the option manually. Enable the option if the user account does not need to be linked with an identity (for example, if several identities use the user account). If attestation approves these user accounts, these user accounts will not be submitted for attestation in the future. In the Web Portal, user accounts that are not linked to an identity can be filtered according to various criteria.
Not linked to an identity	Indicates why the **No link to an identity required** option is enabled for this user account. Possible values: • **`By administrator`**: The option was set manually by the administrator. • **`By attestation`**: The user account was attested. • **`By exclusion criterion`**: The user account is not associated with an identity due to an exclusion criterion. For example, the user account is included in the exclude list for automatic identity assignment (configuration parameter **PersonExcludeList**).
Account definition	Account definition through which the user account was created. Use the account definition to automatically fill user account main data and to specify a manage level for the user account. One Identity Manager finds the IT operating data of the assigned identity and enters it in the corresponding fields in the user account. NOTE: The account definition cannot be changed once the user account has been saved. NOTE: Use the user account's **Remove account definition** task to reset the user account to **Linked** status. This removes the account definition from both the user account and the identity. The user account remains but is not managed by the account definition anymore. The task only removes account definitions that are directly assigned (`XOrigin=1`).
Manage level	Manage level of the user account. Select a manage level from the drop-

Property	Description
	down. You can only specify the manage level can if you have also entered an account definition. All manage levels of the selected account definition are available in the drop-down.
Tenant	Microsoft Entra ID user account's tenant.
User type	Type of user account. Depending on the user type, other mandatory input is required. Permitted values are: • **`Member`**: Normal Microsoft Entra ID user account. • **`Guest`**: User account for guest users. The Microsoft Entra ID connector creates a user account for guest users and ensures that an invitation is sent by email to the given email address. Further configuration of guest users is required in the synchronization project. For more information, see Customizing synchronization projects to invite guest users on page 38.
Creation type	Specifies which method was used to create the user account. Possible values: • **`null`**: Regular school or office account. • **`Invitation`**: External user account. • **`LocalAccount`**: Local user account for a Microsoft Entra ID B2C tenant. • **`EmailVerified`**: Self-service login by an internal user with email verification. • **`SelfServiceSignUp`**: Self-Service login by an external user using a link that is part of a user flow.
Invitation status	(Only for the **`Guest`** user type) Acceptance status of the guest's invitation. Permitted values are: • **`Pending acceptance`**: The user has not accepted the invitation yet. • **`Accepted`**: The user has accepted the invitation. • **`Empty`**: Guest user without invitation.
Last change	(Only for the **`Guest`** user type) Time at which the invitation status was changed.
Domain	User account's domain.
Location	Location where this user account is in use. In the One Identity Manager, if you assign Microsoft Entra ID subscriptions, a location is required.
First name	The user’s first name. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.

Property	Description
Last name	The user’s last name. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Date of birth	The user's date of birth
Age group	The user's age group. Permitted values are **`Minor`**, **`Teenager`**, and **`Adult`**.
Consent for minors	Specifies whether consent must be given for minors. Permitted values are **`Obtained`**, **`Not obtained`**, and **`Not required`**.
User login name	User account login name. The user's login name is made up of the alias and the domain. User login names that are formatted like this correspond to the User Principal Name (UPN) in Microsoft Entra ID.
User identities	Collection of identities with which a user can log in to a user account. For more information, see Deploying user identities for Microsoft Entra ID user accounts on page 230.
Display name	User account display name.
Alias	Email alias for the user account.
Email address	User account's email address.
Preferred language	User's preferred language, for example, **`en-US`**.
Password	Password for the user account. The identity’s central password can be mapped to the user account password. For more information about an identity’s central password, see *One Identity Manager Identity Management Base Module Administration Guide*. If you use a random generated initial password for the user accounts, it is automatically entered when a user account is created. The password is deleted from the database after publishing to the target system. NOTE: One Identity Manager password policies are taken into account when a user password is being verified. Ensure that the password policy does not violate the target system's requirements.
Password confirmation	Reconfirm password.
Change password at next login	Specifies whether the user must change their password the next time they log in.
Password policy	Policies, which only apply to the user account. The available options are: **No restrictions**, **Password never expires**, and **Allow weak passwords**.

Property	Description
Password last changed	Data of last password change. The date is read in from the Microsoft Entra ID system and cannot be changed.
Risk index (calculated)	Maximum risk index value of all assigned groups. The property is only visible if the **QER | CalculateRiskIndex** configuration parameter is set. For more information, see the *One Identity Manager Risk Assessment Administration Guide*.
Category	Categories for the inheritance of groups by the user account. Groups can be selectively inherited by user accounts. To do this, groups and user accounts or contacts are divided into categories. Select one or more categories from the drop-down.
Identity type	User account's identity type Permitted values are: • **`Primary identity`**: Identity's default user account. • **`Organizational identity`**: Secondary user account used for different roles in the organization, for example for subcontracts with other functional areas. • **`Personalized administrator identity`**: User account with administrative permissions, used by one identity. • **`Sponsored identity`**: User account to use for a specific purpose. Training, for example. • **`Shared identity`**: User account with administrative permissions, used by several identities. Assign all identities that use this user account. • **`Service identity`**: Service account.
Privileged user account.	Specifies whether this is a privileged user account.
Disabled service plans can be inherited	Specifies whether the user account can inherit disabled Microsoft Entra ID service plans through the identity. If this option is set, the user account inherits disabled service plans through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned disabled service plans to this department, the user account inherits these disabled service plans. • If an identity has requested a disabled service plan in the IT Shop and the request is granted approval, the identity's user account only inherits the disabled service plan if the option is set.
Subscriptions can be inherited	Specifies whether the user account can inherit Microsoft Entra ID subscriptions through the identity. If this option is set, the user account

Property	Description
	inherits Microsoft Entra ID subscriptions through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned Microsoft Entra ID subscriptions to this department, the user account inherits these Microsoft Entra ID subscriptions. • If an identity has requested a Microsoft Entra ID subscription in the IT Shop and the request is granted approval, the identity's user account only inherits the Microsoft Entra ID subscription if the option is set.
Administrator roles can be inherited	Specifies whether the user account can inherit Microsoft Entra ID administrator roles through the identity If this option is set, the user account inherits administrator roles through hierarchical roles or IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned administrator roles to this department, the user account inherits these administrator roles. • If an identity has requested an administrator role in the IT Shop and the request is granted approval, the identity's user account only inherits the administrator role if the option is set.
Groups can be inherited	Specifies whether the user account can inherit groups through the linked identity. If the option is set, the user account inherits groups through hierarchical roles, in which the identity is a member, or through IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned groups to this department, the user account inherits these groups. • If an identity has requested group membership in the IT Shop and the request is granted approval, the identity's user account only inherits the group if the option is set.
Microsoft 365 groups can be inherited	NOTE: This property is only available if the Exchange Online Module is installed. Specifies whether the user account can inherit Microsoft 365 groups through the linked identity. If the option is set, the user account inherits Microsoft 365 groups through hierarchical roles, in which the identity is a member, or through IT Shop requests. • If you add an identity with a user account to a department, for example, and you have assigned Microsoft 365 groups to this department, the Microsoft Entra ID user account inherits these Microsoft 365 groups.

Property	Description
	• If an identity has requested group membership in the IT Shop and the request is granted approval, the identity's Microsoft Entra ID user account only inherits the Microsoft 365 group if the option is set. For more information about Microsoft 365 groups, see the *One Identity Manager Administration Guide for Connecting to Exchange Online*.
User account is disabled	Specifies whether the user account is disabled. If a user account is not required for a period of time, you can temporarily disable the user account by using the "User account is disabled" option.
Resource account	Specifies whether this user account is a resource account.

Related topics

- Deploying user identities for Microsoft Entra ID user accounts on page 230
- Account definitions for Microsoft Entra ID user accounts on page 64
- Password policies for Microsoft Entra ID user accounts on page 161
- Microsoft Entra ID user identities on page 230
- Microsoft Entra ID group inheritance based on categories on page 115
- Managing Microsoft Entra ID user accounts and identities on page 63
- Supported user account types on page 92
- Disabling Microsoft Entra ID user accounts on page 227
- Prerequisites for indirect assignment of Microsoft Entra ID groups to Microsoft Entra ID user accounts on page 103
- Prerequisites for indirect assignment of Microsoft Entra ID administration roles to Microsoft Entra ID user accounts on page 120
- Prerequisites for indirect assignment of Microsoft Entra ID subscriptions to Microsoft Entra ID user accounts on page 138
- Prerequisites for indirect assignment of disabled Microsoft Entra ID service plans to Microsoft Entra ID user accounts on page 150

Contact data for Microsoft Entra ID user accounts

Enter the following address data for contacting the identity on the **Contact** tab.

Table 32: Contact data

Property	Description
Street	Street or road. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
State	State. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
City	City. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level. Locations can be automatically generated and identities assigned based on the city.
Zip code	Zip code. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Country	The country ID.
Business phones	Business telephone numbers.
Mobile phone	Mobile number. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Fax	Fax number. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Additional email addresses	User email addresses.
Proxy addresses	Other email addresses for the user. You can also add other mail connectors (for example, CCMail, MS) in addition to the standard address type (SMTP, X400). Use the following syntax to set up other proxy addresses: `Address type: <email address>`

Information about the user profile for Microsoft Entra ID user accounts

The following information is displayed on the **User profile** tab.

Table 33: User profile

Property	Description
Preferred name	The user's preferred name.
Legal age group	This is used by Enterprise application to determined the legal age

Property	Description
	groups of users. The property is calculated based on the **Age group** and `Consent for minors` properties.
VoIP SIP addresses	The instant message voice over IP (VoIP) session initiation protocol (SIP) addresses for the user.
Personal site	URL for the user's personal website.
About me	Text field to the user to write a description of themselves.
Responsibilities	List of the user's responsibilities.
Schools	List of schools the user has attended.
Skills and expertise	List of the user's qualifications.
Past projects	List of the user's past projects.
Interests	List of the user's interests.

Organizational data for Microsoft Entra ID user accounts

The following organizational main data is mapped on the **Organizational** tab.

Table 34: Organizational main data

Property	Description
Employee ID	ID of the user within the organization. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Employee type	The hire type of the user, such as employee or supplier.
Hire date	(Planned) Date when the user joins the company.
Leave date	(Planned) Date when the user leaves the company.
Company	Company the user belongs to. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Department	Department the user belongs to. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Division	The user's division within the department.

Property	Description
Cost center	The cost center allocated to the user.
Office	Office. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Job description	Job description. If you have assigned an account definition, the input field is automatically filled out with respect to the manage level.
Account manager	Manager responsible for the user account. ***To specify an account manager*** 1. Click → next to the field. 2. In the **Table** menu, select the table that maps the account manager. 3. In the **Account manager** menu, select the manager. 4. Click **OK**.

Information about the local Active Directory user account

The **Linked** tab shows information about the local Active Directory user account, which is linked to the Microsoft Entra ID user account.

Table 35: Local Active Directory user account data

Property	Description
Synchronization with local Active Directory enabled	Specifies whether synchronization with a local Active Directory is enabled.
Last synchronization	Time of the last Microsoft Entra ID user account synchronization with the local Active Directory.
SID of the local account.	Security ID of the local Active Directory user account.
Immutable identifier	Identifier that is used to maintain the relationship between the Active Directory user account and the Microsoft Entra ID user account. The identifier cannot be changed.
Distinguished name	Active Directory user account's distinguished name.
Full domain name	Full domain name of the user account's Active Directory domain.
Login name (pre	Login name of the Active Directory user account for the previous

Property	Description
Win2000)	version of Active Directory.
User login name (of local account)	Active Directory user account login name.
Attribute extension 01 - attribute extension 15	Additional company-specific information about the Active Directory user account.

Related topics

- Displaying Active Directory user accounts for Microsoft Entra ID user accounts on page 230
- Recommendations for federations on page 269

Audit data for Microsoft Entra ID user accounts

The **Audit** tab shows the following information.

Table 36: Audit

Property	Description
Last non-interactive login	Time of the last user login in the directory with a non-interactive authentication method.
ID last non-interactive login	ID of the last non-interactive login.
Last interactive login	Time of the last user login in the directory with an interactive authentication method.
ID last interactive login	ID of the last interactive login.

Assigning extended properties to Microsoft Entra ID user accounts

Extended properties are meta objects, such as operating codes, cost codes, or cost accounting areas that cannot be mapped directly in One Identity Manager.

For more information about using extended properties, see the *One Identity Manager Compliance Rules Administration Guide*.

To specify extended properties for a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select **Assign extended properties**.
4. In the **Add assignments** pane, assign extended properties.

   TIP: In the **Remove assignments** pane, you can remove assigned extended properties.

   To remove an assignment

   - Select the extended property and double-click ✓.
5. Save the changes.

Disabling Microsoft Entra ID user accounts

The way you disable user accounts depends on how they are managed.

Scenario: The user accounts are linked to identities and are managed through account definitions.

User accounts managed through account definitions are disabled when the identity is temporarily or permanently disabled. The behavior depends on the user account manage level. Accounts with the **Full managed** manage level are disabled depending on the account definition settings. For user accounts with a manage level, configure the required behavior using the template in the `AADUser.AccountDisabled` column.

Scenario: The user accounts are linked to identities. No account definition is applied.

User accounts managed through user account definitions are disabled when the identity is temporarily or permanently disabled. The behavior depends on the **QER | Person | TemporaryDeactivation** configuration parameter

- If the configuration parameter is set, the identity's user accounts are disabled when the identity is permanently or temporarily disabled.
- If the configuration parameter is not set, the identity's properties do not have any effect on the associated user accounts.

To disable the user account when the configuration parameter is disabled

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.

4. On the **General** tab, set the **Account is disabled** option.
5. Save the changes.

Scenario: The user accounts are not linked to identities.

To disable a user account that is no longer linked to an identity

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.
4. On the **General** tab, set the **Account is disabled** option.
5. Save the changes.

For more information about deactivating and deleting identities and user accounts, see the *One Identity Manager Target System Base Module Administration Guide*.

Related topics

- Account definitions for Microsoft Entra ID user accounts on page 64
- Creating manage levels on page 71
- Deleting and restoring Microsoft Entra ID user accounts on page 228

Deleting and restoring Microsoft Entra ID user accounts

As long as an account definition for an identity is valid, the identity retains the user account that was created by it. If the account definition assignment is removed, the user account that was created from this account definition, is deleted. User accounts marked as **Outstanding** are only deleted if the **QER | Person | User | DeleteOptions | DeleteOutstanding** configuration parameter is set.

In the Manager, you can delete a user account that was not created using an account definition in the result list or from the menu bar. After you have confirmed the security alert the user account is marked for deletion in the One Identity Manager. The user account is locked in One Identity Manager and permanently deleted from the One Identity Manager database and the target system depending on the deferred deletion setting.

To delete a user account that is not managed using an account definition

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Click in the result list.
4. Confirm the security prompt with **Yes**.

Deferred deletion is taken into account if a user account is being deleted. You can use deferred deletion to specify how long the user accounts remain in the database after deletion is triggered before they are finally removed. You can reenable the user accounts up until deferred deletion runs.

If the **QER | Person | User | DeleteOptions | ReapplyTemplatesOnRestore** is set, the template is applied again when reenabling a user account marked for deletion that is managed through an account definition. This means that properties dependent on the IT operating data are automatically recreated according to the current configuration.

To restore a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Click in the result list.

For more information about deactivating and deleting identities and user accounts, see the *One Identity Manager Target System Base Module Administration Guide*.

Related topics

- Disabling Microsoft Entra ID user accounts on page 227
- Specifying deferred deletion for Microsoft Entra ID user accounts on page 100

Displaying the Microsoft Entra ID user account overview

Use this task to obtain an overview of the most important information about a user account.

To obtain an overview of a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Microsoft Entra ID user account overview** task.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

Displaying Active Directory user accounts for Microsoft Entra ID user accounts

You can display the Active Directory user account for a Microsoft Entra ID user account on the overview form.

To display the Active Directory user account for a Microsoft Entra ID user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Microsoft Entra ID user account overview** task.

   The **Active Directory user account** form element shows which user account is linked to it.

For more information about Active Directory, see the *One Identity Manager Administration Guide for Connecting to Active Directory*.

Related topics

- Information about the local Active Directory user account on page 225

Microsoft Entra ID user identities

Users can log in to a user account using any of the multiple identities that can be provided. For example, an identity can be provided by Microsoft, organizations, or social identity providers such as Facebook or Google.

Related topics

- Deploying user identities for Microsoft Entra ID user accounts on page 230
- Displaying user identities for Microsoft Entra ID user accounts on page 232
- Displaying Microsoft Entra ID user identities' main data on page 232
- Displaying the Microsoft Entra ID user identity overview on page 233

Deploying user identities for Microsoft Entra ID user accounts

Microsoft Entra ID user identities are imported into One Identity Manager during synchronization. You can create additional user identities for a user account or delete them.

To enter user identities for a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.
4. Next to the **User identities** property, click and enter the identities.

   Syntax:

   ```
   signInType=<type>;issuer=<your
   organization>.onmicrosoft.com;issuerAssignedId=<ID>
   ```

   With:

   - `signInType`: Sign-in type, such as **`emailAddress`** or **`userName`** for local user accounts or **`federated`** for social network identities.
   - `issuer`: Issuer of the identity, `Facebook` for example. Local user accounts use the name of the Microsoft Entra ID B2C tenant's default domain, for example `<your organization>.onmicrosoft.com`.
   - `issuerAssignedId`: ID used to log in depending on the sign-in type. If the sign-in type is **`emailAddress`**, the ID must be a valid email address, if it is **`userName`**, the ID must be a valid local part of an email address, and if it is **`federated`**, the issuer expects a unique identifier of the federated account.

   Example:

   ```
   signInType=emailAddress;issuer=<your
   organization>.onmicrosoft.com;issuerAssignedId=Clara.Harris@<your domain>.com
   ```

5. Save the changes.

To delete a user identity

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Change main data** task.
4. In the **User identities** drop-down, select the identity and click .
5. Save the changes.

Related topics

- General main data of Microsoft Entra ID user accounts on page 216
- Displaying user identities for Microsoft Entra ID user accounts on page 232
- Displaying Microsoft Entra ID user identities' main data on page 232

Displaying user identities for Microsoft Entra ID user accounts

User identities of a Microsoft Entra ID user account are displayed on the user account's overview form.

To obtain an overview of all user identities of a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.
3. Select the **Microsoft Entra ID user account overview** task.

   The **Microsoft Entra ID user identities** form element displays the user identities of the user account.

Related topics

- Microsoft Entra ID user identities on page 230

Displaying Microsoft Entra ID user identities' main data

You cannot edit the main data of an existing user identity. You can create additional user identities or delete user identities.

To display a user identity's main data

1. In the Manager, select the **Microsoft Entra ID > User accounts > User identities** category.
2. Select the user identity in the result list.
3. Select the **Change main data** task.

Table 37: Main data for a user identity

Property	Description
Microsoft Entra ID user account	Name of the Microsoft Entra ID user account used by the identity.
Issuer	Issuer of the identity, `Facebook` for example. Local user accounts use the name of the Microsoft Entra ID B2C tenant's default domain, for example `<your organization>.onmicrosoft.com`.

Property	Description
Assigned ID	ID used to log in depending on sign-in type. If the sign-in type is **`emailAddress`**, the ID must be a valid email address, if it is **`userName`** the ID must be a valid local part of an email address, and if it is **`federated`**, the issuer expects a unique identifier of the federated account.
Sign-in type	Ways in which a user can sign in to the directory. For example, **`emailAddress`** or **`userName`** for local user accounts and **`federated`** for social network identities.

Related topics

- Deploying user identities for Microsoft Entra ID user accounts on page 230

Displaying the Microsoft Entra ID user identity overview

Use this task to obtain an overview of the most important information about a user identity.

To obtain an overview of a user identity

1. In the Manager, select the **Microsoft Entra ID > User accounts > User identities** category.
2. Select the user identity in the result list.
3. Select the **Microsoft Entra ID user identity overview** task.

Related topics

- Displaying user identities for Microsoft Entra ID user accounts on page 232

Microsoft Entra ID groups

Microsoft Entra ID recognizes several group types into which you can organize users and groups to regulate access to resources or email distribution, for example.

Microsoft Entra ID groups are loaded into One Identity Manager by synchronization. You can edit individual main data of the group and you can create new security groups in One Identity Manager. However, you cannot create more group types in One Identity Manager.

To add users to groups, you assign the groups directly to users. This can be assignments of groups to departments, cost centers, locations, business roles, or the IT Shop.

NOTE: Assignments to Microsoft Entra ID groups that are synchronized with the local Active Directory are not allowed in One Identity Manager. These groups cannot be requested through the web portal. You can only manage these groups in your locally. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

The group types supported in One Identity Manager are listed below.

Table 38: Support groups types

Group type	Description
Security group	Resource permissions are distributed through security groups. User accounts and other groups are added to security groups, which makes administration easier. Security groups are loaded into One Identity Manager by synchronization. You can edit security groups in One Identity Manager and also create new ones.
Microsoft 365 group	Microsoft 365 groups are loaded into One Identity Manager by synchronization. You can edit Microsoft 365 groups in One Identity Manager. You can only create new Microsoft 365 groups in One Identity Manager if the Exchange Online Module is installed. For more information, see the *One Identity Manager Administration Guide for Connecting to Exchange Online*.
Distribution group	Distribution groups are used to send emails to group members. Distribution groups are loaded into One Identity Manager by synchronization. You can edit distribution groups in One Identity Manager but you cannot create them in One Identity Manager.
Mail-enabled security groups	Mail-enabled security groups are security groups that are used as distribution groups. Mail-enabled security groups are loaded into One Identity Manager by synchronization. You can edit mail-enabled security groups in One Identity Manager but you can only create new mail-enabled security groups in One Identity Manager if the Exchange Online Module is installed. For more information, see the *One Identity Manager Administration Guide for Connecting to Exchange Online*.
Dynamic group	Members of a dynamic group are not strictly assigned, but determined through defined rules. Dynamic groups are loaded into One Identity Manager by synchronization. You can change dynamic groups in One Identity Manager. You cannot create new dynamic groups in One Identity Manager.

Related topics

- Managing memberships in Microsoft Entra ID groups on page 101
- Editing main data of Microsoft Entra ID groups on page 235

- Adding Microsoft Entra ID groups to Microsoft Entra ID groups on page 238
- Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID groups on page 239
- Assigning owners to Microsoft Entra ID groups on page 240
- Assigning extended properties to Microsoft Entra ID groups on page 240
- Deleting Microsoft Entra ID groups on page 241
- Displaying the Microsoft Entra ID group overview on page 241
- Displaying Active Directory groups for Microsoft Entra ID groups on page 242
- Synchronizing single objects on page 54
- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

Editing main data of Microsoft Entra ID groups

Microsoft Entra ID groups are loaded into One Identity Manager by synchronization. You can create security groups in One Identity Manager. You cannot create distribution group and dynamic groups in One Identity Manager.

You can only create mail-enabled security groups and Microsoft 365 groups in One Identity Manager if the Exchange Online Module is installed. For more information, see the *One Identity Manager Administration Guide for Connecting to Exchange Online*.

the data you can edit depends on the group type.

To edit group main data

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Change main data** task.
4. On the main data form, edit the main data of the group.
5. Save the changes.

Detailed information about this topic

- General main data for Microsoft Entra ID groups on page 236
- Information about local Active Directory groups on page 238

General main data for Microsoft Entra ID groups

Enter the following general main data.

Table 39: General main data

Property	Description
Display name	Name for displaying the group in the user interface of One Identity Manager tools.
Tenant	The group's Microsoft Entra ID tenant.
Alias	Email alias for the group.
Email address	Group's email address
Proxy addresses	Other email addresses for the group. You can also add other mail connectors (for example, CCMail, MS) in addition to the standard address type (SMTP, X400). Use the following syntax to set up other proxy addresses: `Address type: new email address`
Group type	The type of group. This is empty for security and distribution groups. The value is **`Unified`** for Microsoft 365 groups. For dynamic groups, the value is **`DynamicMembership`**.
Security group	Specifies whether this group is a security group. Resource permissions are distributed through security groups. User accounts and other groups are added to security groups, which makes administration easier.
Mail-enabled	Specifies whether the email is enabled for the group. If this option is set for a security group, it is a mail-enabled security group. Otherwise, it is a distribution group.

Assignable to administrator roles	Specifies whether the group can be assigned to administrator roles. The option can be enabled only when creating a new group.
	NOTE: Groups with this option can only be created in Microsoft Entra ID if there is a Microsoft Entra ID Premium license for the Microsoft Entra ID tenant. Otherwise, an error message will be displayed:
	`Code: Authorization_RequestDenied`
	`Message: Only companies who have purchased AAD Premium may perform this operation.`

Read-only memberships	Specifies whether memberships are read-only. For example, dynamic groups. The memberships are regulated by the target system. Manual changes to memberships in One Identity Manager are not permitted.

Property	Description
IT Shop	Specifies whether the group can be requested through the IT Shop. If this option is set, the group can be requested through the Web Portal and allocated by defined approval processes. The group can still be assigned directly to hierarchical roles.
Only for use in IT Shop	Specifies whether the group can only be requested through the IT Shop. If this option is set, the group can be requested through the Web Portal and allocated by defined approval processes. Direct assignment of the group to hierarchical roles or user accounts is not permitted.
Service item	Service item data for requesting the group through the IT Shop.
Risk index	Value for evaluating the risk of assigning the group to user accounts. Set a value in the range `0` to `1`. This input field is only visible if the **QER | CalculateRiskIndex** configuration parameter is activated. For more information about risk assessment, see the *One Identity Manager Risk Assessment Administration Guide*.
Category	Categories for group inheritance. Groups can be selectively inherited by user accounts. To do this, groups and user accounts are divided into categories. Select one or more categories from the drop-down.
Membership rule	Rule that controls how members of the dynamic group are determined in Microsoft Entra ID. The property is only shown for groups with the `DynamicMembership` group type.
Membership rule state	Processing state of the membership rule for a dynamic group. The property is only shown for groups with the `DynamicMembership` group type.
Description	Text field for additional explanation.

Related topics

- Microsoft Entra ID group inheritance based on categories on page 115
- Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID groups on page 239
- Assigning Microsoft Entra ID groups to Microsoft Entra ID administrator roles on page 244
- For more information about preparing groups for requesting through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Information about local Active Directory groups

The **Federation** tab shows information about the local Active Directory user account that is linked to the Microsoft Entra ID user account.

Assignments to Microsoft Entra ID groups that are synchronized with the local Active Directory are not allowed in One Identity Manager. These groups cannot be requested through the web portal. You can only manage these groups in your locally. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

Table 40: Local Active Directory group data

Property	Description
Synchronization with local Active Directory enabled	Specifies whether synchronization with a local Active Directory is enabled.
Last synchronization	Time of the last Microsoft Entra ID group synchronization with the local Active Directory.
SID of local group	Security ID of the local Active Directory group.

Related topics

- Displaying Active Directory groups for Microsoft Entra ID groups on page 242

Adding Microsoft Entra ID groups to Microsoft Entra ID groups

Use this task to add a group to another group. This means that the groups can be hierarchically structured.

To assign groups directly to a group as members

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign groups** category.
4. Select the **Has members** tab.
5. Assign child groups in **Add assignments**.

   TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

To remove an assignment

- Select the group and double-click ✓.

6. Save the changes.

To add a group as a member of other groups

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign groups** task.
4. Select the **Is member of** tab.
5. In the **Add assignments** pane, assign parent groups.

   TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

   To remove an assignment

   - Select the group and double-click ✓.

6. Save the changes.

Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID groups

This task only available for groups with the **Assignable to administrator roles** option enabled.

To assign administrator roles to a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign administrator roles** task.
4. In the **Add assignments** pane, assign administrator roles.

   TIP: In the **Remove assignments** pane, you can remove assigned administrator roles.

   To remove an assignment

   - Select the administrator role and double-click ✓.

5. Save the changes.

Related topics

- General main data for Microsoft Entra ID groups on page 236
- Assigning Microsoft Entra ID groups to Microsoft Entra ID administrator roles on page 244

Assigning owners to Microsoft Entra ID groups

A group owner can edit group properties.

To assign owners to a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Assign owner** task.
4. Select the table containing the owner from the **Table** drop-down at the top of the form. You have the following options:
   - Microsoft Entra ID user accounts
5. In the **Add assignments** pane, assign owners.

   TIP: In the **Remove assignments** pane, you can remove assigned owners.

   To remove an assignment

   - Select the owner and double-click ✓.
6. Save the changes.

Assigning extended properties to Microsoft Entra ID groups

Extended properties are meta objects, such as operating codes, cost codes, or cost accounting areas that cannot be mapped directly in One Identity Manager.

For more information about using extended properties, see the *One Identity Manager Compliance Rules Administration Guide*.

To specify extended properties for a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select **Assign extended properties**.

4. In the **Add assignments** pane, assign extended properties.

   TIP: In the **Remove assignments** pane, you can remove assigned extended properties.

   To remove an assignment

   - Select the extended property and double-click .

5. Save the changes.

Deleting Microsoft Entra ID groups

Groups are deleted permanently from the One Identity Manager database and from Microsoft Entra ID.

To delete a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Click in the result list.
4. Confirm the security prompt with **Yes**.

Displaying the Microsoft Entra ID group overview

Use this task to obtain an overview of the most important information about a group.

To obtain an overview of a group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Microsoft Entra ID group overview** task.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

Displaying Active Directory groups for Microsoft Entra ID groups

The Active Directory group linked to a Microsoft Entra ID group is displayed on the overview form.

To display the Active Directory group for a Microsoft Entra ID group

1. In the Manager, select the **Microsoft Entra ID > Groups** category.
2. Select the group in the result list.
3. Select the **Microsoft Entra ID group overview** task.

   The **Active Directory group** form element shows which group is linked to it.

For more information about Active Directory, see the *One Identity Manager Administration Guide for Connecting to Active Directory*.

Related topics

- Information about local Active Directory groups on page 238

Microsoft Entra ID administrator roles

By using Microsoft Entra ID administrator roles, you can assign administrative permissions to users. Microsoft Entra ID recognizes several administrator roles that fulfill different functions. For more information about administrator roles, see the *Microsoft Entra ID documentation* from Microsoft.

One Identity Manager administrator roles are loaded into Microsoft Entra ID by synchronization. You can edit individual main data of Microsoft Entra ID administrator roles but you cannot create new Microsoft Entra ID administrator roles in One Identity Manager.

To add users to Microsoft Entra ID administrator roles, assign the Microsoft Entra ID administrator roles directly to the user. This may be Microsoft Entra ID administrator role assignments to departments, cost centers, locations, business roles, or the IT Shop.

Related topics

- Managing Microsoft Entra ID administrator roles assignments on page 119
- Editing main data of Microsoft Entra ID administrator roles on page 243
- Assigning Microsoft Entra ID groups to Microsoft Entra ID administrator roles on page 244
- Assigning extended properties to Microsoft Entra ID administrator roles on page 245

- Displaying the Microsoft Entra ID administration role overview on page 245
- Synchronizing single objects on page 54

Editing main data of Microsoft Entra ID administrator roles

One Identity Manager administrator roles are loaded into Microsoft Entra ID by synchronization. You can edit individual main data of Microsoft Entra ID administrator roles but you cannot create new Microsoft Entra ID administrator roles in One Identity Manager.

To edit the main data of a Microsoft Entra ID administrator role

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Change main data** task.
4. Edit the administrator role's main data.
5. Save the changes.

Table 41: Microsoft Entra ID administrator role main data

Property	Description
Display name	The display name is used to display the administrator role in the One Identity Manager tools' user interface.
Tenant	The administrator role's Microsoft Entra ID tenant.
Template ID.	ID of the administrator role template on which this administrator role was based.
IT Shop	Specifies whether the administrator role can be requested through the IT Shop. The administrator role can be requested by your employees through the Web Portal and allocated by defined approval processes. The administrator role can still be assigned directly to user accounts and hierarchical roles.
Only for use in IT Shop	Specifies whether the administrator role can only be requested through the IT Shop. The administrator role can be requested by your employees through the Web Portal and allocated by defined approval processes. You cannot assign an administrator role directly to a hierarchical role.
Service item	Specifies a service item for requesting the administrator role through the IT Shop.
Risk index	Value for assessing the risk of assigning administrator roles to user accounts. Set a value in the range **0** to **1**. This input field is only visible if the **QER | CalculateRiskIndex** configuration parameter is set.

Property	Description
	For more information about risk assessment, see the *One Identity Manager Risk Assessment Administration Guide*.
Category	Categories for inheriting administrator roles. Administrator roles can be selectively inherited by user accounts. To do this, administrator roles and user accounts are divided into categories. Use the drop-down to allocate one or more categories to the administrator role.
Description	Text field for additional explanation.

Related topics

- Microsoft Entra ID administrator role inheritance based on categories on page 130
- For more information about preparing administrator roles for requesting through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Assigning Microsoft Entra ID groups to Microsoft Entra ID administrator roles

Groups can only be assigned if their **Assignable to administrator roles** option is enabled.

To assign groups to an administrator role

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select **Assign groups** category.
4. In the **Add assignments** pane, assign the groups.

   TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

   To remove an assignment

   - Select the group and double-click ✓.
5. Save the changes.

Related topics

- General main data for Microsoft Entra ID groups on page 236
- Assigning Microsoft Entra ID administrator roles to Microsoft Entra ID groups on page 239

Assigning extended properties to Microsoft Entra ID administrator roles

Extended properties are meta objects, such as operating codes, cost codes, or cost accounting areas that cannot be mapped directly in One Identity Manager.

For more information about using extended properties, see the *One Identity Manager Compliance Rules Administration Guide*.

To specify extended properties for an administrator role

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Assign extended properties** task.
4. In the **Add assignments** pane, assign extended properties.

   TIP: In the **Remove assignments** pane, you can remove assigned extended properties.

   To remove an assignment

   - Select the extended property and double-click ✓.
5. Save the changes.

Displaying the Microsoft Entra ID administration role overview

Use this task to obtain an overview of the most important information about an administrator role.

To obtain an overview of a administration role

1. In the Manager, select the **Microsoft Entra ID > Administrator roles** category.
2. Select the administrator role in the result list.
3. Select the **Microsoft Entra ID administrator role overview** task.

Microsoft Entra ID administrative units

Microsoft Entra ID allows you to create and manage administrative units. Administrative units allow you to control and restrict role permissions in areas of your organization that you define. You can use an administrative unit to manage devices, users, and groups.

For more information, see Creating and editing Microsoft Entra ID user accounts and Editing main data of Microsoft Entra ID groups.

To create an administrative unit

1. In the Manager, select the **Microsoft Entra ID > Administrative units** category.
2. Click in the result list.
3. On the main data form, edit the main data of the administrative unit.
4. Save the changes.

To edit main data of administrative units

1. In the Manager, select the **Microsoft Entra ID > Administrative units** category.
2. Select the **Change main data** task.
3. On the main data form, edit the main data of the administrative unit.
4. Save the changes.

Related topics

- Editing main data of administrative units on page 246
- Assigning user accounts to administrative units on page 247
- Assigning groups to administrative units on page 247

Editing main data of administrative units

Enter the following general main data.

Table 42: General main data

Property	Description
Display name	The display name is used to display the administrative unit in the One Identity Manager tools' user interface.
Client	Microsoft Entra ID tenant of the administrative unit.
Owner (Application Role)	Application roles whose members can configure the administrative unit.
Visibility	Specifies which users can see the group.
Description	Text field for additional explanation.

Related topics

- Microsoft Entra ID administrative units on page 245
- Assigning groups to administrative units on page 247
- Assigning user accounts to administrative units on page 247

Assigning user accounts to administrative units

You can assign user accounts to administrative units and use an administrative unit to manage these user accounts' role eligibilities.

To assign user accounts directly to an administrative unit

1. In the Manager, select the **Microsoft Entra ID > Administrative units** category.
2. Select the administrative unit in the result list.
3. In the **Add assignments** pane, assign the user accounts.

   TIP: In the **Remove assignments** pane, you can remove assigned user accounts.

   To remove an assignment

   - Select the user account and double-click ✓.
4. Save the changes.

Related topics

- Microsoft Entra ID administrative units on page 245
- Account definitions for Microsoft Entra ID user accounts on page 64
- Assigning groups to administrative units on page 247

Assigning groups to administrative units

You can assign groups to administrative units and use an administrative unit to manage these groups' role permissions.

To assign groups directly to an administrative unit

1. In the Manager, select the **Microsoft Entra ID > Administrative units** category.
2. Select the administrative unit in the result list.
3. In the **Add assignments** pane, assign the groups.

TIP: In the **Remove assignments** pane, you can remove the assignment of groups.

To remove an assignment

- Select the group and double-click ✓.

4. Save the changes.

Detailed information about this topic

- Microsoft Entra ID administrative units on page 245
- Microsoft Entra ID groups on page 233
- General main data for Microsoft Entra ID groups on page 236

Microsoft Entra ID subscriptions and Microsoft Entra ID service principals

Information about Microsoft Entra ID subscriptions and Microsoft Entra ID service plans within a Microsoft Entra ID tenant is loaded into One Identity Manager during synchronization. In One Identity Manager, you cannot create new Microsoft Entra ID subscriptions or Microsoft Entra ID service plans. However, in One Identity Manager, you can edit certain main data of requesting the Microsoft Entra ID subscription in the IT Shop and for user account assignments.

NOTE: A Microsoft Entra ID user account can also obtain Microsoft Entra ID subscriptions through its Microsoft Entra ID groups. You cannot edit assignments by Microsoft Entra ID groups in One Identity Manager.

Related topics

- Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131
- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Editing Microsoft Entra ID subscription main data on page 249
- Assigning additional properties to Microsoft Entra ID subscriptions on page 250
- Displaying the Microsoft Entra ID subscriptions and service plan overview on page 251
- Synchronizing single objects on page 54
- Disabled Microsoft Entra ID service plans on page 251

Editing Microsoft Entra ID subscription main data

To edit Microsoft Entra ID subscription main data

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select the **Change main data** task.
4. Edit the Microsoft Entra ID subscription's main data.
5. Save the changes.

Table 43: Microsoft Entra ID subscription main data

Property	Description
SKU display name	The SKU display name of the Microsoft Entra ID subscription. For example, AAD_Premium or RMSBASIC.
Tenant	Tenant given for this Microsoft Entra ID subscription.
Subscription status	The Microsoft Entra ID subscription status, such as `enabled` (active).
Purchased licenses	The number of licenses purchased.
Assigned licenses	Number of actively used licenses.
Suspended licenses	Number of suspended licenses.
Warning units	Number of licenses with a warn status.
IT Shop	Specifies whether the Microsoft Entra ID subscription can be requested through the IT Shop. This Microsoft Entra ID subscription can be requested by your employees through the Web Portal and allocated by defined approval processes. The Microsoft Entra ID subscription can still be assigned directly to user accounts and hierarchical roles.
Only for use in IT Shop	Specifies whether the Microsoft Entra ID subscription can only be requested through the IT Shop. This Microsoft Entra ID subscription can be requested by your employees through the Web Portal and allocated by defined approval processes. The Microsoft Entra ID subscription may not be assigned directly to hierarchical roles.
Service item	Service item data for requesting the Microsoft Entra ID subscription through the IT Shop.
Risk index	Value for evaluating the risk of assigning the Microsoft Entra ID

Property	Description
	subscription to Microsoft Entra ID user accounts. Set a value in the range `0` to `1`. This field is only visible if the **QER | CalculateRiskIndex** configuration parameter is set. For more information about risk assessment, see the *One Identity Manager Risk Assessment Administration Guide.*
Category	Category for Microsoft Entra ID subscription inheritance. Microsoft Entra ID subscriptions can be selectively inherited by Microsoft Entra ID user accounts. To do this, the Microsoft Entra ID subscriptions the Microsoft Entra ID user accounts are divided into categories. Use this drop-down to allocate one or more categories to the Microsoft Entra ID subscription.

Related topics

- Microsoft Entra ID group inheritance based on categories on page 115
- For more information about preparing subscriptions for requesting through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Assigning additional properties to Microsoft Entra ID subscriptions

Extended properties are meta objects, such as operating codes, cost codes, or cost accounting areas that cannot be mapped directly in One Identity Manager.

For more information about using extended properties, see the *One Identity Manager Compliance Rules Administration Guide*.

To specify extended properties for a Microsoft Entra ID subscription

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select **Assign extended properties**.
4. In the **Add assignments** pane, assign extended properties.

   TIP: In the **Remove assignments** pane, you can remove assigned extended properties.

   To remove an assignment

   - Select the extended property and double-click ✓.
5. Save the changes.

Displaying the Microsoft Entra ID subscriptions and service plan overview

Use this task to obtain an overview of the most important information about a Microsoft Entra ID subscription and a service plan.

To obtain an overview of a Microsoft Entra ID subscription

1. In the Manager, select the **Microsoft Entra ID > Subscriptions** category.
2. Select a Microsoft Entra ID subscription in the result list.
3. Select the **Microsoft Entra ID subscription overview** task.

To obtain an overview of a Microsoft Entra ID service plan

1. In the Manager, select the **Microsoft Entra ID > Service plans** category.
2. Select the Microsoft Entra ID service plan in the result list.
3. Select the **Microsoft Entra ID service plan overview** task.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

Disabled Microsoft Entra ID service plans

To prevent users from using individual Microsoft Entra ID service plans, so-called "disabled service plans" are mapped in One Identity Manager. Disabled service plans are created automatically in One Identity Manager after synchronizing Microsoft Entra ID subscriptions. Disabled service plans are requested through the IT Shop or assigned to users through departments, cost centers, locations, business roles, or system roles.

> NOTE: A Microsoft Entra ID user accounts can also obtain disabled service plans through its Microsoft Entra ID groups. You cannot edit assignments by Microsoft Entra ID groups in One Identity Manager.

Related topics

- Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131
- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135
- Editing main data of disabled Microsoft Entra ID service plans on page 252

- Assigning extended properties to disabled Microsoft Entra ID service plans on page 253
- Displaying the disabled Microsoft Entra ID service plan overview on page 253
- Synchronizing single objects on page 54

Editing main data of disabled Microsoft Entra ID service plans

To edit disabled Microsoft Entra ID service plan main data

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the disabled service plan from the result list.
3. Select the **Change main data** task.
4. Edit the disabled service plan's main data.
5. Save the changes.

Table 44: Disabled service plan main data

Property	Description
Subscription	Name of the Microsoft Entra ID subscription.
Service plan	Name of the Microsoft Entra ID service plan.
IT Shop	Specifies whether the service plan can be requested through the IT Shop. The disabled service plan can be requested by your employees through the Web Portal and allocated by defined approval processes. The disabled service plan can still be assigned directly to hierarchical roles.
Only for use in IT Shop	Specifies whether the disabled service plan can only be requested through the IT Shop. The disabled service plan can be requested by your employees through the Web Portal and allocated by defined approval processes. The disabled service plan may not be assigned directly to hierarchical roles.
Service item	Service item data for requesting the disabled service plan through the IT Shop.
Category	Categories for disabled service plan inheritance. User accounts can selectively inherit disabled Microsoft Entra ID service plans. To do this, disabled service plans and Microsoft Entra ID user accounts are divided into categories. Use this drop-down to allocate one or more categories to the disabled service plan.

Related topics

- Microsoft Entra ID group inheritance based on categories on page 115
- For more information about preparing service plans for requesting through the IT Shop, see the *One Identity Manager IT Shop Administration Guide*.

Assigning extended properties to disabled Microsoft Entra ID service plans

Extended properties are meta objects, such as operating codes, cost codes, or cost accounting areas that cannot be mapped directly in One Identity Manager.

For more information about using extended properties, see the *One Identity Manager Compliance Rules Administration Guide*.

To specify extended properties for a disabled Microsoft Entra ID service plan

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the disabled service plan from the result list.
3. Select **Assign extended properties**.
4. In the **Add assignments** pane, assign extended properties.

   TIP: In the **Remove assignments** pane, you can remove assigned extended properties.

   To remove an assignment

   - Select the extended property and double-click ✓.
5. Save the changes.

Displaying the disabled Microsoft Entra ID service plan overview

Use this task to obtain an overview of the most important information about a disabled Microsoft Entra ID service plan.

To obtain an overview of a disabled Microsoft Entra ID service plan

1. In the Manager, select the **Microsoft Entra ID > Disabled service plans** category.
2. Select the disabled service plan from the result list.
3. Select the **Disabled Microsoft Entra ID service plan overview** task.

Related topics

- Displaying enabled and disabled Microsoft Entra ID service plans forMicrosoft Entra ID user accounts and Microsoft Entra ID groups on page 135

Microsoft Entra ID app registrations and Microsoft Entra ID service principals

When an application is registered in a Microsoft Entra ID tenant, it creates an associated Microsoft Entra ID service principal. There are so-called app roles defined for app registrations. Microsoft Entra ID users, Microsoft Entra ID groups, or Microsoft Entra ID service principals can use app roles to provide permissions or functions for the application.

For more information about integrating applications in Microsoft Entra ID, see the *Microsoft Entra ID documentation* from Microsoft.

Information about Microsoft Entra ID app registrations, Microsoft Entra ID service principals, and app roles within a Microsoft Entra ID tenant is loaded into One Identity Manager during synchronization.

If an Microsoft Entra ID application is used in a Microsoft Entra ID tenant that is registered in another Microsoft Entra ID tenant, only the Microsoft Entra ID service principal and not the Microsoft Entra ID app registration is loaded into One Identity Manager.

You cannot create new Microsoft Entra ID app registrations, Microsoft Entra ID service principals, and app roles in One Identity Manager but you can specify owners of app registrations and service principals and create or delete app roles in One Identity Manager.

Detailed information about this topic

- Displaying information about Microsoft Entra ID app registrations on page 255
- Assigning owners to Microsoft Entra ID app registrations on page 255
- Displaying Microsoft Entra ID app registration main data on page 256
- Displaying information about Microsoft Entra ID service principals on page 258
- Assigning owners to Microsoft Entra ID service principals on page 259
- Editing authorizations for Microsoft Entra ID service principals on page 259
- Displaying Microsoft Entra ID service principals for enterprise applications on page 260
- Displaying Microsoft Entra ID service principal main data on page 261

Displaying information about Microsoft Entra ID app registrations

The information about the Microsoft Entra ID app registration is loaded into One Identity Manager during synchronization. All the Microsoft Entra ID app registrations with their Microsoft Entra ID service principals that are registered in this Microsoft Entra ID tenant are loaded for a Microsoft Entra ID tenant.

If a Microsoft Entra ID application is used in an Microsoft Entra ID tenant that is registered in another Microsoft Entra ID tenant, only the Microsoft Entra ID service principal and not the Microsoft Entra ID app registration is loaded into One Identity Manager.

You cannot create Microsoft Entra ID app registrations in One Identity Manager.

To display information about a Microsoft Entra ID app registration

1. In the Manager, select the **Microsoft Entra ID > App registrations** category.
2. Select the Microsoft Entra ID app registration in the result list.
3. Select one of the following tasks:
    - **Microsoft Entra ID app registration overview**: This shows you an overview of the Microsoft Entra ID app registration and its dependencies.
    - **Change main data**: Shows the Microsoft Entra ID app registration's main data.
    - **Assign owners**: Shows the Microsoft Entra ID app registration's owners. You can assign owners to an app registration or remove them again.

Related topics

- Assigning owners to Microsoft Entra ID app registrations on page 255
- Displaying Microsoft Entra ID app registration main data on page 256
- Displaying information about Microsoft Entra ID service principals on page 258
- Displaying Microsoft Entra ID service principals for enterprise applications on page 260

Assigning owners to Microsoft Entra ID app registrations

Use this task to assign owners to a Microsoft Entra ID app registration or to remove them from an Microsoft Entra ID application. Microsoft Entra ID app registration owners can display and edit app registrations in Microsoft Entra ID.

To assign owners to a Microsoft Entra ID app registration

1. In the Manager, select the **Microsoft Entra ID > App registrations** category.
2. Select the Microsoft Entra ID app registration in the result list.
3. Select the **Assign owner** task.
4. In the **Table** drop-down, select the `Microsoft Entra ID user accounts (AADUser)` item.
5. In the **Add assignments** pane, assign owners.

   TIP: In the **Remove assignments** pane, you can remove assigned owners.

   To remove an assignment

   - Select the owner and double-click ✓.
6. Save the changes.

Displaying Microsoft Entra ID app registration main data

The information about the Microsoft Entra ID app registration is loaded into One Identity Manager during synchronization. You cannot edit Microsoft Entra ID app registration main data.

To display a Microsoft Entra ID app registration's main data

1. In the Manager, select the **Microsoft Entra ID > App registrations** category.
2. Select the Microsoft Entra ID app registration in the result list.
3. Select **Change main data**.

Table 45: Main data of a Microsoft Entra ID app registration

Property	Description
Display name	Display name of the application.
Client	Microsoft Entra ID tenant of the app registration.
Token issuance policies	Name of the policy for issuing tokens.
Token lifetime policy	Name of the policy for token lifetimes.
Owner (Application Role)	Application roles whose members can configure the app registration.
Registration	Date and time when the application was registered.

Property	Description
date	
Logo URL	Link to the application's logo.
Marketing URL	Link to the application's marketing page.
Privacy statement URL	Link to the application's privacy statement.
Service URL	Link to the application's support page.
Terms of service URL	Link to the application's terms of service.
Publisher domain	Name of the application's verified publisher domain.
Group membership claim	Group membership claim expected by the application. Group types that are included in the access, ID, and SAML tokens.
Supported user accounts	Specifies which Microsoft user accounts for the current application are supported. Permitted values are: • Accounts in this organizational directory only • Accounts in any organizational directory • Accounts in any organizational directory and personal Microsoft accounts • Only personal Microsoft accounts
Tags	User-defined string to use for categorizing and identifying the application.
Fallback public client	Specifies whether the fallback application type is a public client, such as an application installed and running on a mobile device. The default value is false meaning the fallback application type is a confidential client such as a web application. If the option is disabled, it means that the fallback application type is a confidential client, such as a web application (default).

Related topics

- Microsoft Entra ID policies for issuing tokens on page 212
- Microsoft Entra ID policies for token lifetime on page 213

Displaying information about Microsoft Entra ID service principals

When an application is registered in a Microsoft Entra ID tenant in Microsoft Azure Management Portal, it creates an associated Microsoft Entra ID service principal.

The information about the Microsoft Entra ID service principal is loaded into One Identity Manager during synchronization. You cannot create new Microsoft Entra ID service principals in One Identity Manager.

If a Microsoft Entra ID application is used in a Microsoft Entra ID tenant that is registered in another Microsoft Entra ID tenant, only the Microsoft Entra ID service principal and not the Microsoft Entra ID app registration is loaded into One Identity Manager.

To display information about a Microsoft Entra ID service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the Microsoft Entra ID service principal.
3. Select one of the following tasks:
   - **Microsoft Entra ID service principal overview**: This shows you an overview of the Microsoft Entra ID service principal and its dependencies.
   - **Change main data**: This displays the Microsoft Entra ID service principal's main data.
   - **Assign owners**: This displays the Microsoft Entra ID service principals owners. You can assign owners to a service principal or remove them.
   - **Assign authorizations**: This displays user accounts, groups, and service principals with their assigned app roles. You can create more authorizations or removed them.

Related topics

- Assigning owners to Microsoft Entra ID service principals on page 259
- Editing authorizations for Microsoft Entra ID service principals on page 259
- Displaying Microsoft Entra ID service principal main data on page 261
- Displaying information about Microsoft Entra ID app registrations on page 255
- Displaying Microsoft Entra ID service principals for enterprise applications on page 260

Assigning owners to Microsoft Entra ID service principals

Use this task to assign owners to a Microsoft Entra ID service principal or to remove them from a service principal.

To assign owners to a Microsoft Entra ID service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the Microsoft Entra ID service principal.
3. Select the **Assign owner** task.
4. In the **Table** drop-down, select one of the following entries.
   - `Microsoft Entra ID user accounts (AADUser)`
   - `Microsoft Entra ID service principals (AADServicePrincipal)`
5. In the **Add assignments** pane, assign owners.

   TIP: In the **Remove assignments** pane, you can remove assigned owners.

   To remove an assignment

   - Select the owner and double-click ✓.
6. Save the changes.

Editing authorizations for Microsoft Entra ID service principals

There are so-called app roles defined for app registrations. Microsoft Entra ID users, Microsoft Entra ID groups, or Microsoft Entra ID service principals can use app roles to provide permissions or functions for the application.

App roles and their assignments are loaded into One Identity Manager by synchronization. However, you cannot create new app roles in One Identity Manager. In One Identity Manager, you can add or remove authorizations for the service principals and their app registrations respectively.

To assign authorizations to a Microsoft Entra ID service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the Microsoft Entra ID service principal.
3. Select the **Assign authorizations** task.
4. In the **Assignments** pane, click **Add** and enter the following data.

- **Authorized for**: Specify the user account, group, or service principal for the authorization.
  a. Click → next to the field.
  b. Under **Table**, select one of the following tables:
     - To authorize a user account, select `AADUser`.
     - To authorize a group, select `AADGroup`.
     - To authorize a service principal, select `AADServicePrincipal`.
  c. Under **Authorized for**, select the user account, group, or service principal.
  d. Click **OK**.
- **App role**: Select the app role for the authorization.

  NOTE: If there is no app role defined for a service principal, leave this item empty to authorize the user account, group, or service principal.

5. Save the changes.

To remove authorizations from a Microsoft Entra ID service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the Microsoft Entra ID service principal.
3. Select the **Assign authorizations** task.
4. In the **Assignments** pane, select the authorization you want to remove.
5. Click **Remove**.
6. Save the changes.

Displaying Microsoft Entra ID service principals for enterprise applications

This task allows you to display the service principals that represent enterprise applications.

To display enterprise applications

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. Select one of the following entries:
   - **By type > Application > Enterprise applications**
   - **By type > Legacy > Enterprise applications**
3. In the result list, select the Microsoft Entra ID service principal.
4. Select one of the following tasks:

- **Microsoft Entra ID service principal overview**: This shows you an overview of the Microsoft Entra ID service principal and its dependencies.
- **Change main data**: This displays the Microsoft Entra ID service principal's main data.
- **Assign owners**: This displays the Microsoft Entra ID service principals owners. You can assign owners to a service principal or remove them.
- **Assign authorizations**: This displays user accounts, groups, and service principals with their assigned app roles. You can create more authorizations or removed them.

Related topics

- Displaying Microsoft Entra ID service principal main data on page 261
- Assigning owners to Microsoft Entra ID service principals on page 259
- Editing authorizations for Microsoft Entra ID service principals on page 259
- Displaying information about Microsoft Entra ID service principals on page 258

Displaying Microsoft Entra ID service principal main data

The information about the Microsoft Entra ID service principal is loaded into One Identity Manager during synchronization. You cannot edit Microsoft Entra ID service principal main data.

To display a Microsoft Entra ID service principal's main data

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the Microsoft Entra ID service principal.
3. Select the **Change main data** task.

Table 46: General main data for a Microsoft Entra ID service principal

Property	Description
Display name	Name for displaying the service principal.
Tenant	Microsoft Entra ID tenant of the service principal.
Home realm discovery policy	Name of the home realm discovery policy.
Owner (Application Role)	Application roles whose members can configure the service principal.

Property	Description
Delete date	Time at which the service principal was deleted.
Alternative names	Alternative names for the service principal. This is used to call service principals by subscription, to identify resource groups and full resource IDs for managing identities.
Web page	Home page of the Microsoft Entra ID application.
Application display name.	Display name of the associated Microsoft Entra ID application.
Logo URL	Link to the application's logo.
Marketing URL	Link to the application's marketing page.
Privacy statement URL	Link to the application's privacy statement.
Service URL	Link to the application's support page.
Terms of service URL	Link to the application's terms of service.
Login URL	URL that the identity provider uses to reroute the user to Microsoft Entra ID for authentication.
Logout URL	URL that the Microsoft authorization service uses to log out a user using OPENID Connect front channel, OpenID Connect back-channel, or SAML logout protocols.
Notification mail addresses	List of email addresses that Microsoft Entra ID sends a notification to if the active certificate is nearing the expiration date.
Preferred single sign-on mode	Single sign-on mode configured for this Microsoft Entra ID application.
Reply URLs	URLs that user tokens are sent to for logging in with the associated application, or the redirect URIs that OAuth 2.0 authorization codes and access tokens are sent to for the associated application.
Service principal names	Contains the list of URIs that identify the associated Microsoft Entra ID application within its Microsoft Entra ID tenant, or within a verified custom domain, if the Microsoft Entra ID application is an Microsoft Entra ID multi-tenant.
Service principal type	Type of service principal, for example, an application or a managed identity. The type is set internally by Microsoft Entra ID.

Property	Description
Encryption key ID	ID of the public key for logging in using certificates.
Tags	User-defined string to use for categorizing and identifying the application.
Enabled	Specifies whether the service principal is enabled.
App role assignment required	Specifies whether users or other service principals must be assigned an app role for this service principal before they can login or obtain application tokens.

Related topics

- Microsoft Entra ID policies for home realm discovery on page 212

Reports about Microsoft Entra ID objects

One Identity Manager makes various reports available containing information about the selected base object and its relations to other One Identity Manager database objects. The following reports are available for Microsoft Entra ID.

NOTE: Other sections may be available depending on the which modules are installed.

Table 47: Data quality target system report

Report	Published for	Description
Show overview	User account	This report shows an overview of the user account and the assigned permissions.
Show overview including origin	User account	This report shows an overview of the user account and origin of the assigned permissions.
Show overview including history	User account	This report shows an overview of the user accounts including its history. Select the end date for displaying the history (**Min. date**). Older changes and assignments that were removed before this date, are not shown in the report.
License overview	User account	The report contains a summary of assigned and effective subscriptions and service plans for a user account.
License overview	Subscription	The report shows an overview of a subscription

Report	Published for	Description
		license. It shows to which groups and user accounts the subscription is assigned and which service plans effectively apply to the groups and the user accounts.
Overview of all assignments	group Subscription Administrator role	This report finds all roles containing identities who have the selected system entitlement.
Show overview	group	This report shows an overview of the system entitlement and its assignments.
Show overview including origin	group	This report shows an overview of the system entitlement and origin of the assigned user accounts.
Show overview including history	group	This report shows an overview of the system entitlement and including its history. Select the end date for displaying the history (**Min. date**). Older changes and assignments that were removed before this date, are not shown in the report.
Show entitlement drifts	Tenant	This report shows all system entitlements that are the result of manual operations in the target system rather than provisioned by One Identity Manager.
Show user accounts overview (incl. history)	Tenant	This report returns all the user accounts with their permissions including a history. Select the end date for displaying the history (**Min. date**). Older changes and assignments that were removed before this date, are not shown in the report.
Show user accounts with an above average number of system entitle-ments	Tenant	This report contains all user accounts with an above average number of system entitlements.
Show identities with multiple user accounts	Tenant	This report shows all the identities that have multiple user accounts. The report contains a risk assessment.
Show system entitlements	Tenant	This report shows the system entitlements with the assigned user accounts including a history.

Report	Published for	Description
overview (incl. history)		Select the end date for displaying the history (**Min. date**). Older changes and assignments that were removed before this date, are not shown in the report.
Overview of all assignments	Tenant	This report finds all roles containing identities with at least one user account in the selected target system.
Show unused user accounts	Tenant	This report contains all user accounts, which have not been used in the last few months.
Show orphaned user accounts	Tenant	This report shows all user accounts to which no identity is assigned.

Table 48: Additional reports for the target system

Report	Description
Microsoft Entra ID user account and group administration	This report contains a summary of user account and group distribution in all tenants. You can find this report in the **My One Identity Manager** category.
Data quality summary for Microsoft Entra ID user accounts	This report contains different evaluations of user account data quality in all tenants. You can find this report in the **My One Identity Manager** category.

Managing Microsoft Entra ID security attributes

Assign Microsoft Entra ID security attributes to Microsoft Entra ID user accounts and service principals, to allow access to resources to be regulated.

In One Identity Manager, you can assigned Microsoft Entra ID security attributes directly to user accounts and service principals.

Assigning Microsoft Entra ID security attributes to user accounts

To assign a security attribute to a user account

1. In the Manager, select the **Microsoft Entra ID > User accounts** category.
2. Select the user account in the result list.

3. Select the **Assign security attributes** task.
4. Click **Add** and enter the following information:
   - **Microsoft Entra ID security attribute**: In the result list, select the Microsoft Entra ID security attribute.
   - **Value(s)**: Specify whether the value is true or false.

   TIP: In the **Remove** section, remove the assignment of security attributes.

   To remove an assignment

   - Select the security attribute and click **Remove**.
5. Save the changes.

Assigning Microsoft Entra ID security attributes to service principals

To assign a security attribute to a service principal

1. In the Manager, select the **Microsoft Entra ID > Service principals** category.
2. In the result list, select the service principal.
3. Select the **Assign security attributes** task.
4. Click **Add** and enter the following information:
   - **Microsoft Entra ID security attribute**: In the result list, select the Microsoft Entra ID security attribute.
   - **Value(s)**: Specify whether the value is true or false.

   TIP: In the **Remove** section, remove the assignment of security attributes.

   To remove an assignment

   - Select the security attribute and click **Remove**.
5. Save the changes.

10

Handling of Microsoft Entra ID objects in the Web Portal

One Identity Manager enables its users to perform various tasks simply using a Web Portal.

- Managing user accounts and identities

  An account definition can be requested by shop customers in the Web Portal if it is assigned to an IT Shop shelf. The request undergoes a defined approval process. The user account is not created until it has been agreed by an authorized identity, such as a manager.

- Managing assignments of groups, administrator roles, subscriptions, and disabled service plans

  > NOTE: Assignment of Microsoft Entra ID role eligibilities and role assignments is disabled by default. To make these products available on an IT Shop shelf, you must enable them in the Manager. After enabling, the products are only available for user accounts with an RBAC or PIM license.

  In the Web Portal, by assigning groups, administrator roles, subscriptions, and disabled service plans to an IT Shop shelf, you can request these products from shop customers. The request undergoes a defined approval process. The group, administrator role, subscription, or disabled service plan is not assigned until it has been approved by an authorized identity.

  In the IT Shop, the following selves are available: **Identity & Access Lifecycle > Microsoft Entra ID groups**, **Identity & Access Lifecycle > Microsoft Entra ID subscriptions**, and **Identity & Access Lifecycle > Disabled Microsoft Entra ID service plans**.

  In the Web Portal, managers and administrators of organizations can assign groups, administrator roles, subscriptions, and disabled service plans to the departments, cost centers, or locations for which they are responsible. The groups, administrator roles, subscriptions, and disabled service plans are inherited by all identities who are members of these departments, cost centers, or locations.

  If the Business Roles Module is available, in the Web Portal, managers and administrators of business roles can assign groups, administrator roles, subscriptions, and disabled service plans to the business roles for which they are

responsible. The groups, administrator roles, subscriptions, and disabled service plans are inherited by all identities who are members of these business roles.

If the System Roles Module is available, in the Web Portal, supervisors of system roles can assign groups, administrator roles, subscriptions, and disabled service plans to the system roles. The groups, administrator roles, subscriptions, and disabled service plans are inherited by all identities that have these system roles assigned to them.

- Attestation

  If the Attestation Module is available, the correctness of the properties of target system objects and of entitlement assignments can be verified on request. To enable this, attestation policies are configured in the Manager. The attestors use the Web Portal to approve attestation cases.

- Governance administration

  If the Compliance Rules Module is available, you can define rules that identify the invalid entitlement assignments and evaluate their risks. The rules are checked regularly, and if changes are made to the objects in One Identity Manager. Compliance rules are defined in the Manager. Supervisors use the Web Portal to check rule violations and to grant exception approvals.

  If the Company Policies Module is available, company policies can be defined for the target system objects mapped in One Identity Manager and their risks evaluated. Company policies are defined in the Manager. Supervisors use the Web Portal to check policy violations and to grant exception approvals.

- Risk assessment

  You can use the risk index of groups, administrator roles, and subscriptions to evaluate the risk of entitlement assignments for the company.One Identity Manager provides default calculation functions for this. The calculation functions can be modified in the Web Portal.

- Reports and statistics

  The Web Portal provides a range of reports and statistics about the identities, user accounts, and their entitlements and risks.

For more information about the named topics, see Managing Microsoft Entra ID user accounts and identities on page 63, Managing memberships in Microsoft Entra ID groups on page 101, Managing Microsoft Entra ID administrator roles assignments on page 119, Managing Microsoft Entra ID subscription and Microsoft Entra ID service plan assignments on page 131 and in refer to the following guides:

- *One Identity Manager Web Portal User Guide*
- *One Identity Manager Attestation Administration Guide*
- *One Identity Manager Compliance Rules Administration Guide*
- *One Identity Manager Company Policies Administration Guide*
- *One Identity Manager Risk Assessment Administration Guide*

Recommendations for federations

NOTE: The following modules must be installed to support federations in One Identity Manager:

- Active Directory Module
- Microsoft Entra ID Module

In a federation, the local Active Directory user accounts are connected to Microsoft Entra ID user accounts. The connection is established by using the `ms-ds-consistencyGUID` property in the Active Directory user account and the `immutable` property in the Microsoft Entra ID user account. Synchronization of Active Directory and Microsoft Entra ID user accounts is carried out in the federation by Microsoft Entra Connect. For more information about Microsoft Entra Connect, see the *Microsoft Entra ID documentation* from Microsoft.

One Identity Manager maps the connection using the Active Directory user account's Microsoft Entra Connect anchor ID (`ADSAccount.MSDsConsistencyGuid`) and the Microsoft Entra ID user account's immutable identifier (`AADUser.OnPremImmutableId`).

Some of the target system relevant properties of Microsoft Entra ID user accounts that are linked to local Active Directory user account cannot be changed in One Identity Manager. However, assignment of permissions to Microsoft Entra ID user accounts in One Identity Manager is possible.

Assignments to Microsoft Entra ID groups that are synchronized with the local Active Directory are not allowed in One Identity Manager. These groups cannot be requested through the web portal. You can only manage these groups in your locally. For more information, see the *Microsoft Entra ID documentation* from Microsoft.

The One Identity Manager supports the following scenarios for federations.

Scenario 1

1. Active Directory user accounts are created in One Identity Manager and provisioned the local Active Directory environment.
2. Microsoft Entra Connect creates the Microsoft Entra ID user accounts in the Microsoft Entra ID tenant.
3. Microsoft Entra ID synchronization loads the Microsoft Entra ID user accounts in to One Identity Manager.

This is the recommended procedure. Creating a Microsoft Entra ID user account via Microsoft Entra Connect and then importing it into One Identity Manager usually takes some time. Microsoft Entra ID user accounts are not immediately available in One Identity Manager.

Scenario 2

1. Active Directory user accounts and Microsoft Entra ID user accounts are created in One Identity Manager.

   In this case, the connection is established by using the `ADSAccount.MSDsConsistencyGuid` and `AADUser.OnPremImmutableId` columns. This can be carried using custom scripts or custom templates.
2. Active Directory and Microsoft Entra ID user accounts are provisioned independently in their own target systems.
3. Microsoft Entra Connect detects the connection between the user accounts, establishes the connection in the federation and updates the required properties.
4. The next Microsoft Entra ID synchronization updates the Microsoft Entra ID user accounts in One Identity Manager.

With this scenario, the Microsoft Entra ID user accounts are immediately available in One Identity Manager and can be issued their permissions.

NOTE:

- If you work with account definitions, it is recommended you enter the account definition for Active Directory as a required account definition in the account definition for Microsoft Entra ID.
- If you work with account definitions, it is recommended you select the **`Only initially`** value for the **IT operating data overwrites** property in the manage level. Then the data is only determined in the initial case.
- Do not post-process Microsoft Entra ID user accounts using templates because certain target system relevant properties cannot be edited and the following errors may occur:

  ```
  [Exception]: ServiceException occured

  Code: Request_BadRequest

  Message: Unable to update the specified properties for on-
  premises mastered Directory Sync objects or objects currently
  undergoing migration.

  [ServiceException]: Code: Request_BadRequest - Message: Unable to
  update the specified properties for on-premises mastered Directory
  Sync objects or objects currently undergoing migration.
  ```

Related topics

- Information about the local Active Directory user account on page 225
- Account definitions for Microsoft Entra ID user accounts on page 64
- Main data for an account definition on page 66

12

Basic data for managing a Microsoft Entra ID environment

To manage a Microsoft Entra ID environment in One Identity Manager, the following basic data is relevant.

- Target system types

  Target system types are required for configuring target system comparisons. Tables with outstanding objects are maintained with the target system types and settings are configured for provisioning memberships and single objects synchronization. Target system types also map objects in the Unified Namespace.

  For more information, see Post-processing outstanding objects on page 56.

- Target system managers

  A default application role exists for the target system manager in One Identity Manager. Assign identities to this application role who have permission to edit all tenants in One Identity Manager.

  Define additional application roles if you want to limit the permissions for target system managers to individual tenants. The application roles must be added under the default application role.

  For more information, see Target system managers for Microsoft Entra ID on page 273.

- Servers

  Servers must be informed of your server functionality in order to handle Microsoft Entra ID-specific processes in One Identity Manager. For example, the synchronization server.

  For more information about editing Job servers for Microsoft Entra ID components, see the *One Identity Manager Administration Guide for Connecting to Microsoft Entra ID*.

Target system managers for Microsoft Entra ID

A default application role exists for the target system manager in One Identity Manager. Assign identities to this application role who have permission to edit all tenants in One Identity Manager.

Define additional application roles if you want to limit the permissions for target system managers to individual tenants. The application roles must be added under the default application role.

For more information about implementing and editing application roles, see the *One Identity Manager Authorization and Authentication Guide*.

Implementing application roles for target system managers

1. The One Identity Manager administrator allocates identities to be target system administrators.
2. These target system administrators add identities to the default application role for target system managers.

   Target system managers with the default application role are authorized to edit all the tenants in One Identity Manager.
3. Target system managers can authorize other identities within their area of responsibility as target system managers and if necessary, create additional child application roles and assign these to individual tenants.

   NOTE: If no identities are assigned to a child application role for target system administrators, the identities of the parent application role are granted the permissions.

Table 49: Default application roles for target system managers

User	Tasks
Target system managers	Target system managers must be assigned to the **Target systems | Microsoft Entra ID** application role or a child application role. Users with this application role: • Assume administrative tasks for the target system. • Create, change, or delete target system objects. • Edit password policies for the target system. • Prepare groups to add to the IT Shop. • Can add identities that do not have the **Primary identity** identity type. • Configure synchronization in the Synchronization Editor and define

User	Tasks
	the mapping for comparing target systems and One Identity Manager. • Edit the synchronization's target system types and outstanding objects. • Authorize other identities within their area of responsibility as target system managers and create child application roles if required.

To initially specify identities to be target system administrators

1. Log in to the Manager as a One Identity Manager administrator (**Base role | Administrators** application role)
2. Select the **One Identity Manager Administration > Target systems > Administrators** category.
3. Select the **Assign identities** task.
4. Assign the identity and save the changes.

To add the first identities to the default application as target system managers

1. Log in to the Manager as a target system administrator (**Target systems | Administrators** application role).
2. Select the **One Identity Manager Administration > Target systems > Microsoft Entra ID** category.
3. Select the **Assign identities** task.
4. Assign the identities you want and save the changes.

To authorize other identities as target system managers when you are a target system manager

1. Log in to the Manager as a target system manager.
2. Select the application role in the **Microsoft Entra ID > Basic configuration data > Target system managers** category.
3. Select the **Assign identities** task.
4. Assign the identities you want and save the changes.

To specify target system managers for individual tenants

1. Log in to the Manager as a target system manager.
2. Select the **Microsoft Entra ID > Tenants** category.
3. Select the tenant in the result list.
4. Select the **Change main data** task.
5. On the **General** tab, select the application role in the **Target system manager** drop-down.

- OR -

Next to the **Target system manager** drop-down, click to create a new application role.

a. Enter the application role name and assign the **Target systems | Microsoft Entra ID** parent application role.

b. Click **OK** to add the new application role.

6. Save the changes.
7. Assign identities to this application role who are permitted to edit the tenant in One Identity Manager.

Related topics

- One Identity Manager users for managing Microsoft Entra ID on page 13
- Microsoft Entra ID tenant on page 206

Job server for Microsoft Entra ID-specific process handling

Servers must be informed of their server functionality in order to handle Microsoft Entra ID-specific processes in One Identity Manager. For example, the synchronization server.

You have several options for defining a server's functionality:

- In the Designer, create an entry for the Job server in the **Base Data > Installation > Job server** category. For more information about this, see the *One Identity Manager Configuration Guide*.
- In the Manager, select an entry for the Job server in the **Microsoft Entra ID > Basic configuration data > Server** category and edit the Job server main data.

  Use this task if the Job server has already been declared in One Identity Manager and you want to configure special functions for the Job server.

NOTE: One Identity Manager must be installed, configured, and started in order for a server to perform its function in the One Identity Manager Service network. Proceed as described in the *One Identity Manager Installation Guide*.

To edit a Job server and its functions

1. In the Manager, select the **Microsoft Entra ID > Basic configuration data > Server** category.
2. Select the Job server entry in the result list.
3. Select the **Change main data** task.
4. Edit the Job server's main data.

5. Select the **Assign server functions** task and specify server functionality.
6. Save the changes.

Detailed information about this topic

- General main data of Job servers on page 276
- Specifying server functions on page 278

General main data of Job servers

NOTE: All editing options are also available in the Designer under **Base Data > Installation > Job server**.

The following properties are displayed for Job servers.

NOTE: More properties may be available depending on which modules are installed.

Table 50: Job server properties

Property	Meaning
Server	Job server name.
Full server name	Full server name in accordance with DNS syntax. Syntax: `<Name of servers>.<Fully qualified domain name>`
Target system	Computer account target system.
Language	Language of the server.
Server is cluster	Specifies whether the server maps a cluster.
Server belongs to cluster	Cluster to which the server belongs. NOTE: The **Server is cluster** and **Server belongs to cluster** properties are mutually exclusive.
IP address (IPv6)	Internet protocol version 6 (IPv6) server address.
IP address (IPv4)	Internet protocol version 4 (IPv4) server address.
Copy process (source server)	Permitted copying methods that can be used when this server is the source of a copy action. At present, only copy methods that support the `Robocopy` and `rsync` programs are supported. If no method is given, the One Identity Manager Service determines the operating system of the server during runtime. Replication is then performed with the `Robocopy` program

Property	Meaning
	between servers with a Windows operating system or with the `rsync` program between servers with a Linux operating system. If the operating systems of the source and destination servers differ, it is important that the right copy method is applied for successful replication. A copy method is chosen that supports both servers.
Copy process (target server)	Permitted copying methods that can be used when this server is the destination of a copy action.
Coding	Character set coding that is used to write files to the server.
Parent Job server	Name of the parent Job server.
Executing server	Name of the executing server. The name of the server that exists physically and where the processes are handled. This input is evaluated when the One Identity Manager Service is automatically updated. If the server is handling several queues, the process steps are not supplied until all the queues that are being processed on the same server have completed their automatic update.
Queue	Name of the queue to handle the process steps. The process steps are requested by the Job queue using this queue identifier. The queue identifier is entered in the One Identity Manager Service configuration file.
Server operating system	Operating system of the server. This input is required to resolve the path name for replicating software profiles. The values **`Win32`**, **`Windows`**, **`Linux`**, and **`Unix`** are permitted. If no value is specified, **`Win32`** is used.
Service account data	One Identity Manager Service user account information. In order to replicate between non-trusted systems (non-trusted domains, Linux server), the One Identity Manager Service user information has to be declared for the servers in the database. This means that the service account, the service account domain, and the service account password have to be entered for the server.
One Identity Manager Service installed	Specifies whether a One Identity Manager Service is installed on this server. This option is enabled by the `QBM_PJobQueueLoad` procedure the moment the queue is called for the first time.

Property	Meaning
	The option is not automatically removed. If necessary, you can reset this option manually for servers whose queue is no longer enabled.
Stop One Identity Manager Service	Specifies whether the One Identity Manager Service has stopped. If this option is set for the Job server, the One Identity Manager Service does not process any more tasks. You can make the service start and stop with the appropriate administrative permissions in the Job Queue Info program. For more information, see the *One Identity Manager Process Monitoring and Troubleshooting Guide*.
Paused due to unavailability of a target system	Specifies whether task processing for this queue has been stopped because the target system that uses this Job server as a synchronization server is temporarily unavailable. As soon as the target system is available again, processing starts and all outstanding tasks are performed. For more information about offline mode, see the *One Identity Manager Target System Synchronization Reference Guide*.
No automatic software update	Specifies whether to exclude the server from automatic software updating. NOTE: Servers must be manually updated if this option is set.
Software update running	Specifies whether a software update is currently running.
Server function	Server functionality in One Identity Manager. One Identity Manager processes are handled with respect to the server function.

Related topics

- Specifying server functions on page 278

Specifying server functions

NOTE: All editing options are also available in the Designer under **Base Data > Installation > Job server**.

The server function defines the functionality of a server in One Identity Manager. One Identity Manager processes are handled with respect to the server function.

NOTE: More server functions may be available depending on which modules are installed.

Table 51: Permitted server functions

Server function	Remark
Microsoft Entra ID connector (via Microsoft Graph)	Server on which the Microsoft Entra ID connector is installed. This server synchronizes the Microsoft Entra ID target system.
CSV connector	Server on which the CSV connector for synchronization is installed.
Domain controller	The Active Directory domain controller. Servers that are not labeled as domain controllers are considered to be member servers.
Printer server	Server that acts as a print server.
Generic server	Server for generic synchronization with a custom target system.
Home server	Server for adding home directories for user accounts.
Update server	This server automatically updates the software on all the other servers. The server requires a direct connection to the database server that One Identity Manager database is installed on. It can run SQL tasks. The server with the One Identity Manager database installed on it is labeled with this functionality during initial installation of the schema.
SQL processing server	It can run SQL tasks. The server requires a direct connection to the database server that One Identity Manager database is installed on. Several SQL processing servers can be set up to spread the load of SQL processes. The system distributes the generated SQL processes throughout all the Job servers with this server function.
CSV script server	This server can process CSV files using the `ScriptComponent` process component.
Generic database connector	This server can connect to an ADO.Net database.
One Identity Manager database connector	Server on which the One Identity Manager connector is installed. This server synchronizes the One Identity Manager target system.
One Identity Manager Service installed	Server on which a One Identity Manager Service is installed.

Server function	Remark
Primary domain controller	Primary domain controller.
Profile server	Server for setting up profile directories for user accounts.
SAM synchronization Server	Server for synchronizing an SMB-based target system.
SMTP host	Server from which One Identity Manager Service sends email notifications. Prerequisite for sending mails using One Identity Manager Service is SMTP host configuration.
Default report server	Server on which reports are generated.
PowerShell connector	The server can run PowerShell.

Related topics

- General main data of Job servers on page 276

Appendix A

Troubleshooting

Possible errors when synchronizing an Microsoft Entra ID tenant

Issue

An error occurs when loading the Microsoft Entra ID user accounts:

```
[Exception]: ServiceException occured
Code: BadRequest
Message: Tenant does not have a SPO license.
[ServiceException]: Code: BadRequest - Message: Tenant does not have a SPO license.
```

Cause

A Microsoft Entra ID tenant is synchronized that does not have a license for the **SharePoint Online** service.

Possible solutions

- Ensure the Microsoft Entra ID tenant has a license that includes the **SharePoint Online** service. (Recommended)
- If you want to synchronize a Microsoft Entra ID tenant that does not have a license for the **SharePoint Online** service, change the synchronization project with the Synchronization Editor.

  In **Users** mapping, disable the property mapping rules for the following schema properties. To do this, set the mapping direction to the **`Do not map`**.

  - BirthDay
  - PreferedName

- Responsibilities
- Schools
- Skills
- PastProjects
- Interests
- HireDate
- EmployeeID
- AboutMe
- MySite
- ImAddresses
- FaxNumber
- OtherMails

For more information about editing property mapping rules in the Synchronization Editor, see the *One Identity Manager Target System Synchronization Reference Guide*.

Appendix B

Configuration parameters for managing a Microsoft Entra ID environment

The following configuration parameters are available in One Identity Manager after the module has been installed.

Table 52: Configuration parameters

Configuration parameter	Description
TargetSystem | AzureAD	Preprocessor relevant configuration parameter for controlling database model components for Microsoft Entra ID target system administration. If the parameter is set, the target system components are available. Changes to this parameter require the database to be recompiled. If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.
TargetSystem | AzureAD | Accounts	Allows configuration of user account data.
TargetSystem | AzureAD | Accounts | InitialRandomPassword	Specifies whether a random password is generated when a new user account is added. The password must contain at least those character sets that are defined in the password policy.
TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo	Identity to receive an email with the random generated password (manager cost center/department/location/role, identity's manager or `XUserInserted`). If no recipient can be found, the password is sent to the address stored in the **TargetSystem | AzureAD | DefaultAddress** configuration parameter.

Configuration parameter	Description
TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo | MailTemplateAccountName	Mail template name that is sent to supply users with the login credentials for the user account. The **Identity - new user account created** mail template is used.
TargetSystem | AzureAD | Accounts | InitialRandomPassword | SendTo | MailTemplatePassword	Mail template name that is sent to supply users with the initial password. The **Identity - initial password for new user account** mail template is used.
TargetSystem | AzureAD | Accounts | MailTemplateDefaultValues	Mail template used to send notifications about whether default IT operating data mapping values are used for automatically creating a user account. The **Identity - new user account with default properties created** mail template is used.
TargetSystem | AzureAD | Accounts | PrivilegedAccount	Allows configuration of privileged Microsoft Entra ID user account settings.
TargetSystem | AzureAD | Accounts | PrivilegedAccount | AccountName_Postfix	Postfix for formatting the login name of privileged user accounts.
TargetSystem | AzureAD | Accounts | PrivilegedAccount | AccountName_Prefix	Prefix for formatting a login name of privileged user accounts.
TargetSystem | AzureAD | DefaultAddress	Default email address of the recipient for notifications about actions in the target system.
TargetSystem | AzureAD | DeltaTokenDirectory	Directory where the delta token files for the delta synchronization are stored.
TargetSystem | AzureAD | MaxFullsyncDuration	Maximum runtime of a synchronization in minutes. No recalculation of group memberships by the DBQueue Processor can take place during this time. If the maximum runtime is exceeded, group membership are recalculated.
TargetSystem | AzureAD | PersonAutoDefault	Mode for automatic identity assignment for user accounts added to the database outside synchronization.

Configuration parameter	Description
TargetSystem | AzureAD | PersonAutoDisabledAccounts	Specifies whether identities are automatically assigned to disabled user accounts. User accounts are not given an account definition.
TargetSystem | AzureAD | PersonAutoFullSync	Mode for automatic identity assignment for user accounts that are added to or updated in the database by synchronization.
TargetSystem | AzureAD | PersonExcludeList	Listing of all user account without automatic identity assignment. Names are listed in a pipe (|) delimited list that is handled as a regular search pattern. Example: `ADMINISTRATOR|GUEST|KRBTGT|TSINTERNETUSER|IUSR_.*|IWAM_.*|SUPPORT_.*|.* | $`
TargetSystem | AzureAD | PersonUpdate	Specifies whether identities are updated if their user accounts are changed. This configuration parameter is set to allow ongoing update of identities from associated user accounts.
QER | ITShop | AutoPublish | AADGroup	Preprocessor relevant configuration parameter for automatically adding Microsoft Entra ID groups to the IT Shop. If the parameter is set, all groups are automatically assigned as products to the IT Shop. Changes to this parameter require the database to be recompiled. If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.
QER | ITShop | AutoPublish | AADGroup | ExcludeList	List of all Microsoft Entra ID groups that must not be automatically assigned to the IT Shop. Each entry is part of a regular search pattern and supports regular expression notation. Example: `.*Administrator.*|Exchange.*|.*Admins|.*Operators|IIS_IUSRS`
QER | ITShop | AutoPublish | AADSubSku	Preprocessor relevant configuration parameter for automatically adding Microsoft Entra ID subscriptions to the IT Shop. If the parameter is set, all subscriptions are automatically assigned as products to the IT Shop. Changes to this parameter require the database to be recompiled.

Configuration parameter	Description
	If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.
QER | ITShop | AutoPublish | AADSubSku | ExcludeList	List of all Microsoft Entra ID subscriptions that must not be automatically assigned to the IT Shop. Each entry is part of a regular search pattern and supports regular expression notation.
QER | ITShop | AutoPublish | AADDeniedServicePlan	Preprocessor relevant configuration parameter for automatically adding Microsoft Entra ID service plans to the IT Shop. If the parameter is set, all service plans are automatically assigned as products to the IT Shop. Changes to this parameter require the database to be recompiled. If you disable the configuration parameter at a later date, model components and scripts that are no longer required, are disabled. SQL procedures and triggers are still carried out. For more information about the behavior of preprocessor relevant configuration parameters and conditional compiling, see the *One Identity Manager Configuration Guide*.
QER | ITShop | AutoPublish | AADDeniedServicePlan | ExcludeList	List of all Microsoft Entra ID service plans that must not be automatically assigned to the IT Shop. Each entry is part of a regular search pattern and supports regular expression notation.

Appendix C

Default project template for Microsoft Entra ID

A default project template ensures that all required information is added in One Identity Manager. This includes mappings, workflows, and the synchronization base object. If you do not use a default project template you must declare the synchronization base object in One Identity Manager yourself.

Use a default project template for initially setting up the synchronization project. For custom implementations, you can extend the synchronization project with the Synchronization Editor.

Detailed information about this topic

- Project template for Microsoft Entra ID tenants on page 287
- Project template for Microsoft Entra ID B2C tenants on page 288

Project template for Microsoft Entra ID tenants

The project template uses mappings for the following schema types.

Table 53: Microsoft Entra ID schema type mapping

Schema type in Microsoft Entra ID	Table in the One Identity Manager Schema
DirectoryRole	AADDirectoryRole
Group	AADGroup
LicenseAssignments	AADUserHasSubSku
GroupLicenseAssignments	AADGroupHasSubSku

Schema type in Microsoft Entra ID	Table in the One Identity Manager Schema
Organization	AADOrganization
ServicePlanInfo	AADServicePlan
SubscribedSku	AADSubSku
User	AADUser
VerifiedDomain	AADVerifiedDomain
Application	AADApplication
AppRole	AADAppRole
AppRoleAssignment	AADAppRoleAssignment
ServicePrincipal	AADServicePrincipal
ActivityBasedTimeoutPolicy	AADActivityBasedTimeoutPolicy
HomeRealmDiscoveryPolicy	AADHomeRealmDiscoveryPolicy
TokenIssuancePolicy	AADTokenIssuancePolicy
TokenLifetimePolicy	AADTokenLifetimePolicy
AdministrativeUnit	AADAdministrativeUnit

Project template for Microsoft Entra ID B2C tenants

The project template uses mappings for the following schema types.

Table 54: Microsoft Entra ID schema type mapping

Schema type in Microsoft Entra ID	Table in the One Identity Manager Schema
AdministrativeUnit	AADAdministrativeUnit
ActivityBasedTimeoutPolicy	AADActivityBasedTimeoutPolicy
Application	AADApplication
AppRole	AADAppRole
AppRoleAssignment	AADAppRoleAssignment
DirectoryRole	AADDirectoryRole
Group	AADGroup

Schema type in Microsoft Entra ID	Table in the One Identity Manager Schema
GroupLicenseAssignments	AADGroupHasSubSku
HomeRealmDiscoveryPolicy	AADHomeRealmDiscoveryPolicy
Organization	AADOrganization
ServicePrincipal	AADServicePrincipal
TokenIssuancePolicy	AADTokenIssuancePolicy
TokenLifetimePolicy	AADTokenLifetimePolicy
User	AADUser
VerifiedDomain	AADVerifiedDomain

Appendix D

Editing Microsoft Entra ID system objects

The following table describes permitted editing methods of Microsoft Entra ID schema types and names restrictions required by system object processing.

Table 55: Methods available for editing schema types

Type	Read	Add	Delete	Refresh
Subscriptions (`SubscribedSku`)	Yes	No	No	No
Administrator roles (`DirectoryRole`)	Yes	No	No	Yes
User accounts (`User`)	Yes	Yes	Yes	Yes
Service plans (`ServicePlanInfo`)	Yes	No	No	No
Domains (`VerifiedDomain`)	Yes	No	No	No
Groups (`Group`)	Yes	Yes	Yes	Yes
License assignments to user accounts (`LicenseAssignments`)	Yes	Yes	Yes	Yes
License assignments to groups (`GroupLicenseAssignments`)	Yes	No	No	No
Tenants (`Organization`)	Yes	No	No	Yes
Applications (`Application`)	Yes	No	No	Yes
Service principals (`ServicePrincipal`)	Yes	No	No	Yes
App roles (`AppRole`)	Yes	No	No	No
Assignments to app roles (`AppRoleAssignment`)	Yes	Yes	Yes	Yes
Policies on activity-based timeout (`ActivityBasedTimeoutPolicy`)	Yes	No	No	No
Policies on home realm discovery (`HomeRealmDiscoveryPolicy`)	Yes	No	No	No

Type	Read	Add	Delete	Refresh
Policies on token issuance (`TokenIssuancePolicy`)	Yes	No	No	No
Policies on token lifetime (`TokenLifetimePolicy`)	Yes	No	No	No
Classifications (`AADGroupClassificationLbl`)	Yes	No	No	No
Administrative units (`AdministrativeUnit`)	Yes	Yes	Yes	Yes

Appendix E

Microsoft Entra ID connector settings

The following settings are configured for the system connection with the Microsoft Entra ID connector.

Table 56: Microsoft Entra ID connector settings

Setting	Meaning
Client ID	Application ID that was generated during integration of One Identity Manager as a Microsoft Entra ID tenant application. Variable: `CP_ClientID`
Login domain	Base domain or a verified domain of your Microsoft Entra ID tenant. Variable: `CP_OrganizationDomain`
User name	User account name for logging in on Microsoft Entra ID if you have integrated One Identity Manager as a local system client application in for Microsoft Entra ID tenant. Variable: `CP_Username`
Password	The user account's password. Variable: `CP_Password`
Key	Key that was generated during registration of One Identity Manager as a Microsoft Entra ID web application of the tenant. Variable: `CP_Secret`
Organization ID	The Microsoft Entra ID tenant ID. Variable: `OrganizationID`
GuestInviteSendMail	Specifies whether the guest user invitation will be sent. Default: **`True`**

Setting	Meaning
	Variable: `GuestInviteSendMail`
GuestInviteLanguage	Language to use for sending the guest user invitation. Default: **`en-us`** Variable: `GuestInviteLanguage`
GuestInviteCustomMessage	Personal welcome greeting for the guest user. Variable: `GuestInviteCustomMessage`
GuestInviteRedirectUrl	URL to reroute guest users after they have accepted the invitation and registered. Default: **`http://www.office.com`** Variable: `GuestInviteRedirectUrl`

About us

One Identity solutions eliminate the complexities and time-consuming processes often required to govern identities, manage privileged accounts and control access. Our solutions enhance business agility while addressing your IAM challenges with on-premises, cloud and hybrid environments.

Contacting us

For sales and other inquiries, such as licensing, support, and renewals, visit https://www.oneidentity.com/company/contact-us.aspx.

Technical support resources

Technical support is available to One Identity customers with a valid maintenance contract and customers who have trial versions. You can access the Support Portal at https://support.oneidentity.com/.

The Support Portal provides self-help tools you can use to solve problems quickly and independently, 24 hours a day, 365 days a year. The Support Portal enables you to:

- Submit and manage a Service Request
- View Knowledge Base articles
- Sign up for product notifications
- Download software and technical documentation
- View how-to videos at www.YouTube.com/OneIdentity
- Engage in community discussions
- Chat with support engineers online
- View services to assist you with your product

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